Contents

Thesis

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Abstract

We study the stationary queue-length behaviour of a 2-class $M/M/1$ queueing system with non-preemptive priority with two added mechanisms: *jockeying*, in which a custonmer in line switches to the other queue; and *abandonment*, in which a waiting customer may abandon the system entirely. While the former preserves the total number of customers in the system, it is not the case for the latter. Since combining both jockeying and abandonments leads to a very complex model (Model-X), we analyse a series of less mathematically involved submodels: Model-A (baseline), Model B (bidirectional jockeying), Model B_2 (one-way jockeying Class-1 to Class-2), and Model C_2 (class-1 abandonment)

The derivation of an expression for the probability generation function (PGF) $P(x, y)$ of each of these models is carried out using purely analytical techniques and, where possible, by means of probabilistic arguments. Model-A, widely studied, reduces to an algebraic expression solved by the kernel method; Model-B is left open, with a Volterra-type integral equation when it comes to finding the boundary for the queue length of Class-2 that falls beyond the scope of this thesis; However, Model-B_2 reduces to a first-order linear ODE in x, where we use the analyticity of the PGF to derive the boundary function of Class-2 at an interior point; and Model-C_2 reduces to another first-order linear ODE in x where we use the boundedness of our PGF at the boundary of its domain to determine the boundary function.

Lastly, the findings are validated numerically through simulations, showing how the models reduce to the baseline Model-A as the jockeying and/or abandonment parameters approximate 0.

1 Introduction

Motivation

Priority queues arise naturally in a wide range of contexts where a common server must satisfy heterogeneous demand. The primary motivation for this thesis comes from the context of healthcare operations. In particular, the management of waiting lists for access to elderly care facilities, where the medical severity dictates a user's —hereafter referred to as *customer*— priority class. In this context, prolonged waiting times may alter the system's dynamics, given that the lack of appropriate care may deteriorate the health condition of a waiting potential user, leading to a change in their priority status (jockeying) or leaving the system (abandonment), be it due to seeking alternative care or passing away. Despite the initial motivation, this work aims to address the problem from a general modeling, which may be applicable to a broad class of systems: characterising the long-run behaviour of queueing systems when phenomenons such as jockeying and/or abandonments play a role.

In this thesis, an analysis of a 2-class $M/M/1$ queueing system with non-preemptive priority is carried out. In short, this means that arrivals from each priority class are driven by a Poisson process with arrival rates λ_1 and λ_2. exponentially distributed service times at a rate μ —common to both priority classes—, and a single server that completes the current service despite the arrivals of higher-priority customers; with the addition of the mechanisms of jockeying and abandonments. The model and its mechanisms are described in more depth in Section 4.

The interplay of these two mechanisms within a priority discipline and how they affect the difficulty of obtaining a closed-form expression for their Probability Generating Function (PGF) are the pivotal objects of this thesis.

The analytical challenge

The combination of jockeying and abandonment in full generality defines Model X. The bivariate PGF satisfies a functional equation that involves terms of three kinds: algebraic coefficients, a partial derivative term in x —from jockeying and abandonments in Class-1— and a partial derivative term in y —from the same mechanisms in Class-2. When derivative terms of the two kinds are present, the equation belongs to a family of integro-differential equations for which no close-form solution is known, and it becomes a substantially challenging problem.

The strategy adopted here is to explore less complex models that require a use of less advanced mathematical tools by setting some of the jockeying and/or abandonment parameters equal to zero. The table below (Table 1) lists the models studied in this work and provide some insight of the challenge they imply:

Model	Active mechanism	Equation type	Status
A	none	algebraic PGF equation	solved
B	bidirectional jockeying	PDE, Volterra family	open
B_2	one-way jockeying ($1 \to 2$)	ODE in x, Kummer	solved
C_2	class-1 abandonment	ODE in x, Kummer	solved

As one can see, some models —in particular, Model-X and Model B— have been left incomplete, as they involve derivative terms in y, and some other models such a potential Model-B_1, Model-C or Model-C_1 have not even been considered for the same reason: they all reduce to a Volterra integral equation that falls beyond the scope of the work carried out in this thesis. Here Model-B serves as an example of how this sort of problems reduce to such an integral equation, whereas the other models correspond to problems we can solve.

Methods

Three complementary proof strategies are used throughout.

Kernel method / ODE with integrating factor. For Model-A, which does not involve derivative terms whatsoever, the algebraic coefficient of $P(x, y)$ vanishes on a curve (i.e. the *kernel*). Evaluating the equation there lets us determine the rest of the unknowns. Model-B_2 and Model-C_2, which do involve derivative terms in x, reduce to a first-order linear ODE in x and admit an explicit integrating factor. The unknowns there are obtained my means of a regilarity argument at a singularity point of the integrating factor. Both strategies are here explained as one, as the underlying idea is the same. One could say that the latter is the extension of the former when the unknowns are independent from the derivative terms and can be taken outside of the integral.

Probabilistic argument via Pollaczek–Khinchine and the effective service time. For Model-A and Model-C_2, we have Poisson arrivals for Class-2 customers —and therefore the PASTA property holds—, which allows us to utilize the Pollaczek–Khinchine formula, combined with the *effective service time* seen by Class-2 customers: an $M/G/1$ queueing system where the general service times are equivalent to the busy period of the Class-1 subsystem, which is an $M/M/1$ queueing system for Model-A and an $M/M/1 + M$ system for Model-B. In both cases, the probabilistic argument confirms the results obtained analytically, allowing to determine the unknown in an independent way. Since jockeying destroys the Poisson structure, this approach is not valid for the other models we studied.

Partial PGF (PPGF) in the alternative state space $\widetilde{S}$. For the models with no abandonments —Model-A, Model-B and Model-B_2—, we create an alternative space state that looks at these models by taking the number of Class-2 customers —difficult distribution— and the total amount of customers —easy, since the total number of customers is preserved and is basically a regular $M/M/1$ queueing system. This yields a PPGF with a non-homogeneous term. For Model-A, a recursion argument derived by Cohen [Coh56] applied to the regular state space. We explore, without much success, the challenges of such an approach on this alternative space and evaluate whether a rough approximation is feasible using Cohen's trick.

Structure of the thesis

Section 2 describes the relevant literature. Section 3 collects the probabilistic and analytic background used throughout this work. Section 4 defines Model X and derives the general balance equations and the fundamental PGF equations, both for the regular and alternative state space. Sections 6–9 analyse Models A, B, B_2, and C_2 in turn. Section 10 compares the three solved models. Section 11 presents the numerical validation and performance metrics. Section 12 introduces and solves the head-of-line variants.

2 Literature

The study of two-class priority queues with departure mechanisms sits at the intersection of three well-developed streams: exact PGF analysis of non-preemptive priority queues, the theory of queues with customer impatience (abandonment), and the theory of jockeying or priority changes. This thesis contributes an exact, single-server treatment that combines jockeying and abandonment within a unified PGF framework, filling a gap that the multi-server and approximate literature has so far left open.

Non-preemptive priority queues: exact queue-length analysis

Priority queues with abandonment

Jockeying and priority changes

Multi-server extensions and performance design

Position of this thesis

Table 1: Literature overview: models and mechanisms. ✓ = present, – = absent, ≈ = approximate only.

Reference	Servers	Non-preem.	Jockeying	Aband.	Queue-len.	Exact
Cohen (1956)	1	✓	–	–	✓	✓
de Waal (1992)	1	✓	✓	–	–	≈
Jouini & Roubos (2014)	c	✓	–	✓	–	✓
Wang et al. (2015)	c	–	–	–	✓	✓
Stanford et al. (2014)	1	✓	✓	–	–	✓
Zuk & Kirszenblat (2023)	c	✓	–	–	✓	✓
Baron et al. (2025)	c	✓	–	–	–	✓
Hu, Chan & Dong (2022)	c	✓	✓	✓	–	≈
Shmelev & Zychlinski (2025)	c	✓	–	✓	–	≈
This thesis (Model A)	**1**	✓	–	–	✓	✓
This thesis (Model B_2)	**1**	✓	✓	–	✓	✓
This thesis (Model C_2)	**1**	✓	–	✓	✓	✓

3 Preliminaries

This section collects background material referenced throughout the derivations. Introducing these building blocks here allows the subsequent arguments to remain focused on their primary objectives.

The central mathematical object in our derivations will be the (P)PGF of the model in question, as well as the idle probability of the system for some corollaries.

3.1 $M/M/1$ queue

The model described here is the canonical $M/M/1$ queue, well studied in multiple textbooks and manuals and considered the absolute foundation of queueing theory. Here, we will use [AR02] as our reference, but there is a lot of good quality literature and sources about it. Customers arrive at the system according to a Poisson process —i.e., the inter-arrival times are exponentially distributed with parameter λ—, service times are exponentially distributed with parameter μ, and a single server operates in first-come-first-serve (FCFS) —also known as first-in-first-out (FIFO)— order. The system is stable if and only if the system's load $\rho = \lambda/\mu < 1$.

Let π_n be the limiting probability of finding n customers in the system. Under stability, the detailed balance equations $\lambda\,\pi_n = \mu\,\pi_{n+1}$ ($n \geq 0$) together with the normalisation $\sum_{n=0}^{\infty} \pi_n = 1$ give us the geometric steady-state distribution of the number of customers in the system:

$$\pi_n = (1-\rho)\,\rho^n, \quad n \geq 0. \tag{1}$$

By the PASTA property (§**??**), this also corresponds to the distribution seen by the arriving customers. Let L be the random variable that tells us how many customers—including the one that might be in service—are in the system. The expected number of customers in the system is the following:

$$\mathbb{E}[L] = \frac{\rho}{1-\rho}.$$

From this, the expected sojourn time W—the time that elapses between a customer's arrival and their departure after service completion—is derived by Little's Law ($\mathbb{E}[L] = \lambda\mathbb{E}[W]$):

$$\mathbb{E}[W] = \frac{\mathbb{E}[L]}{\lambda} = \frac{1/\mu}{1-\rho}.$$

Another concept, more rare but highly relevant for the content that will be discussed in this thesis, is the Busy Period (BP) of the system, i.e., the time elapsed between the arrival of a customer to an idle system and the departure of a customer who leaves the system empty again after service completion. The Laplace-Stieltjes Transform (LST) and the expectation of this system's busy period are known:

$$\widetilde{BP}(s) = \frac{(\mu+\lambda+s) - \sqrt{(\mu+\lambda+s)^2 - 4\mu\lambda}}{2\lambda}, \tag{2}$$

$$\mathbb{E}[BP] = \frac{1}{\mu(1-\rho)}. \tag{3}$$

3.2 $M/M/1{+}M$ queue (Erlang-A)

Another relevant queueing model for this thesis is the so-called $M/M/1+M$ model, also known as the *Erlang A* model. It's based on the $M/M/1$ queue —Poisson arrivals at a rate λ and exponentially distributed waiting times with parameter μ-, but adds customer impatience to it: each waiting customer can independently abandon at an exponentially distributed waiting time with parameter θ. The key difference of this model with respect to the $M/M/1$ is that the $M/M/1+M$ does not require $\rho = \lambda/\mu \leq 1$ to be stable: it is positive recurrent for all $\lambda, \mu, \theta > 0$.

The steady state distribution follows, naturally, from its balance equations. At state $n \geq 0$ —for n being the number of customers in the *system*— we have

$$\lambda\,\pi_n = (\mu + n\theta)\,\pi_{n+1}, \qquad n \geq 0. \tag{4}$$

Iterating yields

$$\pi_n = \pi_0\,\frac{(\lambda/\theta)^n}{(\mu/\theta)^{(n)}} \quad n \geq 0; \quad \text{where} \quad a^{(n)} = a(a+1)\cdots(a+n-1) \tag{5}$$

Normalisation gives the non-trivial idle probability $\pi_0 = [{}_1F_1(1;\,\mu/\theta;\,\lambda/\theta)]^{-1}$ (see §3.4 for the Kummer function ${}_1F_1$).

The *Busy Period* of the $M/M/1+M$ defined here is far more complex than the one expressed in the $M/M/1$ queueing system. The LST and the expectation are given by

$$\widetilde{B}(s) = \cfrac{\mu}{\lambda_1+\mu+s-\cfrac{\lambda_1(\mu+\theta_1)}{\lambda_1+\mu+\theta_1+s-\cfrac{\lambda_1(\mu+2\theta_1)}{\lambda_1+\mu+2\theta_1+s-\cdots}}}, \tag{6}$$

$$\mathbb{E}[B] = \ldots \tag{7}$$

3.3 Pollaczek–Khinchine formula

The $M/G/1$ queue —explained in [AR02]— is a single-server queue with Poisson arrivals at a rate λ but with generally distributed i.i.d. service times B with mean $\mathbb{E}[B]$

The $M/G/1$ queue [AR02] is a single-server queue with Poisson arrivals at rate λ and generally distributed i.i.d. service times B with mean and LST $\widetilde{B}(s) = \mathbb{E}[e^{-sB}]$. For stability purposes, the system load $\rho = \lambda\,\mathbb{E}[B] < 1$ is required. Under these two conditions —Poisson arrivals and stability— the Pollaczek–Khinchine (PK) formula gives the PGF of the stationary number of customers in the system:

$$L(z) \;=\; \frac{(1-\rho_B)(1-z)\,\widetilde{B}(\lambda(1-z))}{\widetilde{B}(\lambda(1-z))-z}. \tag{8}$$

3.4 Confluent hypergeometric function

The *Kummer confluent hypergeometric function of the first kind* function will come up in some of the derivations of this thesis, and it reads as follows:

$${}_1F_1(a;\,b;\,z) \;=\; \sum_{n=0}^{\infty} \frac{a^{(n)}}{b^{(n)}\,n!}\,z^n, \qquad a^{(n)} = \prod_{k=0}^{n-1}(a+k), \quad a^{(0)} = 1, \tag{9}$$

where b is not 0 or a negative integer —i.e. $b \notin \{0,-1,-2,\ldots\}$— and the series converges for all $z \in \mathbb{C}$. The function ${}_1F_1(a;\,b;\,z)$ is the unique solution of *Kummer's equation* [DLM]:

$$z\,u'' + (b-z)\,u' - a\,u = 0, \tag{10}$$

normalized by ${}_1F_1(a;\,b;\,0) = 1$. The *Kummer series* (Equation (9)) converges for all $z \in \mathbb{C}$ and $b \notin \{0,-1,-2,\ldots\}$.

For this thesis, two identities related to it are used. First, the *Euler integral representation*, for $a, b > 0$ we have:

$${}_1F_1(a;\,a+b;\,z) \;=\; \frac{1}{B(a,b)}\int_0^1 t^{a-1}(1-t)^{b-1}e^{zt}\,dt, \tag{11}$$

where B is the *Beta function* $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, and Γ is the *Gamma function*. Secondly, the *Contiguous-function ratio*:

$$\frac{{}_1F_1(a+1;\,b;\,z)}{{}_1F_1(a;\,b;\,z)}. \tag{12}$$

3.5 First-order linear ODEs and PDEs

Since jockeying and abandonment add state-dependent rates to the balance equations, derivative terms end showing up in the derivations and they require solving first-order linear ODEs and PDEs for the PGF. The three methods below are invoked at specific steps of the analysis.

3.5.1 Integrating factor

This method is standard to solve —for each fixed y— a *first-order linear ODE in x*. Such an equation has standard form

$$\frac{dP}{dx} + p(x;\,y)\,P \;=\; q(x;\,y). \tag{13}$$

The *integrating factor* $\mu_I(x;y) = \exp\big(\int_0^x p(t;y)\,dt\big)$ converts (13) to $\frac{d}{dx}[P \cdot \mu_I] = q \cdot \mu_I$. Integrating from 0 to x gives the general solution

$$P(x,y) \;=\; \frac{1}{\mu_I(x;\,y)}\left[C_0(y) + \int_0^x q(t;\,y)\,\mu_I(t;\,y)\,dt\right], \tag{14}$$

where $C_0(y)$ is an arbitrary function of y, determined by a boundary condition.

3.5.2 Variation of parameters

Variation of parameters provides an alternative derivation of (14) that makes the role of the free function explicit. The homogeneous equation $\frac{dP}{dx} + p(x;\,y)\,P = 0$ has general solution $P_h(x;y) = C/\mu_I(x;y)$ for an arbitrary constant C. To solve the full equation, one *varies* that constant, setting $P(x,y) = C(x;\,y)/\mu_I(x;\,y)$ and substituting into (13). Since the homogeneous part cancels, one obtains the following.

$$\frac{dC}{dx}(x;\,y) \;=\; q(x;\,y)\,\mu_I(x;\,y), \tag{15}$$

and integrating recovers (14) with $C_0(y)$.

3.5.3 Method of Characteristics

This method is standard for solving a *first-order linear PDE in both x and y* simultaneously. It reduces the PDE to a family of ODEs along the so-called *characteristic curves* in the (x,y)-plane; along each curve, the PDE collapses to a single-variable ODE.

A PDE of the form

$$a\frac{\partial P}{\partial x} + b\frac{\partial P}{\partial y} = cP + d \tag{16}$$

is solved by tracing the characteristic curve $(x(t), y(t))$ satisfying

$$\frac{dx}{dt} = a, \qquad \frac{dy}{dt} = b. \tag{17}$$

By the chain rule, $\frac{d}{dt}P(x(t),y(t)) = a\,P_x + b\,P_y$, so (16) reduces to

$$\frac{dP}{dt} = cP + d, \tag{18}$$

a first-order linear ODE in t, solved by the integrating factor (§3.5.1). When a and b are constants, the characteristics are straight lines and the quantity

$$\xi \;:=\; b\,x - a\,y \tag{19}$$

is preserved along each curve —it is the *first integral* that labels each characteristic. The integration constant of (18) therefore varies between characteristics: it is an arbitrary function $F(\xi)$. Re-expressing the solution in the original variables (x,y) via (17) gives the general solution of (16). In the PGF setting, F is identified with a boundary generating function, and analyticity of $P(x,y)$ for $|x|, |y| \le 1$ uniquely determines it.

4 Model description: Model X

This section defines the general two-class system with both jockeying (γ_i) and abandonments (θ_i). We refer to this full model as *Model X*. The simpler variants studied are recovered as special cases by zeroing the appropriate parameters, as summarised in the table below (Table 2):

Model	Jockeying	Abandonment	Parameter restriction
X	both $(1 \leftrightarrow 2)$	both $(1, 2)$	$\gamma_i \geq 0,\ \theta_i \geq 0$ (general)
A	none	none	$\gamma_1 = \gamma_2 = 0,\ \theta_1 = \theta_2 = 0$
B	both $(1 \leftrightarrow 2)$	none	$\gamma_i \geq 0,\ \theta_1 = \theta_2 = 0$
B_2	$1 \rightarrow 2$ only	none	$\gamma_1 \geq 0,\ \gamma_2 = 0,\ \theta_1 = \theta_2 = 0$
C_2	none	class 1 only	$\theta_1 \geq 0,\ \theta_2 = 0,\ \gamma_1 = \gamma_2 = 0$

Table 2: Nested family of models. Model X is the parent; Models A, B, B_2 and C_2 are obtained by restricting the jockeying directions and the abandoning classes.

Figure 1 provides a visual overview of Model X and each of the variants that we will consider:

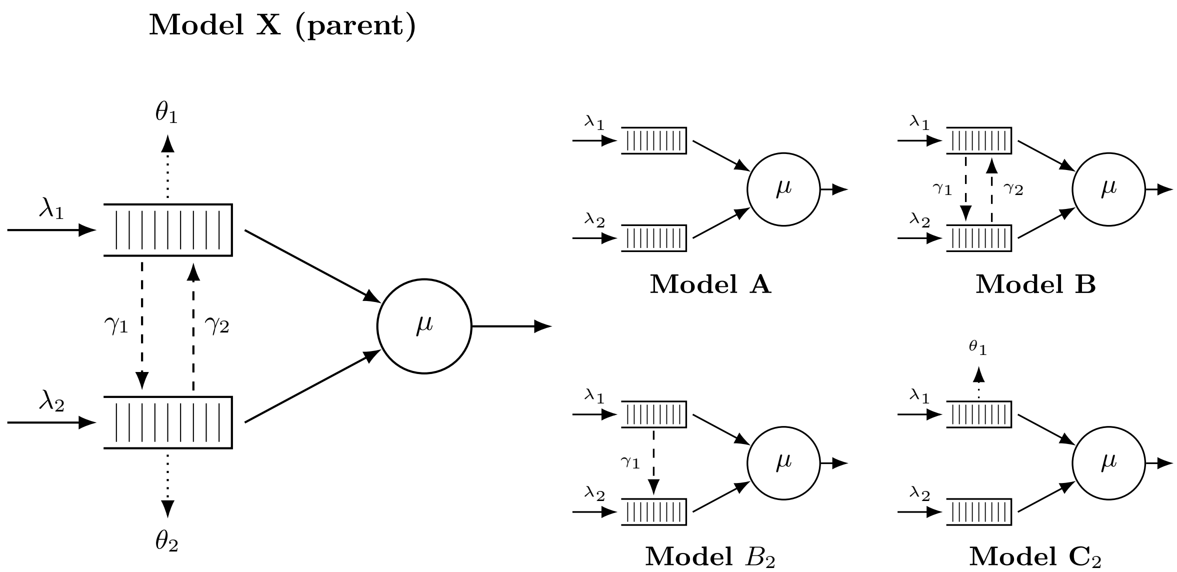

Figure 1: Physical queue layout for Model X (left) and its four specializations (right, 2×2 grid). Queue 1 (top buffer) has non-preemptive priority over Queue 2. Solid arrows: Poisson arrivals (λ_i) and exponential service (μ). Dashed arrows: jockeying (γ_i), which transfers a waiting customer between queues and conserves the total n. Dotted arrows: abandonment (θ_i), a true departure that reduces n.

The model discussed in this section is an $M/M/1$ queueing system with 2 priority classes, where customers are served on a First-Come, First-Served (FCFS) basis within their own class and those coming from the priority class (Class-1) have non-preemptive priority over those of the non-priority class (Class-2); that is, when a Class-2 customer is in service, that service will not be interrupted by a potential arrival of a Class-1 customer. Beyond basic dynamics, Model X incorporates two distinct queue-leaving mechanisms. First, *jockeying*: a waiting customer can abandon its own queue and *join the other queue*, at a per-customer rate γ_1 for direction $1 \rightarrow 2$ and γ_2 for direction $2 \rightarrow 1$, so the total number of customers is preserved. Second, *abandonment* (reneging): a waiting customer may completely leave the *system* at a rate per-customer θ_i, so the total number of customers decreases by one. Concretely, each Class-i customer that is waiting in line (i.e. not the one in service) holds an independent $\mathrm{Exp}(\theta_i)$ patience clock and abandons the system when it expires; consequently, the aggregate Class-i abandonment rate in a state with n_i waiting customers is the linear term $\theta_i n_i$ that appears in the balance equations. This is precisely the linear-reneging mechanism of the $M/M/1+M$ queue introduced in Section 3.2.

This model, being an $M/M/1$ queueing model according to Kendall's notation, features arrivals driven by a Poisson process with arrival rates λ_i for Class-i, exponentially distributed service times with rate μ, and a service capacity of 1 customer at a time. In addition, a waiting customer of Class-1 jockeys in the class-2 queue at the per-customer rate γ_1 and a waiting customer of Class-2 jockeys in the class-1 queue at the per-customer rate γ_2 (transfers that conserve the total), while a waiting customer of Class-i abandons the system at the per-customer rate θ_i (a true departure that reduces the total).

In order to define the stochastic processes that drive this queueing system, it is necessary to introduce the following random variables:

- $\mathcal{B}$: binary indicator for the server being idle or busy, $\mathcal{B} \in \{0, 1\}$;
- N_1, N_2 : number of customers of class 1 and 2, respectively, in their respective queues;
- N : total number of customers in the queues.

Note that the variables N_1, N_2 and N count the number of customers in the queues, as opposed to the customers in the system. This definition is convenient given that we have a common service rate for both priority classes, and hence the states of the system are equivalent regardless of the priority class the customer in service comes from. Since having customers in line directly implies that someone is in service, we will denote the idle state as (0) and the rest of the states by the number of people in each queue as the tuple (n_1, n_2), where n_i denotes the number of customers in the queue for class-i. This is the most conventional way to define the state space, namely S, in such a queueing system. However, we can also look at the total number of customers in the queues. We can define an alternative state space $\widetilde{S}$ using the idle state (0) and the pair (n_2, n) —we could also choose (n_1, n)—, which gives that we have $(n - n_2)$ Class-1 customers in the priority queue. Note that, since n is the total number of customers, we need the constraint $n \geq n_2$. Therefore, we define our state spaces as follows:

$$S := \{(0)\} \cup \{(n_1, n_2) \mid n_1, n_2 \geq 0\}, \tag{20}$$

$$\widetilde{S} := \{(0)\} \cup \{(n_2, n) \mid n \geq n_2 \geq 0\}. \tag{21}$$

For these state spaces, we define the limiting —as opposed to time-dependent— probabilities of finding the system in that state in the long run as follows:

- π_0 for the idle state (0), which belongs to both spaces S and $\widetilde{S}$;
- $\pi(n_1, n_2)$ for states $(n_1, n_2) \in S$, where $n_1, n_2 \geq 0$;
- $\widetilde{\pi}(n_2, n)$ for states $(n_2, n) \in \widetilde{S}$, where $n \geq n_2 \geq 0$.

Regarding the parameters of the system, the arrival process for each class follows a Poisson process (i.e., the interarrival times are exponentially distributed). By the properties of Poisson processes, the total number of arrivals will also form a Poisson process. Furthermore, we have a common service rate for exponentially distributed service times. We can organize our notation into class-specific parameters (alongside our common service rate) and class-blind totals. For the class-specific metrics and service rate, we define:

- λ_1, λ_2: arrival rates of classes 1 and 2, respectively;
- μ: service rate of a customer, regardless of class;
- $\rho_i = \frac{\lambda_i}{\mu}$: system load of class $i \in \{1, 2\}$;
- γ_1, γ_2: per-customer jockeying rates from Class $1 \to 2$ and $2 \to 1$, respectively; jockeying transfers a customer between queues, preserving the total number of customers;
- θ_1, θ_2: per-customer abandonment (reneging) rates from classes 1 and 2, respectively; abandonment removes a customer from the system, reducing the total n by one.

For the aggregate system parameters, we define:

- $\lambda = \lambda_1 + \lambda_2$: total arrival rate;
- $\rho = \frac{\lambda}{\mu}$: total system load.

Stability. The stability conditions depend on the abandonments; jockeying does not affect this condition, since customers are merely relocated. Model-X, having abandonments from both classes, is always stable, as the total number of customers behaves like some sort of $M/M/1+M$ —which is always stable, as stated before 3.2. The rate out is $\mu + \theta_1 n_1 + \theta_2 n_2$, which grows to infinity in every direction of the (n_1, n_2)-plane, so the system is always positive-recurrent. In contrast, Model-A, having no abandonments whatsoever, requires $\rho \leq 1$, since the total number of customers behaves exactly like an $M/M/1$ queueing system (3.2). It is more complex for models where only one type of abandonment is possible, this will be discussed in the dedicated section for Model-C_2 (9) .

Computation of π_0 and $\pi(0,0)$.

The following relation between π_0 and $\pi(0,0)$ holds no only for Model-X but also for every other model we will see in this work. The only transition into state (0) is a service completion from $(0,0)$, and the only transition out from $(0,0)$ is an arrival of a customer from any class to an idle system, leaving us whe following balance equation: $\lambda\pi_0 = \mu\,\pi(0,0)$. Therefore

$$\pi(0,0) \;=\; \frac{\lambda}{\mu}\,\pi_0 \;=\; \rho\,\pi_0. \tag{22}$$

Note that this holds regardless jockeying and abandonments, as in neither of these states we have waiting customers. However, what does depend on the abandonment parameters —not on the jockeying ones— is the value of π_0 itself. For models *without* abandonment, the total number of customers in the system behaves like that of an ordinary $M/M/1$ single-class system, giving

$$\pi_0 = 1 - \rho = 1 - \frac{\lambda_1 + \lambda_2}{\mu}, \qquad \pi(0,0) = \rho\,\pi_0 = (1-\rho)\rho. \tag{23}$$

However, for models *with* abandonments, the total number of customers is no longer conserved. For a busy state, the departure rate is $\mu + \theta_1 n_1 + \theta_2 n_2$, so the rate of state (n_1, n_2) depends on the split between n_1 and n_2. Indeed, the states $(n, 0)$ and $(0, n)$ carry departure rates $\mu + \theta_1 n$ and $\mu + \theta_2 n$, which differ whenever for assymetric abandonment rates $\theta_1 \neq \theta_2$. To determine π_0, we will need our bivariate PGF, which will be defined in Section 4.2.

4.1 Balance equations

This subsection presents the state-transition diagrams of Model X on both state spaces S and $\widetilde{S}$, and derives the corresponding sets of balance equations.

4.1.1 Balance equations for state space S

The state-transition diagram of Model-X on S is given in Figure 2.

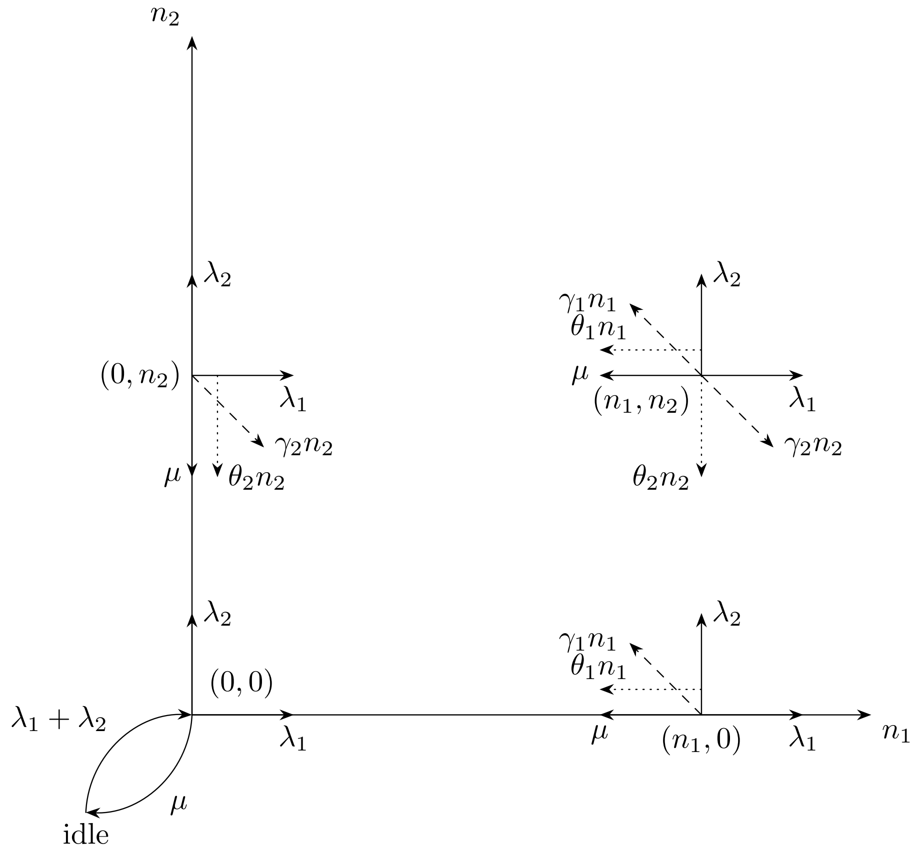

Figure 2: State transition diagram on state space S for the model with jockeying ($\gamma_i n_i$, dashed) and abandonments ($\theta_i n_i$, dotted).

$$
\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)n_1 + (\gamma_2 + \theta_2)n_2]\,\pi(n_1, n_2) \\
&\quad = \quad \mu\pi(n_1 + 1, n_2) \\
&\qquad + \lambda_1\pi(n_1 - 1, n_2) \\
&\qquad + \lambda_2\pi(n_1, n_2 - 1) \\
&\qquad + \gamma_1(n_1 + 1)\pi(n_1 + 1, n_2 - 1) \\
&\qquad + \gamma_2(n_2 + 1)\pi(n_1 - 1, n_2 + 1) \\
&\qquad + \theta_1(n_1 + 1)\pi(n_1 + 1, n_2) \\
&\qquad + \theta_2(n_2 + 1)\pi(n_1, n_2 + 1)
\end{aligned}
\qquad \text{for} \quad n_1 \geq 1, n_2 \geq 1 \tag{24}
$$

$$
\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_2 + \theta_2)n_2]\,\pi(0, n_2) \\
&\quad = \quad \lambda_2\pi(0, n_2 - 1) \\
&\qquad + \mu\pi(1, n_2) \\
&\qquad + \mu\pi(0, n_2 + 1) \\
&\qquad + \gamma_1\pi(1, n_2 - 1) \\
&\qquad + \theta_1\pi(1, n_2) \\
&\qquad + \theta_2(n_2 + 1)\pi(0, n_2 + 1)
\end{aligned}
\qquad \text{for} \quad n_1 = 0, n_2 \geq 1 \tag{25}
$$

$$
\begin{aligned}
&[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n_1]\,\pi(n_1,0)\\
&\quad=\quad \lambda_1\pi(n_1-1,0)\\
&\qquad+\mu\pi(n_1+1,0)\\
&\qquad+\gamma_2\pi(n_1-1,1)\\
&\qquad+\theta_1(n_1+1)\pi(n_1+1,0)\\
&\qquad+\theta_2\pi(n_1,1)
\end{aligned}
\qquad \text{for} \quad n_1\geq 1, n_2=0 \tag{26}
$$

$$[\lambda_1+\lambda_2]\,\pi(0,0)=(\mu+\theta_1)\pi(1,0)+(\mu+\theta_2)\pi(0,1) \tag{27}$$

$$(\lambda_1+\lambda_2)\pi_0=\mu\pi(0,0) \tag{28}$$

4.1.2 Balance equations for state space $\widetilde{S}$

Recall the state space $\widetilde{S}$ defined in (21).

For the sake of readability, we will denote the limiting probabilities associated to this state space with $\widetilde{\pi}(n_2,n)$, for state $(n_2,n)\in\widetilde{S}$.

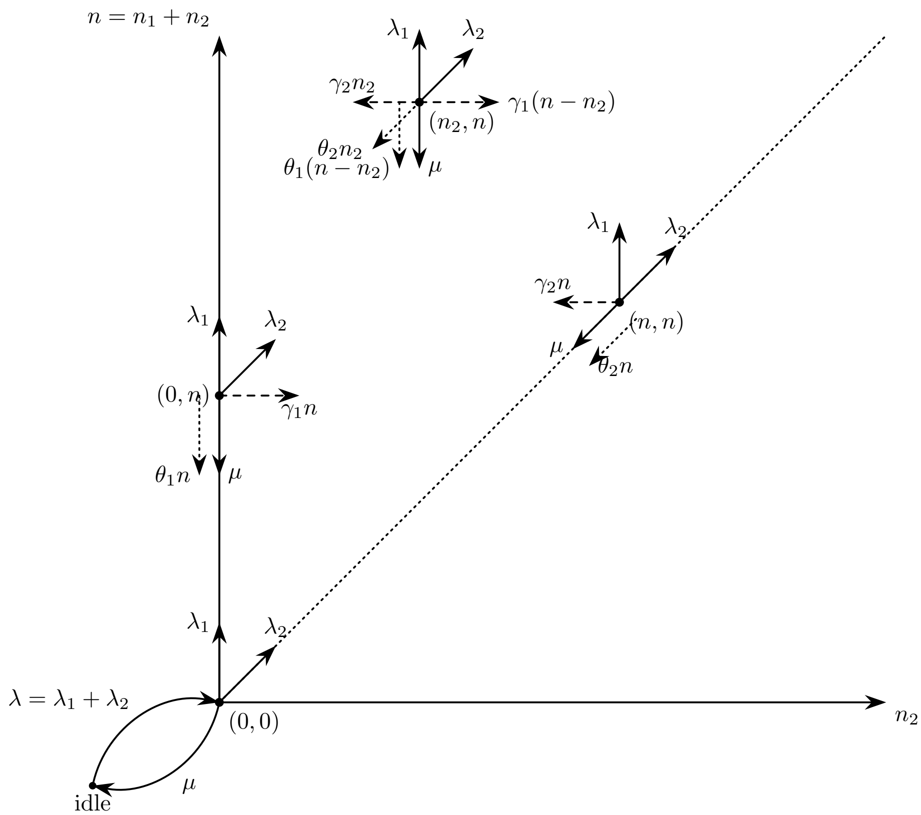

Figure 3: State transition diagram on state space $\widetilde{S}$.

The balance equations, after combining the equations where $\widetilde{\pi}_0$ shows up, are:

$$
\begin{aligned}
&[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)(n-n_2)+(\gamma_2+\theta_2)n_2]\,\widetilde{\pi}(n_2,n)\\
&\quad=\quad \lambda_1\widetilde{\pi}(n_2,n-1)\\
&\qquad+\lambda_2\widetilde{\pi}(n_2-1,n-1)\\
&\qquad+\mu\widetilde{\pi}(n_2,n+1)\\
&\qquad+\gamma_1(n-n_2+1)\widetilde{\pi}(n_2-1,n)\\
&\qquad+\gamma_2(n_2+1)\widetilde{\pi}(n_2+1,n)\\
&\qquad+\theta_1(n+1-n_2)\widetilde{\pi}(n_2,n+1)\\
&\qquad+\theta_2(n_2+1)\widetilde{\pi}(n_2+1,n+1)
\end{aligned}
\qquad \text{for} \quad 1\leq n_2\leq n-1,\ n\geq 2 \tag{29}
$$

$$\begin{aligned}
&[\lambda_1+\lambda_2+\mu+(\gamma_2+\theta_2)n]\,\widetilde{\pi}(n,n) \\
&\quad = \quad \lambda_2\widetilde{\pi}(n-1,n-1) \\
&\qquad + \mu\widetilde{\pi}(n+1,n+1) \\
&\qquad + \mu\widetilde{\pi}(n,n+1) && \text{for} \quad n_2=n,\ n\geq 1 \qquad (30)\\
&\qquad + \gamma_1\widetilde{\pi}(n-1,n) \\
&\qquad + \theta_1\widetilde{\pi}(n,n+1) \\
&\qquad + \theta_2(n+1)\widetilde{\pi}(n+1,n+1) \\
&[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n]\,\widetilde{\pi}(0,n) \\
&\quad = \quad \lambda_1\widetilde{\pi}(0,n-1) \\
&\qquad + \mu\widetilde{\pi}(0,n+1) \\
&\qquad + \gamma_2\widetilde{\pi}(1,n) && \text{for} \quad n_2=0,\ n\geq 1 \qquad (31)\\
&\qquad + \theta_1(n+1)\widetilde{\pi}(0,n+1) \\
&\qquad + \theta_2\widetilde{\pi}(1,n+1) \\
&[\lambda_1+\lambda_2]\,\widetilde{\pi}(0,0) \\
&\quad = \quad (\mu+\theta_1)\widetilde{\pi}(0,1)+(\mu+\theta_2)\widetilde{\pi}(1,1) && (32)\\
&(\lambda_1+\lambda_2)\widetilde{\pi}_0 = \mu\widetilde{\pi}(0,0) && (33)
\end{aligned}$$

4.2 Derivation of equations for (Partial) Probability Generating functions

In order to study the limiting behaviour of this two-class system, we define the bivariate Probability Generating Function (PGF) of the joint number of customers in the queue for each class, N_1 and N_2:

$$P(x,y)=\mathbb{E}[x^{N_1}y^{N_2}]=\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}. \tag{34}$$

The derivation that will be carried out here includes the boundary terms, where at least one of the two queues is empty:

$$P_x(x)=P(x,0)=\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1} \tag{35}$$

$$P_y(y)=P(0,y)=\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}. \tag{36}$$

Note that both of these terms include the state where the server is busy but both queues are empty, which is represented by $\pi(0,0)$, while the state where the entire system is empty (the idle state) is treated separately.

Another term that arises during the application of the balance equations characterizes the boundary dynamics when the class-1 queue contains exactly one waiting customer:

$$P_1(y)=\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}. \tag{37}$$

This term serves as an intermediate step; as the derivation proceeds, it will be possible to express it in terms of the primary boundary functions $P_x(x)$ and $P_y(y)$.

In the same fashion, we introduce the concept of the **Partial Probability Generating Function (PPGF)**. This technique, notably utilized by Cohen [Coh56], consists of transforming only a subset of the random variables while maintaining the discrete indices of the remainder. In the context of a priority queue, this method is particularly advantageous: it allows us to exploit the well-known properties of the priority class (often a standard $M/M/1$ system) and focus the analytical effort on the non-trivial PGF of the non-priority class.

While Cohen originally developed this framework for multi-server systems, we apply it here to the single-server case. To avoid the potential confusion stemming from mid-20th-century notation, we provide a modern formulation consistent with the variables defined above.

We define the PPGF with respect to the second variable N_2, conditioned on the first queue being in state n_1, as:

$$\widetilde{P}(n_1, y) = \mathbb{E}\left[y^{N_2}\mathbb{1}_{\{N_1=n_1\}}\right] = \sum_{n_2=0}^{\infty} \pi(n_1, n_2) y^{n_2}. \tag{38}$$

Similarly, the transform with respect to the first variable N_1, while maintaining the discrete state n_2 for the second queue, is defined as:

$$\widetilde{P}(x, n_2) = \mathbb{E}\left[x^{N_1}\mathbb{1}_{\{N_2=n_2\}}\right] = \sum_{n_1=0}^{\infty} \pi(n_1, n_2) x^{n_1}. \tag{39}$$

These partial transforms are linked to the joint PGF by the following relations:

$$P(x, y) = \sum_{n_1=0}^{\infty} \widetilde{P}(n_1, y) x^{n_1} = \sum_{n_2=0}^{\infty} \widetilde{P}(x, n_2) y^{n_2}. \tag{40}$$

4.2.1 Equation for PGF in state space S

Lemma 1 (Fundamental equation for the PGF on S)**.** *Consider the two-class non-preemptive priority queue on the state space $S = \{(n_1, n_2) : n_1, n_2 \geq 0\}$, with Poisson arrival rates λ_1, λ_2, exponential service rate μ, jockeying rates γ_1, γ_2 and abandonment rates θ_1, θ_2, where n_1 and n_2 count the class-*1 *and class-*2 *customers* waiting *in the queues. Assume the associated CTMC is positive recurrent, so that the stationary distribution $\{\pi(n_1, n_2)\}$ exists and the joint PGF*

$$P(x, y) = \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(n_1, n_2)\, x^{n_1} y^{n_2}$$

is analytic on the closed unit polydisc $|x| \leq 1,\ |y| \leq 1$. Then $P(x, y)$, together with the boundary function $P_y(y) = P(0, y) = \sum_{n_2=0}^{\infty} \pi(0, n_2) y^{n_2}$, satisfies the first-order linear partial differential equation

$$\begin{aligned}
&\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x, y) \\
&+ xy\left[\gamma_1(x - y) + \theta_1(x - 1)\right] \frac{\partial}{\partial x} P(x, y) \\
&+ xy\left[\gamma_2(y - x) + \theta_2(y - 1)\right] \frac{\partial}{\partial y} P(x, y) \\
&\quad = \quad \mu(x - y)P_y(y) - \mu x(1 - y)\, \pi(0, 0).
\end{aligned} \tag{41}$$

Proof. We multiply each balance equation by $x^{n_1} y^{n_2}$ and sum over its domain.

From the interior equation (24):

$$\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&+\gamma_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&+\gamma_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_2\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&+\theta_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&+\theta_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_2\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&\quad= \mu\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1+1,n_2)x^{n_1}y^{n_2} \\
&\quad+\lambda_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1-1,n_2)x^{n_1}y^{n_2} \\
&\quad+\lambda_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)x^{n_1}y^{n_2} \\
&\quad+\gamma_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2-1)x^{n_1}y^{n_2} \\
&\quad+\gamma_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_2+1)\pi(n_1-1,n_2+1)x^{n_1}y^{n_2} \\
&\quad+\theta_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2)x^{n_1}y^{n_2} \\
&\quad+\theta_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_2+1)\pi(n_1,n_2+1)x^{n_1}y^{n_2}
\end{aligned} \tag{42}$$

$$\begin{aligned}
&LHS_1+LHS_2+LHS_3+LHS_4+LHS_5 \\
&\quad= RHS_1+RHS_2+RHS_3+RHS_4+RHS_5+RHS_6+RHS_7
\end{aligned} \tag{43}$$

Evaluating each summand:

$$\begin{aligned}
&LHS_1 \\
&= (\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&= (\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\left[\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\pi(n_1,0)x^{n_1}y^0\right] \\
&= (\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\sum_{n_1=1}^{\infty}\pi(n_1,0)x^{n_1}\right] \\
&= (\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}\right] \\
&- (\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}-\pi(0,0)\right] \\
&= (\lambda_1+\lambda_2+\mu)\left[P(x,y)-P_y(y)-P_x(x)+\pi(0,0)\right]
\end{aligned} \tag{44}$$

LHS_2

$$\begin{aligned}
&= \gamma_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_1\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&= \gamma_1 x \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} \\
&= \gamma_1 x \left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} - \sum_{n_1=1}^{\infty} n_1\pi(n_1,0)x^{n_1-1}\right] \\
&= \gamma_1 x \left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} - \sum_{n_1=0}^{\infty} n_1\pi(n_1,0)x^{n_1-1}\right] \\
&= \gamma_1 x \left[\frac{\partial}{\partial x}P(x,y) - \frac{d}{dx}P_x(x)\right]
\end{aligned} \tag{45}$$

LHS_3

$$\begin{aligned}
&= \gamma_2 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&= \gamma_2 \sum_{n_1=1}^{\infty}\left[\sum_{n_2=0}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2} - 0\cdot\pi(n_1,0)x^{n_1}\right] \\
&= \gamma_2 y \left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2-1} - \sum_{n_2=0}^{\infty} n_2\pi(0,n_2)y^{n_2-1}\right] \\
&= \gamma_2 y \left[\frac{\partial}{\partial y}P(x,y) - \frac{d}{dy}P_y(y)\right]
\end{aligned} \tag{46}$$

LHS_4

$$\begin{aligned}
&= \theta_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_1\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&= \theta_1 x \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} \\
&= \theta_1 x \left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} - \sum_{n_1=1}^{\infty} n_1\pi(n_1,0)x^{n_1-1}\right] \\
&= \theta_1 x \left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} - \sum_{n_1=0}^{\infty} n_1\pi(n_1,0)x^{n_1-1}\right] \\
&= \theta_1 x \left[\frac{\partial}{\partial x}P(x,y) - \frac{d}{dx}P_x(x)\right]
\end{aligned} \tag{47}$$

LHS_5

$$\begin{aligned}
&= \theta_2 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&= \theta_2 y \left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2-1} - \sum_{n_2=0}^{\infty} n_2\pi(0,n_2)y^{n_2-1}\right] \\
&= \theta_2 y \left[\frac{\partial}{\partial y}P(x,y) - \frac{d}{dy}P_y(y)\right]
\end{aligned} \tag{48}$$

$$
\begin{aligned}
&RHS_1 \\
&= \mu \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1+1, n_2) x^{n_1} y^{n_2} \\
&= \mu x^{-1} \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1+1, n_2) x^{n_1+1} y^{n_2} \\
&= \mu x^{-1} \sum_{m=2}^{\infty} \sum_{n_2=1}^{\infty} \pi(m, n_2) x^{m} y^{n_2} \\
&= \mu x^{-1} \sum_{m=2}^{\infty} \left[\sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \pi(m,0) x^m \right] \qquad (49) \\
&= \mu x^{-1} \left[\sum_{m=2}^{\infty} \sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \sum_{m=2}^{\infty} \pi(m,0) x^m \right] \\
&= \mu x^{-1} \left[\sum_{m=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - x \sum_{n_2=0}^{\infty} \pi(1,n_2) y^{n_2} - \sum_{n_2=0}^{\infty} \pi(0,n_2) y^{n_2} \right] \\
&- \mu x^{-1} \left[\sum_{m=0}^{\infty} \pi(m,0) x^m - x\pi(1,0) - \pi(0,0) \right] \\
&= \mu x^{-1} \left[P(x,y) - x P_y(y|n_1=1) - P_y(y) - P_x(x) + x\pi(1,0) + \pi(0,0) \right]
\end{aligned}
$$

$$
\begin{aligned}
&RHS_2 \\
&= \lambda_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1-1, n_2) x^{n_1} y^{n_2} \\
&= \lambda_1 x \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1-1, n_2) x^{n_1-1} y^{n_2} \\
&= \lambda_1 x \sum_{m=0}^{\infty} \sum_{n_2=1}^{\infty} \pi(m, n_2) x^{m} y^{n_2} \qquad (50) \\
&= \lambda_1 x \sum_{m=0}^{\infty} \left[\sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \pi(m,0) x^m \right] \\
&= \lambda_1 x \left[\sum_{m=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \sum_{m=0}^{\infty} \pi(m,0) x^m \right] \\
&= \lambda_1 x \left[P(x,y) - P_x(x) \right]
\end{aligned}
$$

$$
\begin{aligned}
&RHS_3 \\
&= \lambda_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1, n_2-1) x^{n_1} y^{n_2} \\
&= \lambda_2 y \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1, n_2-1) x^{n_1} y^{n_2-1} \qquad (51) \\
&= \lambda_2 y \sum_{n_1=1}^{\infty} \sum_{m=0}^{\infty} \pi(n_1, m) x^{n_1} y^{m} \\
&= \lambda_2 y \left[\sum_{n_1=0}^{\infty} \sum_{m=0}^{\infty} \pi(n_1, m) x^{n_1} y^{m} - \sum_{m=0}^{\infty} \pi(0,m) y^m \right] \\
&= \lambda_2 y \left[P(x,y) - P_y(y) \right]
\end{aligned}
$$

$$
\begin{aligned}
&RHS_4\\
&= \gamma_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2-1)x^{n_1}y^{n_2}\\
&= \gamma_1 y \sum_{n_1=1}^{\infty}\sum_{m_2=0}^{\infty}(n_1+1)\pi(n_1+1,m_2)x^{n_1}y^{m_2}\\
&= \gamma_1 y \sum_{m_1=2}^{\infty}\sum_{m_2=0}^{\infty} m_1\pi(m_1,m_2)x^{m_1-1}y^{m_2}\\
&= \gamma_1 y \left[\sum_{m_1=0}^{\infty}\sum_{m_2=0}^{\infty} m_1\pi(m_1,m_2)x^{m_1-1}y^{m_2} - \sum_{m_2=0}^{\infty}\pi(1,m_2)y^{m_2}\right]\\
&= \gamma_1 y \left[\frac{\partial}{\partial x}P(x,y) - P_y(y|n_1=1)\right]
\end{aligned} \tag{52}
$$

$$
\begin{aligned}
&RHS_5\\
&= \gamma_2 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_2+1)\pi(n_1-1,n_2+1)x^{n_1}y^{n_2}\\
&= \gamma_2 \sum_{n_1=1}^{\infty}\sum_{m_2=2}^{\infty} m_2\pi(n_1-1,m_2)x^{n_1}y^{m_2-1}\\
&= \gamma_2 x \left[\sum_{n_1=1}^{\infty}\sum_{m_2=0}^{\infty} m_2\pi(n_1-1,m_2)x^{n_1-1}y^{m_2-1} - \sum_{n_1=1}^{\infty}\pi(n_1-1,1)x^{n_1-1}\right]\\
&= \gamma_2 x \left[\sum_{m_1=0}^{\infty}\sum_{m_2=0}^{\infty} m_2\pi(m_1,m_2)x^{m_1}y^{m_2-1} - \sum_{m_1=0}^{\infty}\pi(m_1,1)x^{m_1}\right]\\
&= \gamma_2 x \left[\frac{\partial}{\partial y}P(x,y) - P_x(x|n_2=1)\right]
\end{aligned} \tag{53}
$$

$$
\begin{aligned}
&RHS_6\\
&= \theta_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2)x^{n_1}y^{n_2}\\
&= \theta_1 \sum_{m_1=2}^{\infty}\sum_{n_2=1}^{\infty} m_1\pi(m_1,n_2)x^{m_1-1}y^{n_2}\\
&= \theta_1 \left[\sum_{m_1=0}^{\infty}\sum_{n_2=1}^{\infty} m_1\pi(m_1,n_2)x^{m_1-1}y^{n_2} - \sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\right]\\
&= \theta_1 \left[\sum_{m_1=0}^{\infty}\sum_{n_2=0}^{\infty} m_1\pi(m_1,n_2)x^{m_1-1}y^{n_2} - \sum_{m_1=0}^{\infty} m_1\pi(m_1,0)x^{m_1-1}\right]\\
&\quad - \theta_1 \left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2} - \pi(1,0)\right]\\
&= \theta_1 \left[\frac{\partial}{\partial x}P(x,y) - \frac{d}{dx}P_x(x) - P_y(y|n_1=1) + \pi(1,0)\right]
\end{aligned} \tag{54}
$$

$$
\begin{aligned}
& RHS_7 \\
= \quad & \theta_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_2+1)\pi(n_1, n_2+1) x^{n_1} y^{n_2} \\
= \quad & \theta_2 \sum_{n_1=1}^{\infty} \sum_{m_2=2}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} \\
= \quad & \theta_2 \left[\sum_{n_1=1}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} - \sum_{n_1=1}^{\infty} \pi(n_1, 1) x^{n_1} \right] \\
= \quad & \theta_2 \left[\sum_{n_1=0}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} - \sum_{m_2=0}^{\infty} m_2 \pi(0, m_2) y^{m_2-1} \right] \\
& - \theta_2 \left[\sum_{n_1=0}^{\infty} \pi(n_1, 1) x^{n_1} - \pi(0,1) \right] \\
= \quad & \theta_2 \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) - P_x(x|n_2=1) + \pi(0,1) \right]
\end{aligned} \tag{55}
$$

Collecting and multiplying through by xy to clear the x^{-1} factor:

$$
\begin{aligned}
& (\lambda_1 + \lambda_2 + \mu) \left[P(x,y) - P_y(y) - P_x(x) + \pi(0,0) \right] \\
& + \gamma_1 x \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) \right] \\
& + \gamma_2 y \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) \right] \\
& + \theta_1 x \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) \right] \\
& + \theta_2 y \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) \right] \\
& \quad = \quad \mu x^{-1} \left[P(x,y) - x P_y(y|n_1=1) - P_y(y) - P_x(x) + x\pi(1,0) + \pi(0,0) \right] \\
& \qquad + \lambda_1 x \left[P(x,y) - P_x(x) \right] \\
& \qquad + \lambda_2 y \left[P(x,y) - P_y(y) \right] \\
& \qquad + \gamma_1 y \left[\frac{\partial}{\partial x} P(x,y) - P_y(y|n_1=1) \right] \\
& \qquad + \gamma_2 x \left[\frac{\partial}{\partial y} P(x,y) - P_x(x|n_2=1) \right] \\
& \qquad + \theta_1 \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) - P_y(y|n_1=1) + \pi(1,0) \right] \\
& \qquad + \theta_2 \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) - P_x(x|n_2=1) + \pi(0,1) \right]
\end{aligned}
$$

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x-\lambda_2 y\right] P(x,y)\\
&+\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad= \quad \left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x\right] P_x(x)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_2 y\right] P_y(y)\\
&\qquad+\left[-(\lambda_1+\lambda_2+\mu)+\mu x^{-1}\right]\pi(0,0)\\
&\qquad+\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\qquad+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\qquad+\mu\pi(1,0)\\
&\qquad+\theta_1\pi(1,0)\\
&\qquad+\theta_2\pi(0,1)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 x y^2\right] P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad= \quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y\right] P_x(x)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 x y^2\right] P_y(y)\\
&\qquad-\left[(\lambda_1+\lambda_2+\mu)xy-\mu y\right]\pi(0,0)\\
&\qquad+xy\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\qquad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-xy\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-xy\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\qquad+(\mu+\theta_1)xy\,\pi(1,0)\\
&\qquad+\theta_2 xy\,\pi(0,1)
\end{aligned}
\tag{56}
$$

The x-boundary equation (26), treated analogously, gives:

$$
\begin{aligned}
&\sum_{n_1=1}^{\infty}\left[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n_1\right]\pi(n_1,0)x^{n_1}\\
&\quad= \quad \lambda_1\sum_{n_1=1}^{\infty}\pi(n_1-1,0)x^{n_1}+\mu\sum_{n_1=1}^{\infty}\pi(n_1+1,0)x^{n_1}\\
&\qquad+\gamma_2\sum_{n_1=1}^{\infty}\pi(n_1-1,1)x^{n_1}+\theta_1\sum_{n_1=1}^{\infty}(n_1+1)\pi(n_1+1,0)x^{n_1}+\theta_2\sum_{n_1=1}^{\infty}\pi(n_1,1)x^{n_1}
\end{aligned}
$$

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\pi(n_1,0)x^{n_1}+(\gamma_1+\theta_1)x\sum_{n_1=1}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\\
&\quad= \quad \lambda_1 x\sum_{m=0}^{\infty}\pi(m,0)x^m+\mu x^{-1}\sum_{m=2}^{\infty}\pi(m,0)x^m\\
&\qquad+\gamma_2 x\sum_{m=0}^{\infty}\pi(m,1)x^m+\theta_1\sum_{m=2}^{\infty}m\pi(m,0)x^{m-1}+\theta_2\sum_{m=1}^{\infty}\pi(m,1)x^m\\
&(\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}-\pi(0,0)\right]+(\gamma_1+\theta_1)x\sum_{n_1=0}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\\
&\quad= \quad \lambda_1 x\sum_{m=0}^{\infty}\pi(m,0)x^m\\
&\qquad+\mu x^{-1}\left[\sum_{m=0}^{\infty}\pi(m,0)x^m-x\pi(1,0)-\pi(0,0)\right]\\
&\qquad+\gamma_2 x\sum_{m=0}^{\infty}\pi(m,1)x^m\\
&\qquad+\theta_1\left[\sum_{m=0}^{\infty}m\pi(m,0)x^{m-1}-\pi(1,0)\right]\\
&\qquad+\theta_2\left[\sum_{m=0}^{\infty}\pi(m,1)x^m-\pi(0,1)\right]\\
&(\lambda_1+\lambda_2+\mu)\left[P_x(x)-\pi(0,0)\right]+(\gamma_1+\theta_1)x\frac{d}{dx}P_x(x)\\
&\quad= \quad \lambda_1 xP_x(x)+\mu x^{-1}\left[P_x(x)-x\pi(1,0)-\pi(0,0)\right]\\
&\qquad+\gamma_2 xP_x(x|n_2=1)+\theta_1\left[\frac{d}{dx}P_x(x)-\pi(1,0)\right]+\theta_2\left[P_x(x|n_2=1)-\pi(0,1)\right]\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x\right]P_x(x)+\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad= \quad \left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}\right]\pi(0,0)-(\mu+\theta_1)\pi(1,0)-\theta_2\pi(0,1)\\
&\qquad+\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y\right]P_x(x)+xy\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad= \quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y\right]\pi(0,0)\\
&\qquad-(\mu+\theta_1)xy\,\pi(1,0)\\
&\qquad-\theta_2 xy\,\pi(0,1)\\
&\qquad+xy\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)
\end{aligned}
\tag{57}
$$

The y-boundary equation (25) gives:

$$
\begin{aligned}
&\sum_{n_2=1}^{\infty} \left[\lambda_1 + \lambda_2 + \mu + (\gamma_2 + \theta_2) n_2\right] \pi(0, n_2) y^{n_2} \\
&\quad = \quad \lambda_2 \sum_{n_2=1}^{\infty} \pi(0, n_2 - 1) y^{n_2} \\
&\qquad + \mu \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \\
&\qquad + \mu \sum_{n_2=1}^{\infty} \pi(0, n_2 + 1) y^{n_2} \\
&\qquad + \gamma_1 \sum_{n_2=1}^{\infty} \pi(1, n_2 - 1) y^{n_2} \\
&\qquad + \theta_1 \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \\
&\qquad + \theta_2 \sum_{n_2=1}^{\infty} (n_2 + 1) \pi(0, n_2 + 1) y^{n_2} \\
&(\lambda_1 + \lambda_2 + \mu) \sum_{n_2=1}^{\infty} \pi(0, n_2) y^{n_2} \\
&+ (\gamma_2 + \theta_2) y \sum_{n_2=1}^{\infty} n_2 \pi(0, n_2) y^{n_2 - 1} \\
&\quad = \quad \lambda_2 y \sum_{m=0}^{\infty} \pi(0, m) y^{m} \\
&\qquad + \mu \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \\
&\qquad + \mu y^{-1} \sum_{m=2}^{\infty} \pi(0, m) y^{m} \\
&\qquad + \gamma_1 y \sum_{m=0}^{\infty} \pi(1, m) y^{m} \\
&\qquad + \theta_1 \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \\
&\qquad + \theta_2 \sum_{m=2}^{\infty} m \pi(0, m) y^{m-1}
\end{aligned}
$$

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}-\pi(0,0)\right]\\
&+(\gamma_2+\theta_2)y\sum_{n_2=0}^{\infty}n_2\pi(0,n_2)y^{n_2-1}\\
&\quad=\quad \lambda_2 y\sum_{m=0}^{\infty}\pi(0,m)y^m\\
&\qquad+\mu\left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2}-\pi(1,0)\right]\\
&\qquad+\mu y^{-1}\left[\sum_{m=0}^{\infty}\pi(0,m)y^m-y\pi(0,1)-\pi(0,0)\right]\\
&\qquad+\gamma_1 y\sum_{m=0}^{\infty}\pi(1,m)y^m\\
&\qquad+\theta_1\left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2}-\pi(1,0)\right]\\
&\qquad+\theta_2\left[\sum_{m=0}^{\infty}m\pi(0,m)y^{m-1}-\pi(0,1)\right]\\
&(\lambda_1+\lambda_2+\mu)\left[P_y(y)-\pi(0,0)\right]+(\gamma_2+\theta_2)y\frac{d}{dy}P_y(y)\\
&\quad=\quad \lambda_2 yP_y(y)+\mu\left[P_y(y|n_1=1)-\pi(1,0)\right]\\
&\qquad+\mu y^{-1}\left[P_y(y)-y\pi(0,1)-\pi(0,0)\right]+\gamma_1 yP_y(y|n_1=1)\\
&\qquad+\theta_1\left[P_y(y|n_1=1)-\pi(1,0)\right]+\theta_2\left[\frac{d}{dy}P_y(y)-\pi(0,1)\right]\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-(\mu+\theta_1)\pi(1,0)-(\mu+\theta_2)\pi(0,1)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}\right]\pi(0,0)\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-(\lambda_1+\lambda_2)\right]\pi(0,0)\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)+\left[\mu-\mu y^{-1}\right]\pi(0,0)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu x-\lambda_2 xy^2\right]P_y(y)+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]xy\,P_y(y|n_1=1)-\mu x(1-y)\,\pi(0,0)
\end{aligned}
\tag{58}
$$

Substituting (57) into (56) eliminates all P_x terms:

$$\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 xy^2\right]P_y(y)\\
&\qquad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-xy\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)
\end{aligned}\tag{59}$$

Isolating $[\mu+\gamma_1 y+\theta_1]\,xy\,P_y(y|n_1=1)$ from (58) and substituting into (59):

$$\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 xy^2\right]P_y(y)\\
&\qquad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-\Big\{\left[(\lambda_1+\lambda_2+\mu)xy-\mu x-\lambda_2 xy^2\right]P_y(y)\\
&\qquad\quad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)+\mu x(1-y)\,\pi(0,0)\Big\}\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\mu(x-y)P_y(y)-\mu x(1-y)\,\pi(0,0)
\end{aligned}\quad.\tag{60}$$

□

4.2.2 Equation for PPGF in state space $\widetilde{S}$

Lemma 2 (Fundamental equation for the PPGF on $\widetilde{S}$)**.** *Consider the same system on the state space $\widetilde{S}=\{(n_2,n):0\le n_2\le n\}$, where n is the total number of customers* waiting *in the queues and n_2 the number of waiting class-2 customers. Assume positive recurrence, so that the stationary distribution $\{\widetilde{\pi}(n_2,n)\}$ exists, and define the partial PGF (transforming only n_2, keeping n discrete)*

$$\widetilde{P}(y,n)=\sum_{n_2=0}^{n}\widetilde{\pi}(n_2,n)\,y^{n_2},$$

a polynomial of degree n in y. Then, for every $n\ge 1$, the family $\{\widetilde{P}(y,n)\}_{n\ge 0}$ satisfies the differential–difference equation

$$\begin{aligned}
&\left[-(\gamma_2+\gamma_1 y)(1-y)+(\theta_2-\theta_1)y\right]\frac{d}{dy}\widetilde{P}(y,n)+\left[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n\right]\widetilde{P}(y,n)\\
&\quad=\quad(\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+\left[\mu+\theta_1(n+1)\right]\widetilde{P}(y,n+1)\\
&\qquad+(\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1)+\mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1).
\end{aligned}\tag{61}$$

Proof. Multiply the interior equation (29) by y^{n_2} and sum over $1 \leq n_2 \leq n-1$:

$$\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_1\sum_{n_2=1}^{n-1}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_2\sum_{n_2=1}^{n-1}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_1\sum_{n_2=1}^{n-1}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_2\sum_{n_2=1}^{n-1}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&\quad=\quad \lambda_1\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n-1)y^{n_2}\\
&\qquad+\lambda_2\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2-1,n-1)y^{n_2}\\
&\qquad+\mu\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\qquad+\gamma_1\sum_{n_2=1}^{n-1}(n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2}\\
&\qquad+\gamma_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2}\\
&\qquad+\theta_1\sum_{n_2=1}^{n-1}(n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\qquad+\theta_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2}
\end{aligned}$$

Adding the diagonal equation (30) multiplied by y^n extends most summation limits to n; the λ_1-

and γ_2-terms on the RHS remain restricted to $n_2 \leq n-1$, and a spare $\mu\widetilde{\pi}(n+1,n+1)y^n$ survives:

$$\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_2\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_2\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&= \quad \lambda_1\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n-1)y^{n_2}\\
&\quad +\lambda_2\sum_{n_2=1}^{n}\widetilde{\pi}(n_2-1,n-1)y^{n_2}\\
&\quad +\mu\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\quad +\mu\widetilde{\pi}(n+1,n+1)y^{n}\\
&\quad +\gamma_1\sum_{n_2=1}^{n}(n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2}\\
&\quad +\gamma_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2}\\
&\quad +\theta_1\sum_{n_2=1}^{n}(n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\quad +\theta_2\sum_{n_2=1}^{n}(n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2}.
\end{aligned}$$

Evaluating each summand:

$$\begin{aligned}
&LHS_1\\
&= \quad (\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}\\
&= \quad (\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{n}\widetilde{\pi}(n_2,n)y^{n_2}-\widetilde{\pi}(0,n)y^0\right]\\
&= \quad (\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\right]
\end{aligned} \tag{62}$$

$$
\begin{aligned}
&LHS_2\\
&= \gamma_1 \sum_{n_2=1}^{n} (n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&= \gamma_1 \left[n\sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n)y^{n_2} - \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2}\right]\\
&= \gamma_1 \left[n\left(\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n)y^{n_2} - \widetilde{\pi}(0,n)y^0\right) - y\sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2-1}\right]\\
&= \gamma_1 \left[n\big(\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\big) - y\frac{d}{dy}\widetilde{P}(y,n)\right]
\end{aligned} \tag{63}
$$

$$
\begin{aligned}
&LHS_3\\
&= \gamma_2 \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&= \gamma_2 \left[y\sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2-1}\right]\\
&= \gamma_2 y\frac{d}{dy}\widetilde{P}(y,n)
\end{aligned} \tag{64}
$$

$$
\begin{aligned}
&LHS_4\\
&= \theta_1 \sum_{n_2=1}^{n} (n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&= \theta_1 \left[n\big(\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\big) - y\frac{d}{dy}\widetilde{P}(y,n)\right]
\end{aligned} \tag{65}
$$

$$
\begin{aligned}
&LHS_5\\
&= \theta_2 \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&= \theta_2 y\frac{d}{dy}\widetilde{P}(y,n)
\end{aligned} \tag{66}
$$

$$
\begin{aligned}
&RHS_1\\
&= \lambda_1 \sum_{n_2=1}^{n-1} \widetilde{\pi}(n_2,n-1)y^{n_2}\\
&= \lambda_1 \left[\sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n-1)y^{n_2} - \widetilde{\pi}(n,n-1)y^n\right]\\
&= \lambda_1 \left[\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n-1)y^{n_2} - \widetilde{\pi}(n,n-1)y^n - \widetilde{\pi}(0,n-1)y^0\right]\\
&= \lambda_1 \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1)y^n - \widetilde{\pi}(0,n-1)\right]
\end{aligned} \tag{67}
$$

$$
\begin{aligned}
&RHS_2 \\
&= \quad \lambda_2 \sum_{n_2=1}^{n} \widetilde{\pi}(n_2-1,n-1)y^{n_2} \\
&= \quad \lambda_2 \left[y \sum_{n_2=1}^{n} \widetilde{\pi}(n_2-1,n-1)y^{n_2-1} \right] \\
&= \quad \lambda_2 \left[y \sum_{n_2=0}^{n-1} \widetilde{\pi}(n_2,n-1)y^{n_2} \right] \\
&= \quad \lambda_2 \left[y \left(\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n-1)y^{n_2} - \widetilde{\pi}(n,n-1)y^{n} \right) \right] \\
&= \quad \lambda_2 y \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1)y^{n} \right]
\end{aligned} \tag{68}
$$

$$
\begin{aligned}
&RHS_3 \\
&= \quad \mu\widetilde{\pi}(n+1,n+1)y^n + \mu \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1)y^{n_2} \\
&= \quad \mu \left[\widetilde{\pi}(n+1,n+1)y^n + \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1)y^{n_2} \right] \\
&= \quad \mu \left[\widetilde{\pi}(n+1,n+1)y^n + \sum_{n_2=0}^{n+1} \widetilde{\pi}(n_2,n+1)y^{n_2} - \widetilde{\pi}(0,n+1)y^0 - \widetilde{\pi}(n+1,n+1)y^{n+1} \right] \\
&= \quad \mu \left[\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1) + \widetilde{\pi}(n+1,n+1)y^n(1-y) \right]
\end{aligned} \tag{69}
$$

$$
\begin{aligned}
&RHS_4 \\
&= \quad \gamma_1 \sum_{n_2=1}^{n} (n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2} \\
&= \quad \gamma_1 \left[y \sum_{n_2=1}^{n} (n-(n_2-1))\widetilde{\pi}(n_2-1,n)y^{n_2-1} \right] \\
&= \quad \gamma_1 \left[y \sum_{k=0}^{n-1} (n-k)\widetilde{\pi}(k,n)y^{k} \right] \\
&= \quad \gamma_1 \left[y \sum_{k=0}^{n} (n-k)\widetilde{\pi}(k,n)y^{k} \right] \quad \text{(since the } k=n \text{ term is 0)} \\
&= \quad \gamma_1 y \left[n \sum_{k=0}^{n} \widetilde{\pi}(k,n)y^{k} - y \sum_{k=1}^{n} k\widetilde{\pi}(k,n)y^{k-1} \right] \\
&= \quad \gamma_1 y \left[n\widetilde{P}(y,n) - y\frac{d}{dy}\widetilde{P}(y,n) \right]
\end{aligned} \tag{70}
$$

$$
\begin{aligned}
&RHS_5 \\
&= \quad \gamma_2 \sum_{n_2=1}^{n-1} (n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2} \\
&= \quad \gamma_2 \sum_{k=2}^{n} k\widetilde{\pi}(k,n)y^{k-1} \quad \text{(letting } k=n_2+1\text{)} \\
&= \quad \gamma_2 \left[\sum_{k=1}^{n} k\widetilde{\pi}(k,n)y^{k-1} - 1\cdot\widetilde{\pi}(1,n)y^0 \right] \\
&= \quad \gamma_2 \left[\frac{d}{dy}\widetilde{P}(y,n) - \widetilde{\pi}(1,n) \right]
\end{aligned} \tag{71}
$$

$$
\begin{aligned}
&RHS_6 \\
=\quad & \theta_1 \sum_{n_2=1}^{n} (n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2} \\
=\quad & \theta_1 \left[(n+1)\sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1)y^{n_2} - \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n+1)y^{n_2}\right] \\
=\quad & \theta_1 \left[(n+1)\left(\sum_{n_2=0}^{n+1} \widetilde{\pi}(n_2,n+1)y^{n_2} - \widetilde{\pi}(0,n+1) - \widetilde{\pi}(n+1,n+1)y^{n+1}\right)\right] \\
& -\theta_1 \left[y\left(\sum_{n_2=0}^{n+1} n_2\widetilde{\pi}(n_2,n+1)y^{n_2-1} - (n+1)\widetilde{\pi}(n+1,n+1)y^{n}\right)\right] \\
=\quad & \theta_1 \left[(n+1)\big(\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1)\big) - (n+1)\widetilde{\pi}(n+1,n+1)y^{n+1}\right] \\
& -\theta_1 \left[y\frac{d}{dy}\widetilde{P}(y,n+1) - (n+1)\widetilde{\pi}(n+1,n+1)y^{n+1}\right] \\
=\quad & \theta_1 \left[(n+1)\big(\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1)\big) - y\frac{d}{dy}\widetilde{P}(y,n+1)\right]
\end{aligned}
\tag{72}
$$

$$
\begin{aligned}
&RHS_7 \\
=\quad & \theta_2 \sum_{n_2=1}^{n} (n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2} \\
=\quad & \theta_2 \sum_{k=2}^{n+1} k\widetilde{\pi}(k,n+1)y^{k-1} \quad \text{(letting } k = n_2+1\text{)} \\
=\quad & \theta_2 \left[\sum_{k=1}^{n+1} k\widetilde{\pi}(k,n+1)y^{k-1} - 1\cdot\widetilde{\pi}(1,n+1)y^{0}\right] \\
=\quad & \theta_2 \left[\frac{d}{dy}\widetilde{P}(y,n+1) - \widetilde{\pi}(1,n+1)\right]
\end{aligned}
\tag{73}
$$

Collecting:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\right] \\
&+\gamma_1\left[n\big(\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\big)-y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&+\gamma_2 y\frac{d}{dy}\widetilde{P}(y,n) \\
&+\theta_1\left[n\big(\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\big)-y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&+\theta_2 y\frac{d}{dy}\widetilde{P}(y,n) \\
&\quad = \quad \lambda_1\left[\widetilde{P}(y,n-1)-\widetilde{\pi}(n,n-1)y^n-\widetilde{\pi}(0,n-1)\right] \\
&\qquad +\lambda_2 y\left[\widetilde{P}(y,n-1)-\widetilde{\pi}(n,n-1)y^n\right] \\
&\qquad +\mu\left[\widetilde{P}(y,n+1)-\widetilde{\pi}(0,n+1)+\widetilde{\pi}(n+1,n+1)y^n(1-y)\right] \\
&\qquad +\gamma_1 y\left[n\widetilde{P}(y,n)-y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&\qquad +\gamma_2\left[\frac{d}{dy}\widetilde{P}(y,n)-\widetilde{\pi}(1,n)\right] \\
&\qquad +\theta_1\left[(n+1)\big(\widetilde{P}(y,n+1)-\widetilde{\pi}(0,n+1)\big)-y\frac{d}{dy}\widetilde{P}(y,n+1)\right] \\
&\qquad +\theta_2\left[\frac{d}{dy}\widetilde{P}(y,n+1)-\widetilde{\pi}(1,n+1)\right]
\end{aligned}
$$

Regrouping:

$$
\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y)+(\theta_2-\theta_1)y]\,\frac{d}{dy}\widetilde{P}(y,n)+[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n) \\
&\quad = \quad (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+[\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1)+(\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1) \\
&\qquad +\{[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n]\,\widetilde{\pi}(0,n)-\lambda_1\widetilde{\pi}(0,n-1)-[\mu+\theta_1(n+1)]\,\widetilde{\pi}(0,n+1) \\
&\qquad\quad -\gamma_2\widetilde{\pi}(1,n)-\theta_2\widetilde{\pi}(1,n+1)\} \\
&\qquad -y^n(\lambda_1+\lambda_2 y)\widetilde{\pi}(n,n-1)+\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}
$$

The y-boundary equation (31) makes the bracketed terms vanish; since $\widetilde{\pi}(n,n-1)=0$ (outside $\widetilde{S}$):

$$
\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y)+(\theta_2-\theta_1)y]\,\frac{d}{dy}\widetilde{P}(y,n)+[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n) \\
&\quad = \quad (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+[\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1)+(\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1) \\
&\qquad +\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}
\tag{74}
$$

□

5 Analysis of Model X $(\gamma_1,\gamma_2>0,\ \theta_1,\theta_2>0)$

Model X is the full model: it retains *both* bidirectional jockeying $(\gamma_1,\gamma_2>0)$ *and* two-class abandonment $(\theta_1,\theta_2>0)$. It is the parent of the hierarchy, and every other model analysed in this work is recovered from it by zeroing parameters. Jockeying conserves the total number of customers while abandonment removes them; the second mechanism makes the chain positive recurrent for all loads, so the stationary distribution always exists. As in Model B, jockeying breaks the per-class Poisson/renewal structure — a Class-2 customer can no longer identify its effective service time with a

Class-1 busy period — and abandonment, although it leaves each arrival stream Poisson, couples the two queues through the state-dependent rates $\theta_i n_i$; in either case the Pollaczek–Khinchine probabilistic route is unavailable. Applying the method of characteristics to the fundamental equation still yields a general solution, but closing it — i.e., determining the unknown boundary function $P_y(y)$ — requires solving a Volterra-type integral equation that does not admit a closed form. In fact the obstruction is strictly more severe than in Model B, on two counts developed below: the characteristics are no longer straight lines but integral curves of a nonlinear ODE, and the diagonal value $P(z,z)$ that anchored the reduction in Model B is itself no longer known. Model X is therefore intractable and is left open; its role is to expose the precise obstructions that its tractable specialisations A, B, B_2 and C_2 each circumvent. The queue layout is the left panel of Figure 1.

Stability. Model X has abandonment from both classes, so the total departure rate in state (n_1,n_2) is $\mu+\theta_1 n_1+\theta_2 n_2$, which grows without bound in every direction of the (n_1,n_2)-plane. The chain is therefore positive recurrent for *all* $\lambda_1,\lambda_2,\mu>0$ and all $\theta_1,\theta_2>0$; no traffic-intensity restriction is required (cf. the Erlang-A queue, §3.2). Setting $\theta_1=\theta_2=0$ removes this and returns the Model B/Model A condition $\rho=(\lambda_1+\lambda_2)/\mu<1$.

Corollary 1. *The empty-system probability π_0 (server idle, no one waiting) and the both-queues-empty busy probability $\pi(0,0)$ (server busy, $N_1=N_2=0$) of Model X satisfy the universal idle balance*

$$\pi(0,0)=\rho\,\pi_0,\qquad \rho=\frac{\lambda_1+\lambda_2}{\mu},\tag{75}$$

as in every model of this work (22). Unlike Models A and B, however, abandonment makes the total-customer process a state-dependent birth–death chain rather than an $M/M/1$, so $\pi_0\neq 1-\rho$: the idle probability depends on θ_1,θ_2 and is fixed only by the global normalisation $P(1,1)=1-\pi_0$ once P is known. Correspondingly, the diagonal of the PGF no longer collapses to $\pi(0,0)/(1-\rho z)$; setting $x=y=z$ in the fundamental equation leaves the abandonment-corrected relation

$$(\mu-\lambda z)\,P(z,z)+z\big[\theta_1\,\partial_x P+\theta_2\,\partial_y P\big]_{(z,z)}=\mu\,\pi(0,0),\qquad \lambda=\lambda_1+\lambda_2.\tag{76}$$

As $\theta_1,\theta_2\to 0^+$ the gradient terms drop out and (76) returns the Model B identities $P(z,z)=\pi(0,0)/(1-\rho z)$, $\pi_0=1-\rho$ and $\pi(0,0)=\rho(1-\rho)$ (150), (108).

Proof. Setting $x=y$ in the fundamental equation (41), both jockeying terms carry the factor $(x-y)$ and vanish, as does the boundary term $\mu(x-y)P_y(y)$; the abandonment terms, carrying instead the factor $(x-1)$, survive. The coefficient of P becomes $(\lambda_1+\lambda_2+\mu)z^2-\mu z-(\lambda_1+\lambda_2)z^3=z(z-1)(\mu-\lambda z)$, and the right-hand side becomes $-\mu z(1-z)\pi(0,0)=\mu z(z-1)\pi(0,0)$; dividing the surviving terms by the common factor $z(z-1)$ (for $z\in(0,1)$) leaves

$$z(z-1)(\mu-\lambda z)P(z,z)+z^2(z-1)\big[\theta_1\,\partial_x P+\theta_2\,\partial_y P\big]_{(z,z)}=\mu z(z-1)\,\pi(0,0),\tag{77}$$

which is (76). The idle state (0) exchanges probability flux only with the both-queues-empty busy state, so global balance across the idle–busy cut reads $\lambda\pi_0=\mu\pi(0,0)$ (28) (an arrival at total rate λ leaves (0); a service completion at rate μ returns to it), whence $\pi(0,0)=\rho\pi_0$, which is (75). In Models A and B the gradient terms in (76) are absent ($\theta_1=\theta_2=0$), so $P(z,z)=\pi(0,0)/(1-\rho z)$; evaluating at $z=1$ with $P(1,1)=1-\pi_0$ then forces $(1-\pi_0)(1-\rho)=\rho\pi_0$, i.e. $\pi_0=1-\rho$. With abandonment those gradient terms are present and *not* known — exactly the phenomenon isolated for Model C_2 (260) — so the diagonal no longer determines $P(z,z)$, and π_0 cannot be read off in closed form: it follows only from the full solution via normalisation. □

Claim 1. *Under positive recurrence, the bivariate PGF of Model X satisfies, on each characteristic curve $\Phi(x,y)=C_1$ of the fundamental equation, the integral representation*

$$P(x,y(x))\;=\;P_h(x,y(x))\int_0^x\frac{h(t;C_1)}{P_h\big(t,\,y(t)\big)}\,dt,\tag{78}$$

where P_h is the homogeneous solution (90), $y(t)=y(t;C_1)$ is the characteristic parametrisation, Φ is the first integral (88) (a transcendental function of (x,y), not the linear form of Model B), and $h(t;C_1)$ is a forcing function depending on the unknown boundary trace $P_y(y)=P(0,y)$. Evaluating (78) along *the characteristic through a diagonal point* (z,z) *reduces the problem to a* Volterra integral equation of the first kind *for* P_y*:*

$$\mathcal{F}(z) \;=\; \int_0^z \mathcal{K}(z,t)\, P_y\big(y(t;z)\big)\, dt, \tag{79}$$

whose kernel $\mathcal{K}$ *and right-hand side* $\mathcal{F}$ *are defined in* (101)–(103) *below. The kernel* $\mathcal{K}$ *is built from the non-elementary homogeneous solution* P_h *and does not simplify for general* $(\gamma_1, \gamma_2, \theta_1, \theta_2)$*; moreover, in contrast to Model B, the right-hand side* $\mathcal{F}$ *is* not *fully known — it is coupled through* (76) *to the diagonal gradient* $[\theta_1\partial_x P + \theta_2\partial_y P]_{(z,z)}$*. Equation* (79) *therefore does not reduce to a closed-form solution for* P_y*, and Model X is left open.*

Proof of Claim 1. The proof follows the five steps of Model B: (i) homogeneous solution via characteristics, (ii) behaviour of P_h at the boundaries, (iii) particular solution via variation of parameters, (iv) pinning the free function from the boundary condition, and (v) diagonal evaluation yielding the Volterra equation. At each step we point out where the abandonment terms θ_i obstruct the closed-form algebra available in Model B.

Step 1: Homogeneous solution via the method of characteristics.

The fundamental equation (41) is a first-order linear PDE in $P(x,y)$. Setting the right-hand side to zero and dividing through by y yields the homogeneous PDE

$$x\big[\gamma_1(x-y) + \theta_1(x-1)\big]\,\frac{\partial P}{\partial x} + x\big[\gamma_2(y-x) + \theta_2(y-1)\big]\,\frac{\partial P}{\partial y} = -A(x,y)\,P, \tag{80}$$

with the same coefficient polynomial as Model B,

$$A(x,y) := (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy; \tag{81}$$

abandonment changes only the two convection (drift) coefficients, adding the terms $\theta_1(x-1)$ and $\theta_2(y-1)$. Following the method of characteristics (§3.5.3), we associate to (80) the characteristic system

$$\frac{dx}{x\big[\gamma_1(x-y)+\theta_1(x-1)\big]} = \frac{dy}{x\big[\gamma_2(y-x)+\theta_2(y-1)\big]} = \frac{dP}{-A(x,y)\,P}. \tag{82}$$

Taking the ratio of the first two differentials, the common factor x cancels:

$$\frac{dy}{dx} = \frac{\gamma_2(y-x)+\theta_2(y-1)}{\gamma_1(x-y)+\theta_1(x-1)} = \frac{-\gamma_2 x + (\gamma_2+\theta_2)y - \theta_2}{(\gamma_1+\theta_1)x - \gamma_1 y - \theta_1}. \tag{83}$$

In Model B ($\theta_1 = \theta_2 = 0$) both affine forms shared the factor $(x-y)$ and the ratio collapsed to the constant $-\gamma_2/\gamma_1$, integrating to the straight lines $\gamma_2 x + \gamma_1 y = \text{const}$. With abandonment this collapse no longer occurs: (83) is a genuine first-order ODE whose right-hand side is a ratio of two affine forms, and its integral curves are not straight lines.

Both affine forms in (83) vanish simultaneously only at the single point $(x,y) = (1,1)$ (the unique intersection of the lines $(\gamma_1+\theta_1)x - \gamma_1 y - \theta_1 = 0$ and $-\gamma_2 x + (\gamma_2+\theta_2)y - \theta_2 = 0$), which is therefore the lone singular point of the characteristic field. Shifting to it via $X = x-1,\ Y = y-1$ removes the constant terms and renders (83) homogeneous:

$$\frac{dY}{dX} = \frac{-\gamma_2 X + (\gamma_2+\theta_2)Y}{(\gamma_1+\theta_1)X - \gamma_1 Y}. \tag{84}$$

The substitution $v = Y/X$ then separates the variables,

$$\frac{(\gamma_1+\theta_1) - \gamma_1 v}{Q(v)}\,dv = \frac{dX}{X}, \qquad Q(v) := \gamma_1 v^2 + (\gamma_2+\theta_2-\gamma_1-\theta_1)\,v - \gamma_2. \tag{85}$$

The quadratic Q has discriminant $D = (\gamma_2+\theta_2-\gamma_1-\theta_1)^2 + 4\gamma_1\gamma_2 > 0$ and real roots of opposite sign (their product is $-\gamma_2/\gamma_1 < 0$),

$$v_\pm = \frac{-(\gamma_2+\theta_2-\gamma_1-\theta_1) \pm \sqrt{D}}{2\gamma_1}, \qquad v_- < 0 < v_+. \tag{86}$$

Decomposing the left-hand side of (85) by partial fractions over $Q(v) = \gamma_1(v - v_+)(v - v_-)$,

$$\frac{(\gamma_1 + \theta_1) - \gamma_1 v}{\gamma_1(v - v_+)(v - v_-)} = \frac{a_+}{v - v_+} + \frac{a_-}{v - v_-}, \qquad a_\pm = \frac{(\gamma_1 + \theta_1) - \gamma_1 v_\pm}{\gamma_1\,(v_\pm - v_\mp)}, \qquad a_+ + a_- = -1, \tag{87}$$

(the relation $a_+ + a_- = -1$ is the residue of the integrand at $v = \infty$), and integrating gives $a_+ \ln(v - v_+) + a_- \ln(v - v_-) = \ln X + \text{const}$. Exponentiating and returning to (x, y) via $v - v_\pm = (Y - v_\pm X)/X$ and $a_+ + a_- = -1$, the powers of X cancel and we obtain the first integral

$$\Phi(x, y) := \big[(y - 1) - v_+(x - 1)\big]^{a_+} \big[(y - 1) - v_-(x - 1)\big]^{a_-} = C_1, \tag{88}$$

constant along each characteristic. Unlike Model B's linear constant $\gamma_2 x + \gamma_1 y$, the first integral (88) is a product of non-integer powers of two linear forms — a transcendental level curve. As a consistency check, letting $\theta_1, \theta_2 \to 0^+$ gives $Q(v) \to (\gamma_1 v + \gamma_2)(v - 1)$, so $v_+ \to 1$, $v_- \to -\gamma_2/\gamma_1$, and $(a_+, a_-) \to (0, -1)$; then $\Phi(x, y) \to \big[(y - 1) + (\gamma_2/\gamma_1)(x - 1)\big]^{-1}$, whose level curves are exactly the straight lines $\gamma_2 x + \gamma_1 y = \text{const}$ of Model B.

To find P along a characteristic we take the ratio of the first and third differentials in (82),

$$\frac{dP}{P} = \frac{-A\big(x, y(x)\big)}{x\big[\gamma_1(x - y(x)) + \theta_1(x - 1)\big]}\, dx, \tag{89}$$

where $y(x) = y(x; C_1)$ is the characteristic (88). Integrating yields the general homogeneous solution

$$P_h(x, y) = F(C_1)\, \exp\!\left(- \int_{x_\star}^{x} \frac{A\big(s, y(s)\big)}{s\big[\gamma_1(s - y(s)) + \theta_1(s - 1)\big]}\, ds \right), \qquad C_1 = \Phi(x, y), \tag{90}$$

where F is an arbitrary differentiable function of the characteristic constant and $x_\star$ is an arbitrary base point (its choice is absorbed into F). Here the parallel with Model B ends in closed form: because $y(s)$ is the transcendental characteristic (88), the integrand in (90) is not a rational function of s, and the quadrature is not elementary. Model B, by contrast, could substitute the linear $y(s)$, partial-fraction the rational integrand, and read off the explicit factors $x^{-\mu/C_1}(x - y)^E$ with a closed-form exponent E. No such explicit P_h exists for Model X.

Step 2: Boundary behaviour of P_h.

Although the quadrature (90) is not elementary, the two qualitative features needed downstream survive, and a third feature of Model B is lost.

(a) Blow-up at $x = 0$. A characteristic meets the axis $x = 0$ at a finite height $y_0 := y(0; C_1) \in (0, 1]$, where $A(0, y_0) = -\mu$ and the convection coefficient is $\gamma_1(0 - y_0) + \theta_1(0 - 1) = -(\gamma_1 y_0 + \theta_1) < 0$. Hence the integrand in (90) behaves like $-\mu/[(\gamma_1 y_0 + \theta_1)\, s]$ as $s \to 0^+$, so

$$P_h\big(x, y(x)\big) \;\sim\; x^{-\mu/(\gamma_1 y_0 + \theta_1)} \;\longrightarrow\; +\infty \qquad (x \to 0^+), \tag{91}$$

with a strictly negative exponent because $\gamma_1 y_0 + \theta_1 > 0$. This is exactly the factor $x^{-\mu/C_1}$ of Model B, now with the denominator shifted by θ_1; it drives the pinning of F in Step 4.

(b) Regularity on the diagonal. In Model B the convection coefficient was $\gamma_1 x(x - y)$, which *vanishes* on the diagonal $x = y$; the resulting factor $(x - y)^E$ with $E > 0$ made $P_h \to 0$ there and forced the delicate Hadamard finite-part regularisation of Model B's Step 5. For Model X the coefficient is $x[\gamma_1(x - y) + \theta_1(x - 1)]$, which at $x = y = z$ equals $z\, \theta_1(z - 1) \neq 0$ for $z \in (0, 1)$: the abandonment term $\theta_1(x - 1)$ keeps the characteristic field away from zero on the diagonal. Consequently the integrand in (90) is regular as $x \to z$ along the diagonal-crossing characteristic, and

$$P_h^{\text{diag}}(z) := \lim_{x \to z} P_h\big(x, y(x; z)\big) \quad \text{is finite and non-zero.} \tag{92}$$

The diagonal singularity of Model B thus disappears; as we will see in Step 5, the obstruction migrates instead to the right-hand side of the Volterra equation.

Branch convention. As in Model B, the non-integer powers in (88) and the quadrature in (90) may carry a complex phase off the real region; this phase is absorbed into the arbitrary function $F(C_1)$ and cancels exactly in the ratio $P_h(x, y(x))/P_h(t, y(t))$ appearing in (78), since numerator and denominator lie on the same characteristic (same C_1).

Step 3: Particular solution via variation of parameters.

We now restrict the full (inhomogeneous) fundamental equation (41) to a fixed characteristic. Along $y = y(x; C_1)$ the total derivative is $\frac{d}{dx}P(x, y(x)) = \partial_x P + y'(x)\,\partial_y P$, and by the characteristic ODE (83) the bracket multiplying $\partial_y P$ equals $[\gamma_1(x-y)+\theta_1(x-1)]\,y'(x)$, so the two convection terms combine:

$$xy\big[\gamma_1(x-y)+\theta_1(x-1)\big]\partial_x P + xy\big[\gamma_2(y-x)+\theta_2(y-1)\big]\partial_y P = xy\big[\gamma_1(x-y)+\theta_1(x-1)\big]\frac{d}{dx}P(x, y(x)).$$

Substituting into (41) and dividing by $xy\big[\gamma_1(x-y)+\theta_1(x-1)\big]$ reduces the PDE to the first-order linear ODE (for fixed C_1)

$$\frac{dP}{dx} = f(x; C_1)\,P + h(x; C_1), \tag{93}$$

with coefficient and forcing

$$f(x; C_1) = -\frac{A\big(x, y(x)\big)}{x\big[\gamma_1(x-y(x))+\theta_1(x-1)\big]}, \tag{94}$$

$$h(x; C_1) = \frac{G\big(x, y(x)\big)}{x\,y(x)\big[\gamma_1(x-y(x))+\theta_1(x-1)\big]}, \tag{95}$$

where A is as in (81) and the right-hand side of the fundamental equation is, as in Model B,

$$G(x, y) := \mu(x-y)\,P_y(y) - \mu x(1-y)\,\pi(0, 0). \tag{96}$$

Comparing (94)–(95) with their Model B counterparts, the denominator $\gamma_1 x(x-y)$ is replaced throughout by $x[\gamma_1(x-y)+\theta_1(x-1)]$ — the only footprint of abandonment, and the source of all the difficulty. Since P_h from (90) satisfies the homogeneous equation $\frac{d}{dx}P_h = f(x; C_1)P_h$, it is the fundamental solution of (93); applying variation of parameters (§3.5.2), the general solution of the inhomogeneous ODE is

$$P(x, y(x)) = P_h(x, y(x))\left[F(C_1) + \int_{x_0}^{x}\frac{h(t; C_1)}{P_h(t, y(t))}\,dt\right]. \tag{97}$$

Step 4: Pinning the free function from the boundary condition at $x = 0$.

Along any characteristic the endpoint at $x = 0$ is $y(0) = y_0$, giving $P(0, y_0) = P_y(y_0)$, which is finite. However, by (91) the homogeneous factor blows up, $P_h(x, y(x)) \to +\infty$ as $x \to 0^+$, while the integral in (97) vanishes as $x_0 \to 0$ (the integration interval collapses). Therefore

$$\lim_{x\to 0^+} P(x, y(x)) = \lim_{x\to 0^+} P_h(x, y(x)) \cdot F(C_1) = +\infty \cdot F(C_1),$$

which can be finite only if $F(C_1) = 0$. Setting $x_0 = 0$ and $F(C_1) = 0$ in (97) gives the integral representation of Claim 1,

$$P(x, y(x)) = P_h(x, y(x)) \int_0^x \frac{h(t; C_1)}{P_h(t, y(t))}\,dt. \tag{98}$$

This argument is identical to Model B's; abandonment shifts the blow-up exponent in (91) from μ/C_1 to $\mu/(\gamma_1 y_0 + \theta_1)$ but leaves it strictly positive, so the conclusion $F \equiv 0$ is unchanged.

Step 5: Reduction to the Volterra equation.

We simplify the forcing h on the characteristic. Substituting G from (96) into (95) and cancelling one factor of t in the second term,

$$h(t; C_1) = \frac{\mu\,(t-y(t))}{t\,y(t)\big[\gamma_1(t-y(t))+\theta_1(t-1)\big]}\,P_y\big(y(t)\big) - \frac{\mu\,(1-y(t))}{y(t)\big[\gamma_1(t-y(t))+\theta_1(t-1)\big]}\,\pi(0, 0). \tag{99}$$

We evaluate (98) along the characteristic through a diagonal point (z, z), that is, the curve with $C_1 = \Phi(z, z)$; its parametrisation $y(t; z)$ satisfies $y(z; z) = z$. Setting $x = z$,

$$P(z, z) = P_h(z, z) \int_0^z \frac{h\big(t; \Phi(z, z)\big)}{P_h\big(t, y(t; z)\big)}\,dt. \tag{100}$$

By the regularity (92), $P_h(z,z) = P_h^{\mathrm{diag}}(z)$ is finite and non-zero and the integrand is regular as $t \to z$ (its numerator $t - y(t;z) \to 0$), so (100) is a *proper* integral — no finite-part regularisation is needed, in contrast to Model B. Substituting the two pieces of (99) and separating the term containing the unknown P_y from the known remainder, define the kernel and remainder

$$\mathcal{K}(z,t) := \frac{\mu\,\big(t - y(t;z)\big)\,P_h^{\mathrm{diag}}(z)}{t\,y(t;z)\big[\gamma_1(t - y(t;z)) + \theta_1(t-1)\big]\,P_h\big(t, y(t;z)\big)}, \tag{101}$$

$$\mathcal{R}(z) := -\,\mu\,\pi(0,0)\,P_h^{\mathrm{diag}}(z)\int_0^z \frac{1 - y(t;z)}{y(t;z)\big[\gamma_1(t - y(t;z)) + \theta_1(t-1)\big]\,P_h\big(t, y(t;z)\big)}\,dt, \tag{102}$$

so that (100) reads $P(z,z) = \int_0^z \mathcal{K}(z,t)\,P_y(y(t;z))\,dt + \mathcal{R}(z)$. Setting

$$\mathcal{F}(z) := P(z,z) - \mathcal{R}(z) = \underbrace{\frac{\pi(0,0)}{1-\rho z}}_{\text{Model-}B\text{ value}} - \underbrace{\frac{z}{\mu - \lambda z}\big[\theta_1\partial_x P + \theta_2\partial_y P\big]_{(z,z)}}_{\text{abandonment coupling}} - \mathcal{R}(z), \tag{103}$$

where the expression for $P(z,z)$ is the abandonment-corrected diagonal (76), equation (100) becomes the Volterra integral equation of the first kind (79), confirming Claim 1. Two features distinguish (79) from its Model B analogue, and both make Model X strictly less tractable. First, the kernel (101) is built from the non-elementary homogeneous solution P_h along transcendental characteristics, so it has no closed form for general parameters (in Model B, P_h and the straight-line characteristics were explicit). Second, and decisively, the right-hand side (103) is *not* known data: through (76) it is coupled to the diagonal gradient $[\theta_1\partial_x P + \theta_2\partial_y P]_{(z,z)}$, which is precisely the quantity that abandonment prevents the diagonal from determining (260). Where Model B left a clean first-kind Volterra equation with explicit kernel and known forcing (itself already beyond closed-form inversion), Model X leaves one whose kernel is implicit and whose forcing is entangled with the unknown solution. Resolving it lies well beyond the scope of this thesis. □

Model X is therefore left open at this point. Its role in the hierarchy is to make the two obstructions explicit and to show how each tractable specialisation removes them. The limit $\theta_1 = \theta_2 \to 0$ (Model B) restores both at once: the characteristics (88) straighten to the lines $\gamma_2 x + \gamma_1 y = \text{const}$, so the kernel (101) becomes explicit, and the gradient coupling in (103) vanishes, so $\mathcal{F}(z) = \pi(0,0)/(1-\rho z) - \mathcal{R}(z)$ becomes known data — leaving the clean (still unsolved) Volterra equation of Section 7. Removing one mechanism at a time then closes the problem entirely: setting in addition $\gamma_2 = 0$ (Model B_2) collapses the characteristic family to vertical lines and degenerates the PDE to an ODE in x alone, whose boundary function $P_y(y)$ is pinned by an analyticity argument at the interior singular point $x = y$ (Section 8); removing jockeying entirely while keeping only class-1 abandonment (Model C_2) likewise degenerates the PDE to an ODE in x, whose $P_y(y)$ is pinned by a boundedness argument at $x = 1$ (Section 9); and zeroing all four parameters returns the fully solvable baseline Model A (Section 6). The full model X, retaining all mechanisms, retains both obstructions simultaneously and is the one case that does not close.

6 Analysis of Model A ($\gamma_1 = \gamma_2 = 0$, $\theta_1 = \theta_2 = 0$)

Model A is the baseline of this collection of models: all jockeying and abandonment parameters are set to zero, leaving an ordinary two-class $M/M/1$ queueing model with non-preemptive priority. The analysis is carried out in both state spaces S and $\widetilde{S}$: on the former, two complete proofs are presented; while on the latter, an approximate solution and its verification are provided.

The balance equations are obtained by plugging $\gamma_1 = \gamma_2 = 0$ and $\theta_1 = \theta_2 = 0$ into Equations (24), (26), (25), (27), and (28) for state space S, and Equations (29), (30), (31), (32), and (33) for state space $\widetilde{S}$. The corresponding starting points are Lemma 1 for S and Lemma 2 for $\widetilde{S}$.

Stability. Since there is no abandonment, the system is positive recurrent if and only if the total traffic intensity satisfies

$$\rho = \frac{\lambda_1 + \lambda_2}{\mu} < 1. \tag{104}$$

The figure below corresponds to the queueing diagram of Model A.

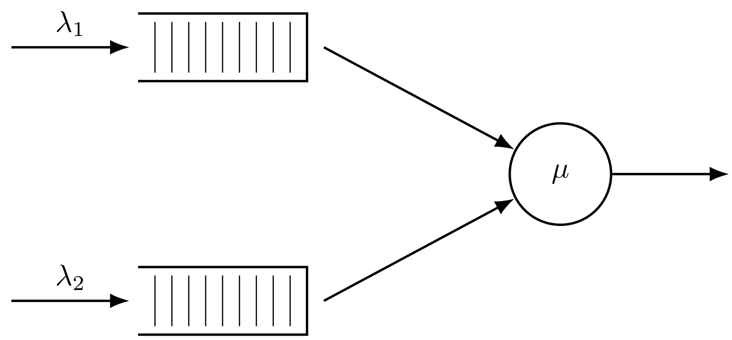

Figure 4: Queue layout for Model A. Only Poisson arrivals and exponential service are present; all jockeying and abandonment rates are zero.

6.1 Analysis of Model A: state space S

We begin from Equation (41) with $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$, in which the marginal $P_y(y) = P(0, y)$ is still to be determined:

$$\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x, y) = \mu(x - y)P_y(y) - \mu x(1 - y)\pi(0, 0). \tag{105}$$

The following theorem summarises the main result of this subsection; two standalone proofs are given, one analytical and one probabilistic.

Theorem 1. *Consider a queueing system with two priority classes with Poisson arrival rates λ_1 and λ_2, where class-1 customers have non-preemptive priority over those of class 2, and exponentially distributed service times with rate μ independently of the class. Under the stability condition (104), the joint PGF $P(x, y) = \mathbb{E}[x^{N_1} y^{N_2}]$ is*

$$P(x, y) = \left[(1 + \rho_1 + \rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1} \cdot y(1 - y)\left[\frac{x - x^*(y)}{x^*(y) - y}\right] \cdot \rho(1 - \rho), \tag{106}$$

where

$$x^*(y) = \frac{[\mu + \lambda_1 + \lambda_2(1 - y)] - \sqrt{[\mu + \lambda_1 + \lambda_2(1 - y)]^2 - 4\lambda_1\mu}}{2\lambda_1}. \tag{107}$$

Corollary 2. *Under the stability condition (104), the empty-state probabilities for Model A are*

$$\pi_0 = 1 - \rho, \qquad \pi(0, 0) = \rho(1 - \rho). \tag{108}$$

In particular, $P(1, 1) = 1 - \pi_0 = \rho$ and the diagonal generating function satisfies $P(z, z) = \pi(0, 0)/(1 - \rho z)$.

Proof of Corollary 2. Setting $x = y = z$ in the fundamental equation (105), the boundary term $\mu(x - y)P_y(y)$ vanishes:

$$\left[(\lambda_1 + \lambda_2 + \mu)z^2 - \mu z - \lambda_1 z^3 - \lambda_2 z^3\right] P(z, z) = -\mu z(1 - z)\pi(0, 0).$$

Dividing both sides by z and factoring the bracket:

$$-(1 - z)\left[\mu - (\lambda_1 + \lambda_2)z\right] P(z, z) = -\mu(1 - z)\pi(0, 0).$$

Cancelling $(1 - z) \neq 0$ for $z \in [0, 1)$ and rearranging:

$$P(z, z) = \frac{\mu\,\pi(0, 0)}{\mu - \lambda z} = \frac{\pi(0, 0)}{1 - \rho z}, \tag{109}$$

where $\lambda = \lambda_1 + \lambda_2$ and the second equality follows by dividing numerator and denominator by μ.

Setting $z = 1$ and using the normalisation $P(1,1) = 1 - \pi_0$:

$$1 - \pi_0 = \frac{\pi(0,0)}{1-\rho}. \tag{110}$$

The global balance equation for the idle state gives $\lambda\pi_0 = \mu\pi(0,0)$, i.e. $\pi(0,0) = \rho\pi_0$. Substituting into (110):

$$1 - \pi_0 = \frac{\rho\pi_0}{1-\rho} \implies (1-\pi_0)(1-\rho) = \rho\pi_0 \implies \pi_0 = 1-\rho,$$

and therefore $\pi(0,0) = \rho(1-\rho)$. □

6.1.1 Analytical approach for Model A in state space S

Proof. We begin from the fundamental equation for Model A (105):

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) = \mu(x-y)P_y(y) - \mu x(1-y)\pi(0,0).$$

We apply the *Kernel Method*: fix $y \in (0,1]$ and find $x^*(y)$ such that the LHS of the expression above is zero. Since $P(x,y)$ is a PGF, it is bounded. At any such root the LHS vanishes, so the RHS must also vanish:

$$\mu\left[(x^*(y) - y)\, P_y(y) - x^*(y)(1-y)\,\pi(0,0)\right] = 0.$$

To find the roots of the coefficient of $P(x,y)$, divide by $y \neq 0$ and set

$$\begin{aligned}
0 &= (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 x y \\
0 &= \lambda_1 x^2 - \left[\mu + \lambda_1 + \lambda_2(1-y)\right] x + \mu,
\end{aligned}$$

whose roots are

$$x^{\pm}(y) = \frac{\left[\mu+\lambda_1+\lambda_2(1-y)\right] \pm \sqrt{\left[\mu+\lambda_1+\lambda_2(1-y)\right]^2 - 4\mu\lambda_1}}{2\lambda_1}.$$

We label the root with the negative sign $x^*(y)$ and the root with the positive sign $x^+(y)$. At $y = 1$:

$$x^*(1) = 1 \quad \text{and} \quad x^+(1) = \mu/\lambda_1 > 1.$$

We require our root to lie inside the closed unit disk for $y \in [0,1)$. Computing the derivative of $x^*(y)$:

$$\frac{d}{dy}x^*(y) = \frac{\lambda_2}{2\lambda_1}\left[-1 + \frac{\mu+\lambda_1+\lambda_2(1-y)}{\sqrt{(\mu+\lambda_1+\lambda_2(1-y))^2 - 4\lambda_1\mu}}\right] > 0, \tag{111}$$

since $(\mu+\lambda_1+\lambda_2(1-y))^2 > (\mu+\lambda_1+\lambda_2(1-y))^2 - 4\lambda_1\mu$ whenever $\lambda_1\mu > 0$. Therefore $x^*(y)$ is strictly increasing on $[0,1]$, and since $x^*(1) = 1$ we have $x^*(y) < 1$ for $y \in [0,1)$. By Vieta's formulas for the kernel polynomial $\lambda_1 x^2 - [\mu+\lambda_1+\lambda_2(1-y)]x + \mu = 0$, the product of the two roots satisfies $x^*(y)\, x^+(y) = \mu/\lambda_1 > 1$, so $x^+(y) = \mu/(\lambda_1 x^*(y)) > 1$ throughout $[0,1)$. Hence the only root strictly inside the unit disk for $y \in [0,1)$ is $x^*(y)$:

$$x^*(y) = \frac{(\mu+\lambda_1+\lambda_2(1-y)) - \sqrt{(\mu+\lambda_1+\lambda_2(1-y))^2 - 4\lambda_1\mu}}{2\lambda_1}. \tag{112}$$

Since subsequent steps involve L'Hôpital's rule at $y \to 1$, we evaluate $\frac{d}{dy}x^*(y)$ from (111) at $y = 1$. Under stability, $\rho_1 = \lambda_1/\mu < \rho < 1$ implies $\mu > \lambda_1$, so $\sqrt{(\mu-\lambda_1)^2} = \mu - \lambda_1$:

$$\frac{d}{dy}x^*(1) = \frac{\lambda_2}{2\lambda_1}\left[-1 + \frac{\mu+\lambda_1}{\sqrt{(\mu+\lambda_1)^2 - 4\lambda_1\mu}}\right] = \frac{\lambda_2}{2\lambda_1}\left[-1 + \frac{\mu+\lambda_1}{\sqrt{(\mu-\lambda_1)^2}}\right] = \frac{\lambda_2}{2\lambda_1}\left[\frac{\mu+\lambda_1-(\mu-\lambda_1)}{\mu-\lambda_1}\right] = \frac{\lambda_2}{\mu-\lambda_1}.$$

For future reference:

$$\frac{d}{dy}x^*(1) = \frac{\lambda_2}{\mu-\lambda_1} = \frac{\rho_2}{1-\rho_1}. \tag{113}$$

Returning to the fundamental equation (105) at $x = x^*(y)$, the RHS also equals zero, giving

$$P_y(y) = \frac{x^*(y)(1-y)}{x^*(y)-y}\pi(0,0). \tag{114}$$

By Corollary 2, $\pi(0,0) = \rho(1-\rho)$. Substituting this and Equation (114) into the fundamental equation (105):

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 xy^2\right] P(x,y) = \mu(x-y)\frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho) - \mu x(1-y)\rho(1-\rho)$$

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 xy^2\right] P(x,y) = \mu(1-y)\left[\frac{x^*(y)(x-y)}{x^*(y)-y} - x\right]\rho(1-\rho)$$

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 xy^2\right] P(x,y) = \mu(1-y)\left[\frac{y(x-x^*(y))}{x^*(y)-y}\right]\rho(1-\rho)$$

$$\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 xy^2\right] P(x,y) = y(1-y)\left[\frac{x-x^*(y)}{x^*(y)-y}\right]\rho(1-\rho).$$

Multiplying both sides by $\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 xy^2\right]^{-1}$ isolates $P(x,y)$ and concludes the proof. □

6.1.2 Probabilistic approach for Model A in state space S

Proof. The proof is similar to the analytical approach, but deriving the Class-2 marginal PGF $P_y(y)$ by means of probabilistic arguments. We begin by stating the fundamental blobal balance equation for Model-A (Equation (105)):

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 xy^2\right] P(x,y) = \mu\left[(x-y)P_y(y) - x(1-y)\pi(0,0)\right]. \tag{115}$$

Class-2 conditional distribution via $M/G/1$. Since class 1 has non-preemptive priority, class-2 customers are will not be taken into service whenever $N_1 > 0$; their *effective service time*, namely B, is therefore the duration of a complete Busy Period, of the single-class $M/M/1$ sub-queue with arrival rate λ_1 and service rate μ, so from class-2's viewpoint the system is an $M/G/1$ queue with arrivale rate λ_2 and generally distributed service times B with mean $\mathbb{E}[B]$. The LST and mean of B are presented in §3.1 (Equations (2)–(3)), now with λ_1 instead of λ:

$$\widetilde{B}(s) = \frac{(\mu+\lambda_1+s) - \sqrt{(\mu+\lambda_1+s)^2 - 4\mu\lambda_1}}{2\lambda_1}, \tag{116}$$

$$\mathbb{E}[B] = \frac{1}{\mu(1-\rho_1)}. \tag{117}$$

Applying the Pollaczek–Khinchine (PK) formula (§3.3, Equation (8)) to this $M/G/1$ system, with $\widetilde{B}(\lambda_2(1-y)) = x^*(y)$ (cf. (112)) and $\lambda_2\mathbb{E}[B] = \rho_2/(1-\rho_1)$:

$$L_2(y) = \frac{(1-\lambda_2\mathbb{E}[B])(1-y)\widetilde{B}(\lambda_2(1-y))}{\widetilde{B}(\lambda_2(1-y)) - y} = \frac{1-\rho}{1-\rho_1}\cdot\frac{x^*(y)(1-y)}{x^*(y)-y}. \tag{118}$$

By the PASTA property, the time-average distribution of N_2 conditional on $(\text{busy}, N_1 = 0)$ coincides with the $M/G/1$ steady-state distribution, so $\mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0] = L_2(y)$.

By the law of total expectation, $P_y(y) = \mathbb{P}(\text{busy}, N_1 = 0)\cdot\mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0]$. So

$$P_y(y) = \mathbb{P}(\text{busy}, N_1 = 0)\cdot\mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0] = \mathbb{P}(\text{busy}, N_1 = 0)\cdot L_2(y).$$

We now can obtain $\mathbb{P}(\text{busy}, N_1 = 0)$, from $\mathbb{P}(\text{busy}) = \rho$ and $\mathbb{P}(N_1 = 0 \mid \text{busy}) = 1-\rho_1$. So

$$\mathbb{P}(\text{busy}, N_1 = 0) = \rho(1-\rho_1)$$

and therefore

$$P_y(y) = \rho(1-\rho_1)\cdot L_2(y) = \rho(1-\rho_1)\cdot\frac{1-\rho}{1-\rho_1}\cdot\frac{x^*(y)(1-y)}{x^*(y)-y} = \frac{x^*(y)(1-y)}{x^*(y)-y}\cdot(1-\rho)\rho \tag{119}$$

Substituting (119) and $\pi(0,0) = \rho(1-\rho)$ (Corollary 2) into (115):

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) = \mu(x-y)\frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho) - \mu x(1-y)\rho(1-\rho)$$

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) = \mu(1-y)\left[\frac{x^*(y)(x-y)}{x^*(y)-y} - x\right]\rho(1-\rho)$$

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) = \mu(1-y)\left[\frac{y(x-x^*(y))}{x^*(y)-y}\right]\rho(1-\rho)$$

$$\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right] P(x,y) = y(1-y)\left[\frac{x-x^*(y)}{x^*(y)-y}\right]\rho(1-\rho).$$

Multiplying both sides by $\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1}$ isolates $P(x,y)$ and concludes the proof. $\square$

6.1.3 Partial-PGF recurrence approach for Model A in state space S

This proof follows Cohen [Coh56]. It consists in keeping the class-1 index n_1 as a discrete parameter and forming a generating function only with respect to N_2, namely the PPGF $\widetilde{P}(n_1,y) = \sum_{n_2\geq 0}\pi(n_1,n_2)\,y^{n_2}$ introduced in (38). Multiplying the balance equations by y^{n_2} and summing over n_2 turns the two-dimensional balance equations into a second-order linear recurrence in n_1, whose characteristic roots fix the geometric structure of the solution in the priority class index.

Proof. We start by stating the balance equations for Model A, with all jockeying and abandonment parameters set to zero (i.e. $\gamma_1=\gamma_2=\theta_1=\theta_2=0$) in the interior balance equation (Equation (24)) and our x-boundary equation (Equation (26)). We have

$$(\lambda_1+\lambda_2+\mu)\,\pi(n_1,n_2) = \mu\,\pi(n_1+1,n_2) + \lambda_1\,\pi(n_1-1,n_2) + \lambda_2\,\pi(n_1,n_2-1), \quad n_1\geq 1,\ n_2\geq 1, \tag{120}$$

$$(\lambda_1+\lambda_2+\mu)\,\pi(n_1,0) = \mu\,\pi(n_1+1,0) + \lambda_1\,\pi(n_1-1,0), \qquad n_1\geq 1,\ n_2=0. \tag{121}$$

$$\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{\infty}\pi(n_1,n_2)y^{n_2}\\
&\quad= \quad \mu\sum_{n_2=1}^{\infty}\pi(n_1+1,n_2)y^{n_2} + \lambda_1\sum_{n_2=1}^{\infty}\pi(n_1-1,n_2)y^{n_2} + \lambda_2\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)y^{n_2}\\
&LHS_1 \quad = \quad RHS_1 + RHS_2 + RHS_3\\
&LHS_1\\
&\quad= \quad (\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{\infty}\pi(n_1,n_2)y^{n_2}\\
&\quad= \quad (\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{\infty}\pi(n_1,n_2)y^{n_2} - \pi(n_1,0)y^0\right]\\
&\quad= \quad (\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(n_1,y) - \pi(n_1,0)\right]\\
&RHS_1\\
&\quad= \quad \mu\sum_{n_2=1}^{\infty}\pi(n_1+1,n_2)y^{n_2}\\
&\quad= \quad \mu\left[\sum_{n_2=0}^{\infty}\pi(n_1+1,n_2)y^{n_2} - \pi(n_1+1,0)y^0\right]\\
&\quad= \quad \mu\left[\widetilde{P}(n_1+1,y) - \pi(n_1+1,0)\right]
\end{aligned}$$

$$
\begin{aligned}
&RHS_2 \\
&\quad = \quad \lambda_1 \sum_{n_2=1}^{\infty} \pi(n_1-1, n_2) y^{n_2} \\
&\quad = \quad \lambda_1 \left[\sum_{n_2=0}^{\infty} \pi(n_1-1, n_2) y^{n_2} - \pi(n_1-1, 0) y^0 \right] \\
&\quad = \quad \lambda_1 \left[\widetilde{P}(n_1-1, y) - \pi(n_1-1, 0) \right] \\
&RHS_3 \\
&\quad = \quad \lambda_2 \sum_{n_2=1}^{\infty} \pi(n_1, n_2-1) y^{n_2} \\
&\quad = \quad \lambda_2 y \sum_{n_2=1}^{\infty} \pi(n_1, n_2-1) y^{n_2-1} \\
&\quad = \quad \lambda_2 y \sum_{m=0}^{\infty} \pi(n_1, m) y^{m} \\
&\quad = \quad \lambda_2 y \, \widetilde{P}(n_1, y) \\
&LHS_1 \quad = \quad RHS_1 + RHS_2 + RHS_3 \\
&(\lambda_1+\lambda_2+\mu)\big[\widetilde{P}(n_1, y) - \pi(n_1, 0)\big] \\
&\quad = \quad \mu\big[\widetilde{P}(n_1+1, y) - \pi(n_1+1, 0)\big] \\
&\qquad + \ \lambda_1\big[\widetilde{P}(n_1-1, y) - \pi(n_1-1, 0)\big] \\
&\qquad + \ \lambda_2 y \, \widetilde{P}(n_1, y)
\end{aligned}
$$

$$
\begin{aligned}
&\big[\mu+\lambda_1+\lambda_2(1-y)\big]\widetilde{P}(n_1, y) - \mu\widetilde{P}(n_1+1, y) - \lambda_1\widetilde{P}(n_1-1, y) \\
&\quad = \quad (\lambda_1+\lambda_2+\mu)\pi(n_1, 0) - \mu\pi(n_1+1, 0) - \lambda_1\pi(n_1-1, 0)
\end{aligned}
\tag{122}
$$

Note that, using our x-boundary balance equation (Equation (121)), the right-hand side of the expression above (Equation (122)) vanishes completely. We obtain the homogeneous recurrence

$$
\left[\mu+\lambda_1+\lambda_2(1-y)\right]\widetilde{P}(n_1, y) = \mu\widetilde{P}(n_1+1, y) + \lambda_1\widetilde{P}(n_1-1, y), \qquad n_1 \geq 1. \tag{123}
$$

Characteristic roots and mode selection. Equation (123) is a *second-order linear recurrence* in n_1 with constant coefficients in n_1. So its general solution is a linear combination of two geometric sequences $[\widetilde{x}^{\pm}(y)]^{n_1}$, where $\widetilde{x}^{\pm}(y)$ solve the characteristic equation

$$
\mu\,\widetilde{x}^2 - \left[\mu+\lambda_1+\lambda_2(1-y)\right]\widetilde{x} + \lambda_1 = 0, \tag{124}
$$

that is,

$$
\widetilde{x}^{\pm}(y) = \frac{\big(\mu+\lambda_1+\lambda_2(1-y)\big) \pm \sqrt{\big(\mu+\lambda_1+\lambda_2(1-y)\big)^2 - 4\lambda_1\mu}}{2\mu}. \tag{125}
$$

The *characteristic polynomial* (Equation (124)) is the reciprocal to that of the bivariate PGF approaches (i.e. the reciprocal of $\lambda_1 x^2 - [\mu+\lambda_1+\lambda_2(1-y)]x + \mu$), thus its roots are the reciprocals of that kernel's roots: the positive root is the reciprocal of the negative one ($\widetilde{x}^{+}(y) = 1/x^{*}(y)$), and conversely $\widetilde{x}^{-}(y) = 1/x^{+}(y)$. By Vieta's formulas, $\widetilde{x}^{-}(y)\,\widetilde{x}^{+}(y) = \lambda_1/\mu = \rho_1 < 1$. Recall from §6.1.1 that $x^{*}(y) < 1 < x^{+}(y)$ for $y \in [0, 1)$. The selection of the negative-sign root is motivated by the fact that we need the sequence $\{\widetilde{P}(n_1, y)\}_{n_1 \geq 0}$ to be summable. In fact, we need

$$
\sum_{n_1=0}^{\infty} \widetilde{P}(n_1, y) = P(1, y) \leq P(1, 1) = 1 - \pi_0 = \rho < \infty.
$$

At $y = 1$, we have

$$
\widetilde{x}^{-}(1) = \rho_1 \quad \text{and} \quad \widetilde{x}^{+}(1) = 1,
$$

the latter making the geometric sequence $[\widetilde{x}^+(y)]^{n_1}$ non-summable. We therefore define $\widetilde{x}^*(y) := \widetilde{x}^-(y)$,

$$\widetilde{x}^*(y) := \widetilde{x}^-(y) = \frac{\big(\mu+\lambda_1+\lambda_2(1-y)\big) - \sqrt{\big(\mu+\lambda_1+\lambda_2(1-y)\big)^2 - 4\lambda_1\mu}}{2\mu} = \rho_1\, x^*(y). \tag{126}$$

The last identity holds because $\widetilde{x}^*$ and x^* share the same numerator and differ only in the denominator (2μ versus $2\lambda_1$). Next,

$$\widetilde{P}(n_1, y) = \widetilde{P}(0, y)\,[\widetilde{x}^*(y)]^{n_1}, \qquad n_1 \geq 0, \tag{127}$$

where the relation also holds at $n_1 = 0$ trivially. Recall that at $y = 1$, we have $\widetilde{x}^*(1) = \lambda_1/\mu = \rho_1$. Thus, $\widetilde{P}(n_1, 1) = \widetilde{P}(0, 1)\,\rho_1^{n_1}$ and $\widetilde{P}(0, 1) = P_y(1) = \rho(1-\rho_1)$. The constant $\widetilde{P}(0, y)$ is resolved by its definition

$$\widetilde{P}(0, y) = \sum_{n_2=0}^{\infty} \pi(0, n_2)\, y^{n_2} = P(0, y) = P_y(y),$$

as determined in the bivariate PGF approaches. Summing the geometric law (Equation (127)) over $n_1 \geq 0$ for $x \in (0, 1]$ recovers

$$P(x, y) = \sum_{n_1=0}^{\infty} \widetilde{P}(n_1, y)\, x^{n_1} = P_y(y) \sum_{n_1=0}^{\infty} \big[\widetilde{x}^*(y)\, x\big]^{n_1} = \frac{P_y(y)}{1 - \widetilde{x}^*(y)\, x}. \tag{128}$$

Now, in order to bring this to the form stated in Theorem 1, we define the kernel of the fundamental equation divided by μ as

$$K(x, y) := (1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2 = -y\rho_1\big(x - x^*(y)\big)\big(x - x^+(y)\big), \tag{129}$$

where $x^*(y)$ and $x^+(y)$ are the roots found in §6.1.1. By Vieta's product formula, $x^*(y)\, x^+(y) = \mu/\lambda_1 = 1/\rho_1$. This, together with $\widetilde{x}^*(y)$ (Equation (126)), gives $\widetilde{x}^*(y) = \rho_1 x^*(y) = 1/x^+(y)$. Now, looking at the denominator of our reconstructed PGF (Equation (128)), and using the factorisation of $K(x, y)$ (Equation (129)), we see that

$$1 - \widetilde{x}^*(y)\, x = 1 - \frac{x}{x^+(y)} = \frac{x^+(y) - x}{x^+(y)} = \frac{K(x, y)}{y\rho_1\, x^+(y)\big(x - x^*(y)\big)}.$$

Substituting this into our reconstructed $P(x, y)$ (Equation (128)), using $P_y(y) = \frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho)$ and also noting that $\rho_1\, x^*(y)\, x^+(y) = 1$, we obtain

$$\begin{aligned} P(x, y) &= \frac{x^*(y)(1-y)}{x^*(y) - y}\,\rho(1-\rho) \cdot \frac{y\rho_1\, x^+(y)\big(x - x^*(y)\big)}{K(x, y)} \\ &= \underbrace{\rho_1\, x^*(y)\, x^+(y)}_{=1}\ \frac{1}{K(x, y)}\, y(1-y)\, \frac{x - x^*(y)}{x^*(y) - y}\,\rho(1-\rho) \\ &= \big[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\big]^{-1} \cdot y(1-y)\, \frac{x - x^*(y)}{x^*(y) - y}\,\rho(1-\rho), \end{aligned}$$

which recovers Equation (106) from Theorem 1 and concludes the proof. □

6.2 Analysis of Model A: state space $\widetilde{S}$

Setting $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$ in the $\widetilde{S}$ balance equations (29)–(33) and forming the PPGF $\widetilde{P}(y, n) = \sum_{n_2=0}^{n} \widetilde{\pi}(n_2, n)\, y^{n_2}$ yields the inhomogeneous recurrence:

$$\begin{aligned} &[\lambda_1+\lambda_2+\mu]\, \widetilde{P}(y, n) \\ &\quad= \quad (\lambda_1+\lambda_2 y)\widetilde{P}(y, n-1) + \mu\widetilde{P}(y, n+1) + \mu y^n (1-y)\widetilde{\pi}(n+1, n+1) \end{aligned} \tag{130}$$

Theorem 2 (Approximate PPGF on $\widetilde{S}$, heuristic)**.** (The following result rests on an unproved decay assumption; see the proof.) *Under the approximation that neglects the boundary-diagonal forcing term* $\mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1)$ *in* (130)*, the homogeneous recurrence admits the geometric solution*

$$\widetilde{P}(y,n) \;\approx\; \rho(1-\rho)\,[\widetilde{y}^*(y)]^n, \tag{131}$$

where

$$\widetilde{y}^*(y) = \frac{(\lambda+\mu) - \sqrt{(\lambda+\mu)^2 - 4\mu(\lambda_1+\lambda_2 y)}}{2\mu}, \qquad \lambda = \lambda_1+\lambda_2. \tag{132}$$

At $y=1$ this recovers $\widetilde{P}(1,n) = (1-\rho)\rho^{n+1}$, consistent with the $M/M/1(\lambda,\mu)$ steady-state marginal for n customers waiting.

Proof of Theorem 2. The inhomogeneous term $\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)$ prevents the direct application of a geometric-sequence ansatz. The diagonal probabilities $\widetilde{\pi}(n,n) = \pi(0,n)$ correspond to states where all n waiting customers are class 2; from the exact formula in Theorem 1 one can verify that $\pi(0,n)$ decays geometrically in n, but this is not derived here. *Assuming* this geometric decay, the forcing term is exponentially small and we obtain an approximation by neglecting it. The resulting homogeneous recurrence is

$$(\lambda_1+\lambda_2+\mu)\widetilde{P}(y,n) = (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1) + \mu\widetilde{P}(y,n+1). \tag{133}$$

The characteristic polynomial of this recurrence is

$$\mu[\widetilde{y}(y)]^2 - (\lambda_1+\lambda_2+\mu)\widetilde{y}(y) + (\lambda_1+\lambda_2 y) = 0, \tag{134}$$

with roots

$$\widetilde{y}^{\pm}(y) = \frac{(\lambda_1+\lambda_2+\mu) \pm \sqrt{(\lambda_1+\lambda_2+\mu)^2 - 4\mu(\lambda_1+\lambda_2 y)}}{2\mu}. \tag{135}$$

At $y=1$ the roots are $\widetilde{y}_+(1) = 1$ and $\widetilde{y}_-(1) = \rho$. The root $\widetilde{y}_+ = 1$ would produce a non-summable geometric series as $n\to\infty$, so only the negative-sign root is admissible:

$$\widetilde{y}^*(y) = \frac{(\lambda_1+\lambda_2+\mu) - \sqrt{(\lambda_1+\lambda_2+\mu)^2 - 4\mu(\lambda_1+\lambda_2 y)}}{2\mu}. \tag{136}$$

The geometric ansatz gives $\widetilde{P}(y,n) = \widetilde{P}(y,0)\cdot[\widetilde{y}^*(y)]^n$. Using $\widetilde{P}(y,0) = \widetilde{\pi}(0,0) = \pi(0,0) = \rho(1-\rho)$ (Corollary 2):

$$\widetilde{P}(y,n) = \rho(1-\rho)\cdot[\widetilde{y}^*(y)]^n. \tag{137}$$

Since n counts customers *waiting* (the customer in service is not counted), $n+1$ is the total number in the system, so evaluating at $y=1$:

$$\widetilde{P}(1,n) = \rho(1-\rho)\cdot\rho^n = (1-\rho)\rho^{n+1}, \tag{138}$$

□

which matches the $M/M/1(\lambda,\mu)$ stationary distribution for $n+1$ customers in the system [AR02].

6.3 Mean queue lengths and mean waiting times

6.3.1 Class-1 marginal PGF and mean queue length

Setting $y=1$ in the fundamental equation (105) gives

$$\left[-\lambda_1(x-1)(x-1/\rho_1)\right] P(x,1) \;=\; \mu(x-1)\,P_y(1). \tag{139}$$

Cancelling the common factor $(x-1)$ for $x\neq 1$ and rearranging:

$$P(x,1) \;=\; \frac{\mu\,P_y(1)}{\mu-\lambda_1 x}. \tag{140}$$

The constant $P_y(1)$ is pinned by the normalisation $P(1,1) = 1-\pi_0 = \rho$:

$$\rho \;=\; \frac{\mu\,P_y(1)}{\mu-\lambda_1} \quad\Longrightarrow\quad P_y(1) \;=\; \rho\,(1-\rho_1), \tag{141}$$

so that

$$P(x,1) \;=\; \frac{\rho(1-\rho_1)}{1-\rho_1 x}. \tag{142}$$

Differentiating by the quotient rule and evaluating at $x=1$:

$$\mathbb{E}[N_1] \;=\; \frac{d}{dx}P(x,1)\bigg|_{x=1} \;=\; \frac{\rho(1-\rho_1)\cdot\rho_1}{(1-\rho_1 x)^2}\bigg|_{x=1} \;=\; \frac{\rho(1-\rho_1)\,\rho_1}{(1-\rho_1)^2} \;=\; \frac{\rho\rho_1}{1-\rho_1}. \tag{143}$$

6.3.2 Total queue length from the diagonal

Setting $x=y=z$ in the kernel polynomial $\lambda_1 x^2-(\lambda_1+\lambda_2+\mu)x+\mu$ shows that $x^*(z)=z$ iff $z\in\{1,\,1/\rho\}$; hence $x^*(z)\neq z$ for all $z\in(0,1)$ with $\rho<1$. Therefore the ratio $(z-x^*(z))/(x^*(z)-z)=-1$ throughout $(0,1)$, and the formula of Theorem 1 reduces to:

$$P(z,z) \;=\; \frac{z(1-z)\,\rho(1-\rho)}{D(z,z)}\cdot(-1). \tag{144}$$

The denominator at $x=y=z$ is $D(z,z)=(1+\rho)z^2-z-\rho z^3=-\rho z(z-1)(z-1/\rho)$, so:

$$\begin{aligned} P(z,z) \;&=\; \frac{-z(1-z)\,\rho(1-\rho)}{-\rho z(z-1)(z-1/\rho)} \\ &=\; \frac{(1-z)(1-\rho)}{(z-1)(z-1/\rho)} \;=\; \frac{-(1-\rho)}{z-1/\rho} \;=\; \frac{\rho(1-\rho)}{1-\rho z}, \end{aligned} \tag{145}$$

confirming the diagonal identity of Corollary 2. Since $\frac{d}{dz}P(z,z)\big|_{z=1}=\frac{\partial P}{\partial x}(1,1)+\frac{\partial P}{\partial y}(1,1)=\mathbb{E}[N_1]+\mathbb{E}[N_2]$, differentiating by the quotient rule and evaluating at $z=1$ gives:

$$\mathbb{E}[N_1]+\mathbb{E}[N_2] \;=\; \frac{d}{dz}\,\frac{\rho(1-\rho)}{1-\rho z}\bigg|_{z=1} \;=\; \frac{\rho(1-\rho)\cdot\rho}{(1-\rho z)^2}\bigg|_{z=1} \;=\; \frac{\rho^2(1-\rho)}{(1-\rho)^2} \;=\; \frac{\rho^2}{1-\rho}. \tag{146}$$

This is the $M/M/1(\lambda,\mu)$ queue-length formula for the combined system [AR02], as expected: the two classes together behave as a single $M/M/1$ queue.

6.3.3 Class-2 mean queue length by subtraction

$$\begin{aligned} \mathbb{E}[N_2] \;&=\; \big(\mathbb{E}[N_1]+\mathbb{E}[N_2]\big)-\mathbb{E}[N_1] \;=\; \frac{\rho^2}{1-\rho}-\frac{\rho\rho_1}{1-\rho_1} \\ &=\; \frac{\rho^2(1-\rho_1)-\rho\rho_1(1-\rho)}{(1-\rho)(1-\rho_1)} \;=\; \frac{\rho\big[\rho(1-\rho_1)-\rho_1(1-\rho)\big]}{(1-\rho)(1-\rho_1)} \\ &=\; \frac{\rho(\rho-\rho_1)}{(1-\rho)(1-\rho_1)} \;=\; \frac{\rho\rho_2}{(1-\rho_1)(1-\rho)}. \end{aligned} \tag{147}$$

6.3.4 Mean waiting times via Little's Law

Little's Law applied to the *waiting sub-system* of each class: since N_i counts only customers *waiting* in queue (excluding the customer in service), Little's Law gives $\mathbb{E}[N_i]=\lambda_i\,\mathbb{E}[W_i]$, where $\mathbb{E}[W_i]$ is the mean waiting time in queue (time from arrival until service start, not including the service duration):

$$\mathbb{E}[W_1] \;=\; \frac{\mathbb{E}[N_1]}{\lambda_1} \;=\; \frac{\rho\rho_1}{\lambda_1(1-\rho_1)} \;=\; \frac{\rho\cdot(\lambda_1/\mu)}{\lambda_1(1-\rho_1)} \;=\; \frac{\rho}{\mu(1-\rho_1)}, \tag{148}$$

$$\mathbb{E}[W_2] \;=\; \frac{\mathbb{E}[N_2]}{\lambda_2} \;=\; \frac{\rho\rho_2}{\lambda_2(1-\rho_1)(1-\rho)} \;=\; \frac{\rho\cdot(\lambda_2/\mu)}{\lambda_2(1-\rho_1)(1-\rho)} \;=\; \frac{\rho}{\mu(1-\rho_1)(1-\rho)}. \tag{149}$$

Class-1 benefits from priority: $\mathbb{E}[W_2]$ carries the additional penalty factor $1/(1-\rho)$ relative to $\mathbb{E}[W_1]$, reflecting the delay that class-2 customers incur while class-1 customers in the queue are served first. Note that both $\mathbb{E}[W_1]$ and $\mathbb{E}[W_2]$ depend on ρ_2 through the total load $\rho=\rho_1+\rho_2$; class-2 affects class-1 waiting times only via server utilisation, not through a direct queueing penalty.

7 Analysis of Model B ($\gamma_1, \gamma_2 > 0$, $\theta_1 = \theta_2 = 0$)

Model B introduces bidirectional jockeying ($\gamma_1, \gamma_2 > 0$) while retaining no abandonment ($\theta_1 = \theta_2 = 0$). Jockeying conserves the total number of customers in the system, so the stationary distribution still exists under the same condition as Model A. However, jockeying breaks the per-class Poisson/renewal structure: a Class-2 customer can no longer identify its effective service time with a Class-1 busy period, so the Pollaczek–Khinchine probabilistic route is unavailable. Applying the method of characteristics to the fundamental equation yields a general solution, but closing it — i.e., determining the unknown boundary function $P_y(y)$ — requires solving a Volterra-type integral equation that does not admit a closed form. Model B is therefore intentionally left open, in structural parallel with Model X; its role is to motivate the tractable specialisation Model B_2.

Stability. $\rho = (\lambda_1 + \lambda_2)/\mu < 1$. Jockeying conserves the total customer count and does not affect this condition.

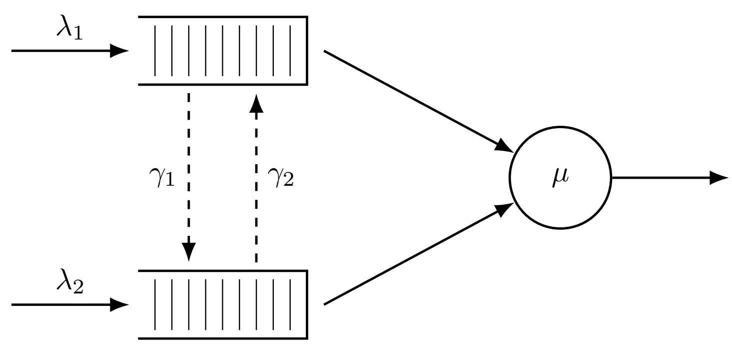

Figure 5: Queue layout for Model B. Dashed arrows indicate bidirectional jockeying: γ_1 moves a Class-1 customer down to Queue 2 and γ_2 moves a Class-2 customer up to Queue 1. Both transfers conserve the total n.

Corollary 3. *The empty-system probability π_0 (server idle, no one waiting) and the both-queues-empty probability $\pi(0,0)$ (server busy, $N_1 = N_2 = 0$) of Model B coincide with the Model A baseline,*

$$\pi_0 = 1 - \rho, \qquad \pi(0,0) = \rho(1-\rho), \tag{150}$$

and the diagonal of the PGF is $P(z,z) = \pi(0,0)/(1-\rho z)$.

Proof. Setting $x = y$ in the fundamental equation (41) with $\theta_1 = \theta_2 = 0$, both jockeying terms carry the factor $(x-y)$ and vanish, as does the boundary term $\mu(x-y)P_y(y)$; dividing the surviving terms by the common factor x leaves

$$\big[(\lambda_1 + \lambda_2 + \mu)x - \mu - (\lambda_1+\lambda_2)x^2\big] P(x,x) = -\mu(1-x)\,\pi(0,0). \tag{151}$$

Writing $\lambda = \lambda_1 + \lambda_2$, the bracket factors as $(\mu - \lambda x)(x-1)$; cancelling $(x-1)$ against $-(1-x) = (x-1)$ on the right gives

$$P(x,x) = \frac{\mu\,\pi(0,0)}{\mu - \lambda x} = \frac{\pi(0,0)}{1-\rho x}. \tag{152}$$

The idle state (0) exchanges probability flux only with the both-queues-empty busy state, so the global balance across the idle–busy cut reads $\lambda\pi_0 = \mu\pi(0,0)$ (an arrival at total rate λ leaves (0); a service completion at rate μ returns to it), whence $\pi(0,0) = \rho\pi_0$. Evaluating (152) at $x = 1$ with $P(1,1) = 1-\pi_0$ gives $1-\pi_0 = \rho\pi_0/(1-\rho)$, so $(1-\pi_0)(1-\rho) = \rho\pi_0$, i.e. $\pi_0 = 1-\rho$ and $\pi(0,0) = \rho(1-\rho)$. This matches Model A (108): jockeying conserves the total-customer process and cannot change the marginal idle probability. The same identities hold in Models B_2 and C_2 for the same conservation reason (23). □

Claim 2. *Under stability, $\rho < 1$, the bivariate PGF of Model B satisfies, on each characteristic curve $\gamma_2 x + \gamma_1 y = C_1$ of the fundamental equation, the integral representation*

$$P(x, y(x)) \;=\; P_h(x, y(x)) \int_0^x \frac{h(t;\,C_1)}{P_h\big(t,\,y(t)\big)}\,dt, \tag{153}$$

where P_h is the homogeneous solution (165), $y(t) = (C_1 - \gamma_2 t)/\gamma_1$ is the characteristic parametrisation, and $h(t; C_1)$ is an explicit forcing function depending on the unknown boundary trace $P_y(y) = P(0, y)$. Evaluating (153) along the diagonal $x = y = z$ and using the conservation identity $P(z, z) = \pi(0, 0)/(1 - \rho z)$ reduces the problem to a Volterra integral equation of the first kind *for P_y:*

$$\mathcal{F}(z) \;=\; \int_0^z \mathcal{K}(z,t)\, P_y\big(y(t;z)\big)\, dt, \tag{154}$$

where $\mathcal{K}$ and $\mathcal{F}$ are fully explicit (defined in (181)–(183) below). The kernel $\mathcal{K}$ does not factorise for general (γ_1, γ_2), so (154) does not reduce to a closed-form solution for P_y. Model B is therefore left open.

Proof of Claim 2. The proof proceeds in five steps: (i) homogeneous solution via characteristics, (ii) rigorous sign analysis of the exponent E, (iii) particular solution via variation of parameters, (iv) pinning the free function from the boundary condition, and (v) diagonal evaluation yielding the Volterra equation.

Step 1: Homogeneous solution via the method of characteristics.
The fundamental equation (41) with $\theta_1 = \theta_2 = 0$ is a first-order linear PDE in $P(x, y)$. Setting the right-hand side to zero and dividing through by y yields the homogeneous PDE

$$\gamma_1 x(x-y)\,\frac{\partial P}{\partial x} - \gamma_2 x(x-y)\,\frac{\partial P}{\partial y} = -\big[(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy\big]P. \tag{155}$$

Following the method of characteristics (§3.5.3), we associate to this PDE the characteristic system

$$\frac{dx}{\gamma_1 x(x-y)} = \frac{dy}{-\gamma_2 x(x-y)} = \frac{dP}{-\big[(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy\big]P}. \tag{156}$$

Taking the ratio of the first two differentials, the $x(x-y)$ factors cancel, giving $dx/\gamma_1 = dy/(-\gamma_2)$. Integrating yields the first integral

$$C_1 = \gamma_2 x + \gamma_1 y \quad \text{(constant along each characteristic).} \tag{157}$$

The characteristics are therefore the straight lines of slope $-\gamma_2/\gamma_1$ in the (x, y)-plane. Fix one such line and parametrise y by $y(x) = (C_1 - \gamma_2 x)/\gamma_1$. To extract the equation for P along this curve we need the coefficient $A(x, y)$ and the denominator $x(x - y)$ after substitution. Setting

$$A(x,y) := (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy, \tag{158}$$

and substituting $y = (C_1 - \gamma_2 x)/\gamma_1$ so that $\lambda_2 xy = (\lambda_2 C_1/\gamma_1)x - (\lambda_2\gamma_2/\gamma_1)x^2$, we obtain

$$A\big(x, y(x)\big) = -\frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1}\,x^2 + \frac{\gamma_1(\lambda_1+\lambda_2+\mu) - \lambda_2 C_1}{\gamma_1}\,x - \mu. \tag{159}$$

Similarly, $x - y(x) = [(\gamma_1 + \gamma_2)x - C_1]/\gamma_1$, so $\gamma_1 x(x - y(x)) = x[(\gamma_1 + \gamma_2)x - C_1]$. The dP/P equation from the characteristic system therefore becomes

$$\frac{dP}{P} = \frac{-A(x,y(x))}{\gamma_1 x(x-y(x))}\,dx = \frac{(\gamma_1\lambda_1 - \lambda_2\gamma_2)x^2 - \big[\gamma_1(\lambda_1+\lambda_2+\mu) - \lambda_2 C_1\big]x + \gamma_1\mu}{\gamma_1 x\big[(\gamma_1+\gamma_2)x - C_1\big]}\,dx. \tag{160}$$

To integrate this, we decompose the integrand via partial fractions. Writing the integrand as $(1/\gamma_1)\, I(x)\, dx$ with $I(x) = K + M/x + N(C_1)/[(\gamma_1 + \gamma_2)x - C_1]$, the partial fraction identity requires

$$K(\gamma_1+\gamma_2)x^2 + \big[M(\gamma_1+\gamma_2) + N - KC_1\big]x - MC_1 = (\gamma_1\lambda_1 - \lambda_2\gamma_2)x^2 - \big[\gamma_1(\lambda_1+\lambda_2+\mu) - \lambda_2 C_1\big]x + \gamma_1\mu.$$

Matching the x^2, x^0, and x^1 coefficients in turn:

$$K = \frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1+\gamma_2}, \qquad M = -\frac{\gamma_1\mu}{C_1}, \qquad N(C_1) = \gamma_1\left[\frac{\lambda_1+\lambda_2}{\gamma_1+\gamma_2}C_1 - (\lambda_1+\lambda_2+\mu) + \frac{\mu(\gamma_1+\gamma_2)}{C_1}\right]. \tag{161}$$

(For N: substituting K and M into the x^1 match gives $N = -\gamma_1(\lambda_1+\lambda_2+\mu)+\lambda_2 C_1+KC_1-M(\gamma_1+\gamma_2)$; expanding KC_1 and $-M(\gamma_1+\gamma_2)$ and collecting yields (161).) Integrating $dP/P = (1/\gamma_1)\, I(x)\, dx$ term by term gives

$$\ln P = \frac{K}{\gamma_1}x - \frac{\mu}{C_1}\ln x + \frac{N(C_1)}{\gamma_1(\gamma_1+\gamma_2)}\ln\big[(\gamma_1+\gamma_2)x - C_1\big] + \ln f(C_1), \tag{162}$$

where $f(C_1)$ is an arbitrary function of the characteristic constant. Now observe that

$$(\gamma_1+\gamma_2)x - C_1 = (\gamma_1+\gamma_2)x - (\gamma_2 x + \gamma_1 y) = \gamma_1(x-y), \tag{163}$$

so $\ln[(\gamma_1+\gamma_2)x - C_1] = \ln\gamma_1 + \ln(x-y)$. The constant $\ln\gamma_1$ is absorbed into a redefined arbitrary function $F(C_1)$. Reading off the exponent of $(x-y)$:

$$E(x,y) := \frac{N(C_1)}{\gamma_1(\gamma_1+\gamma_2)}\bigg|_{C_1=\gamma_2 x+\gamma_1 y} = \frac{\lambda_1+\lambda_2}{(\gamma_1+\gamma_2)^2}(\gamma_2 x+\gamma_1 y) - \frac{\lambda_1+\lambda_2+\mu}{\gamma_1+\gamma_2} + \frac{\mu}{\gamma_2 x+\gamma_1 y}. \tag{164}$$

Exponentiating (162) yields the general homogeneous solution

$$P_h(x,y) = F(\gamma_2 x+\gamma_1 y)\cdot\exp\!\left(\frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1(\gamma_1+\gamma_2)}x\right)\cdot x^{-\mu/(\gamma_2 x+\gamma_1 y)}\cdot (x-y)^{E(x,y)}, \tag{165}$$

where F is an arbitrary differentiable function to be determined.

Step 2: The exponent $E(x,y)$ is strictly positive on $(0,1)^2$.

Write $\lambda = \lambda_1+\lambda_2$ and $\Gamma = \gamma_1+\gamma_2$. Along any fixed characteristic, $C_1 = \gamma_2 x + \gamma_1 y$ is constant, so the exponent (164) depends only on $u := C_1 \in (0,\Gamma)$:

$$E(u) = \frac{\lambda}{\Gamma^2}\,u - \frac{\lambda+\mu}{\Gamma} + \frac{\mu}{u}. \tag{166}$$

(For $(x,y)\in(0,1)^2$ we have $u = \gamma_2 x+\gamma_1 y \in (0,\Gamma)$; $E(u)$ is continuous there, with $E(u)\to+\infty$ as $u\to 0^+$ and $E(u)\to 0^+$ as $u\to\Gamma^-$, so it is finite at every interior point although not uniformly bounded.) Multiplying through by $u\Gamma^2>0$ gives the equivalent quadratic

$$q(u) := \lambda u^2 - (\lambda+\mu)\Gamma u + \mu\Gamma^2. \tag{167}$$

The sign of E on $(0,\Gamma)$ equals the sign of $q(u)$ there. We factor q by finding its roots. The discriminant is $\Delta = (\lambda+\mu)^2\Gamma^2 - 4\lambda\mu\Gamma^2 = \Gamma^2(\mu-\lambda)^2$, so (using $\rho<1 \Rightarrow \mu>\lambda \Rightarrow \mu-\lambda>0$)

$$u_{1,2} = \frac{(\lambda+\mu)\Gamma \pm \Gamma(\mu-\lambda)}{2\lambda} \quad\Longrightarrow\quad u_1 = \frac{2\lambda\Gamma}{2\lambda} = \Gamma,\quad u_2 = \frac{2\mu\Gamma}{2\lambda} = \frac{\Gamma}{\rho} > \Gamma. \tag{168}$$

Hence

$$q(u) = \lambda(u-\Gamma)\left(u - \tfrac{\Gamma}{\rho}\right). \tag{169}$$

For every $u\in(0,\Gamma)$ both factors are strictly negative ($(u-\Gamma)<0$ and $(u-\Gamma/\rho)<0$ since $\Gamma/\rho>\Gamma$), so $q(u) = \lambda\cdot(\text{neg})\cdot(\text{neg})>0$, and therefore

$$E(x,y) > 0 \quad \text{for all } (x,y)\in(0,1)^2. \tag{170}$$

Note that $E\to 0$ as $u\to\Gamma^-$ (i.e. as $(x,y)\to(1,1)$), while E is bounded and bounded away from zero in any compact subset of $(0,1)^2$.

Sign convention and diagonal limit. Because $E>0$, the factor $(x-y)^E$ in P_h carries the complex phase $e^{i\pi E}$ whenever $x<y$. This phase is absorbed into the arbitrary function $F(C_1)$, and the ratio $P_h(x,y(x))/P_h(t,y(t))$ that appears in the integral representation is always real and positive: both numerator and denominator lie on the same characteristic (same C_1, same constant value of E), so the phases cancel exactly. Crucially, since $E>0$ the factor $(y(t;z)-t)^E \to 0$ as $t\to z^-$, meaning P_h formally vanishes on the diagonal $x=y$. The diagonal equation that appears in Step 5 below is therefore understood as the $\lim_{x\to z^-}$ of (176); an asymptotic analysis near $t=z$ (expanding $y(t;z)-t \sim (\Gamma/\gamma_1)(z-t)$) confirms that the $0\cdot\infty$ limit is finite and equals $\pi(0,0)/(1-\rho z)$.

Step 3: Particular solution via variation of parameters.

We now work with the full (inhomogeneous) fundamental equation restricted to a fixed characteristic. Along the curve $y(x) = (C_1 - \gamma_2 x)/\gamma_1$, the total derivative satisfies $\frac{d}{dx}P(x, y(x)) = \partial_x P + \frac{dy}{dx}\partial_y P = \partial_x P - \frac{\gamma_2}{\gamma_1}\partial_y P$, so the jockeying terms in (41) become

$$\gamma_1 xy(x-y)\,\partial_x P + \gamma_2 xy(y-x)\,\partial_y P = \gamma_1 xy(x-y)\Big(\partial_x P - \tfrac{\gamma_2}{\gamma_1}\partial_y P\Big) = \gamma_1 xy(x-y)\,\frac{d}{dx}P(x, y(x)).$$

Substituting into the fundamental equation and dividing by $\gamma_1 x(x - y(x))$ reduces the PDE to the first-order linear ODE (for fixed C_1)

$$\frac{dP}{dx} = f(x; C_1)\,P + h(x; C_1), \tag{171}$$

where the coefficient f and the forcing term h are

$$f(x; C_1) = -\frac{A(x,\, y(x))}{\gamma_1 x\big(x - y(x)\big)}, \tag{172}$$

$$h(x; C_1) = \frac{G(x,\, y(x))}{\gamma_1 x\, y(x)\big(x - y(x)\big)}, \tag{173}$$

with A as in (158) and

$$G(x, y) := \mu(x-y)\,P_y(y) - \mu x(1-y)\,\pi(0,0). \tag{174}$$

Since $P_h(x, y(x))$ satisfies the homogeneous equation $\frac{d}{dx}P_h = f(x; C_1)\,P_h$, it is the fundamental solution of ODE (171). Applying variation of parameters (§3.5.2), the general solution of the inhomogeneous ODE is

$$P(x, y(x)) = P_h(x, y(x))\left[F(C_1) + \int_{x_0}^{x}\frac{h(t; C_1)}{P_h(t, y(t))}\,dt\right]. \tag{175}$$

Step 4: Pinning the free function from the boundary condition at $x = 0$.

Along any characteristic with constant $C_1 > 0$, the endpoint at $x = 0$ is $y(0) = C_1/\gamma_1$, giving

$$P\left(0,\, \frac{C_1}{\gamma_1}\right) = P_y\left(\frac{C_1}{\gamma_1}\right),$$

which is finite. However, P_h contains the factor $x^{-\mu/C_1} \to +\infty$ as $x \to 0^+$ (since $\mu/C_1 > 0$ for $C_1 > 0$), while the integral in (175) vanishes as $x_0 \to 0$ (the integration interval collapses). Therefore

$$\lim_{x\to 0^+} P(x, y(x)) = \lim_{x\to 0^+} P_h(x, y(x)) \cdot F(C_1) = +\infty \cdot F(C_1),$$

which can only be finite if $F(C_1) = 0$. Setting $x_0 = 0$ and $F(C_1) = 0$ in (175) gives the integral representation of Claim 2:

$$P(x, y(x)) = P_h(x, y(x))\int_0^x \frac{h(t; C_1)}{P_h(t, y(t))}\,dt. \tag{176}$$

Step 5: Reduction to the Volterra equation.

By Corollary 3 the empty-state probabilities are $\pi_0 = 1 - \rho$ and $\pi(0,0) = \rho(1-\rho)$, with diagonal value $P(z, z) = \pi(0,0)/(1 - \rho z)$ by (152); these are the inputs to the diagonal evaluation below.

We simplify the forcing function h on the characteristic. Substituting $G(t, y(t))$ from (174) into (173):

$$\begin{aligned} h(t; C_1) &= \frac{\mu(t - y(t))\,P_y(y(t)) - \mu t(1 - y(t))\,\pi(0,0)}{\gamma_1\, t\, y(t)\,(t - y(t))} \\ &= \frac{\mu\, P_y(y(t))}{\gamma_1\, t\, y(t)} - \frac{\mu(1 - y(t))\,\pi(0,0)}{\gamma_1\, y(t)\,(t - y(t))}. \end{aligned} \tag{177}$$

We will evaluate (176) along the diagonal characteristic, which passes through (z, z) and has $C_1 = (\gamma_1 + \gamma_2)z$. The parametrisation along that curve is $y(t; z) = ((\gamma_1 + \gamma_2)z - \gamma_2 t)/\gamma_1$, so $y(0; z) =$

$(\gamma_1+\gamma_2)z/\gamma_1 > z$ and $y(z;z) = z$. Therefore $y(t;z) > t$ for all $t \in [0,z)$, which means $t - y(t;z) < 0$ throughout the integration interval. Writing $t - y(t;z) = -(y(t;z) - t)$ in the denominator of (177):

$$h\big(t;\,(\gamma_1+\gamma_2)z\big) = \frac{\mu}{\gamma_1\, t\, y(t;z)}\, P_y\big(y(t;z)\big) + \frac{\mu\,(1-y(t;z))}{\gamma_1\, y(t;z)\,(y(t;z)-t)}\,\pi(0,0). \tag{178}$$

As established in Step 2, $E > 0$, so the homogeneous factor $(x-y)^E$ vanishes on the diagonal and $P_h\big(x, y(x;z)\big) \to 0$ as $x \to z^-$; simultaneously the integrand in (176) develops a non-integrable singularity $\sim (y(t;z)-t)^{-1-E}$ as $t \to z^-$, so the integral diverges. Their product is the finite $0\cdot\infty$ limit

$$\frac{\pi(0,0)}{1-\rho z} = \lim_{x\to z^-} P_h\big(x, y(x;z)\big) \int_0^x \frac{h\big(t;(\gamma_1+\gamma_2)z\big)}{P_h\big(t, y(t;z)\big)}\, dt, \tag{179}$$

whose value is $P(z,z) = \pi(0,0)/(1-\rho z)$ by Corollary 3. Substituting (178) and separating the term containing the unknown P_y from the known remainder, this limit splits as

$$\begin{aligned}\frac{\pi(0,0)}{1-\rho z} = \lim_{x\to z^-} & \frac{\mu\, P_h\big(x, y(x;z)\big)}{\gamma_1} \int_0^x \frac{P_y\big(y(t;z)\big)}{t\cdot y(t;z)\cdot P_h\big(t, y(t;z)\big)}\, dt \\ &+ \lim_{x\to z^-} \frac{\mu\, P_h\big(x, y(x;z)\big)\,\pi(0,0)}{\gamma_1} \int_0^x \frac{1-y(t;z)}{y(t;z)\cdot(y(t;z)-t)\cdot P_h\big(t, y(t;z)\big)}\, dt.\end{aligned} \tag{180}$$

Each limit is a Hadamard finite part: the vanishing prefactor $P_h^{\mathrm{diag}}(z) := \lim_{x\to z^-} P_h\big(x, y(x;z)\big)$ regularises the divergent integral, and neither factor is meaningful in isolation. Writing the kernel and known term *formally* as

$$\mathcal{K}(z,t) := \frac{\mu\cdot P_h^{\mathrm{diag}}(z)}{\gamma_1\cdot t\cdot y(t;z)\cdot P_h\big(t, y(t;z)\big)}, \tag{181}$$

$$\mathcal{B}(z) := \frac{\mu\cdot P_h^{\mathrm{diag}}(z)\cdot\pi(0,0)}{\gamma_1} \int_0^z \frac{1-y(t;z)}{y(t;z)\cdot(y(t;z)-t)\cdot P_h\big(t, y(t;z)\big)}\, dt, \tag{182}$$

to be read through the regularised limit (180) *rather than pointwise*, and setting $\mathcal{F}(z) = \pi(0,0)/(1-\rho z) - \mathcal{B}(z)$, equation (180) becomes the Volterra integral equation of the first kind

$$\mathcal{F}(z) = \int_0^z \mathcal{K}(z,t)\, P_y\big(y(t;z)\big)\, dt, \tag{183}$$

confirming Claim 2. The kernel $\mathcal{K}$ depends on P_h through both arguments and does not simplify for general (γ_1, γ_2), so (183) does not yield a closed form. Resolving it would require Volterra inversion methods well beyond the scope of this thesis. □

Model B is therefore left open at this point. Its role in the hierarchy is structural: it identifies the precise obstruction — the Volterra equation for P_y — that the tractable special cases B_2 and C_2 circumvent, each by a different mechanism. Setting $\gamma_2 = 0$ collapses the characteristic family from a two-parameter set of oblique lines to a one-parameter set of vertical lines, degenerating the PDE to an ODE in x alone; the boundary function $P_y(y)$ is then pinned by an analyticity argument at the interior singular point $x = y$. Model C_2 removes jockeying entirely and pins P_y by a boundedness argument at the boundary of the domain. Both specialisations are analysed in the following sections.

8 Analysis of Model B_2 ($\gamma_1 > 0$, $\gamma_2 = 0$, $\theta_1 = \theta_2 = 0$)

Model B_2 restricts jockeying to the direction $1 \to 2$ only ($\gamma_1 > 0$, $\gamma_2 = 0$) and retains no abandonment ($\theta_1 = \theta_2 = 0$). Setting $\gamma_2 = 0$ eliminates the $\partial P/\partial y$ term from the fundamental equation, reducing it to a first-order linear ODE in x for each fixed $y \in [0,1]$. An integrating factor then yields a formal integral expression for $P(x,y)$; the boundary function $P_y(y)$ is pinned by requiring $P(\cdot, y)$ to be holomorphic at the interior singular point $x = y$, where the integrating factor carries a branch singularity with negative exponent. This analyticity condition replaces the boundedness argument used for Model C_2.

Stability. As in Models A and B, $\rho = (\lambda_1 + \lambda_2)/\mu < 1$. Jockeying conserves the total customer count and does not affect the stability condition.

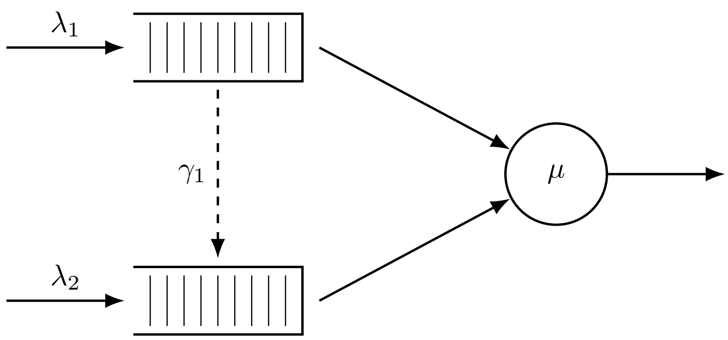

Figure 6: Queue layout for Model B_2. Only the $1 \to 2$ jockeying direction is active (dashed, downward): Class-1 customers may defect to Queue 2, but no customer can move in the reverse direction ($\gamma_2 = 0$).

8.1 Reduced fundamental equation

Setting $\gamma_2 = 0$ and $\theta_1 = \theta_2 = 0$ in the general fundamental equation (41) (Lemma 1) annihilates the $\partial P/\partial y$ term, whose coefficient $xy\big[\gamma_2(y-x) + \theta_2(y-1)\big]$ vanishes identically, and leaves only the γ_1 contribution to the $\partial P/\partial x$ term:

$$\begin{aligned}\big[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\big] P + \gamma_1 xy(x-y)\,\frac{\partial P}{\partial x} \\ = \mu(x-y)\,P_y(y) - \mu x(1-y)\,\pi(0,0).\end{aligned} \tag{184}$$

For each fixed $y \in (0,1]$ this is a first-order linear ODE in x; the entire analysis of this section is built on it.

Theorem 3. *Consider the two-class non-preemptive priority queueing system of Model B_2 with per-customer jockeying rate $\gamma_1 > 0$ from Queue 1 to Queue 2 ($\gamma_2 = 0$, $\theta_1 = \theta_2 = 0$). Under the stability condition $\rho = \lambda/\mu < 1$, $\lambda = \lambda_1 + \lambda_2$, the boundary generating function is*

$$P_y(y) = \frac{\mu\,\pi(0,0)}{\mu - \lambda y} \cdot \frac{{}_1F_1\big(\alpha(y)+1;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha(y);\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}, \tag{185}$$

and the bivariate PGF, for $0 \le x \le y$, is

$$P(x,y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}\, x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Big[P_y(y)\,\mathcal{I}_1(x,y) + (1-y)\pi(0,0)\,\mathcal{I}_2(x,y)\Big], \tag{186}$$

where ${}_1F_1$ is Kummer's confluent hypergeometric function, the exponents are

$$\alpha(y) = \frac{\mu}{\gamma_1 y}, \qquad \beta(y) = \frac{(1-y)(\lambda y - \mu)}{\gamma_1 y}, \qquad b^\star(y) = \alpha + \beta + 1 = \frac{\mu + \lambda(1-y) + \gamma_1}{\gamma_1}, \tag{187}$$

the auxiliary integrals are

$$\mathcal{I}_1(x,y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\alpha-1}(y-t)^{\beta}\, dt, \tag{188}$$

$$\mathcal{I}_2(x,y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\alpha}(y-t)^{\beta-1}\, dt, \tag{189}$$

and the empty-state probability $\pi(0,0) = \rho(1-\rho)$ is given by Corollary 4 below. On $y < x \le 1$ the same expression (186) holds with $\mathcal{I}_1, \mathcal{I}_2$ replaced by their finite-part continuations.

Corollary 4. *The empty-system probability π_0 (server idle, no one waiting) and the both-queues-empty probability $\pi(0,0)$ (server busy, $N_1 = N_2 = 0$) of Model B_2 are*

$$\pi_0 = 1 - \rho, \qquad \pi(0,0) = \rho(1-\rho), \tag{190}$$

identical to the Model A baseline.

Proof. Because one-way jockeying conserves the total customer count and there is no abandonment, the total-count process L (equal to the number in queue N plus the single customer in service whenever the server is busy) is itself a birth–death chain: it increases at rate $\lambda = \lambda_1 + \lambda_2$ on any arrival and decreases at rate μ on any service completion, irrespective of class. Thus L is an $M/M/1$ queue, and by the memorylessness of the exponential clocks its stationary law is geometric, $\mathbb{P}(L = k) = (1-\rho)\rho^k$; in particular $\pi_0 = \mathbb{P}(L = 0) = 1 - \rho$.

For $\pi(0,0)$, set $x = y$ in the reduced fundamental equation (184). The convection term $\gamma_1 xy(x - y)\partial_x P$ and the boundary term $\mu(x-y)P_y(y)$ both carry the factor $(x-y)$ and vanish on the diagonal; dividing the surviving terms by the common factor x leaves

$$\big[(\lambda_1 + \lambda_2 + \mu)x - \mu - (\lambda_1 + \lambda_2)x^2\big]P(x,x) = -\mu(1-x)\,\pi(0,0). \tag{191}$$

Factoring the bracket and cancelling $(x-1)$ against $-(1-x) = (x-1)$ on the right:

$$\begin{aligned}\big[-\lambda x^2 + (\lambda+\mu)x - \mu\big]P(x,x) &= (\mu - \lambda x)(x-1)\,P(x,x)\\ &= \mu(x-1)\,\pi(0,0),\end{aligned} \tag{192}$$

so that

$$P(x,x) = \frac{\mu\,\pi(0,0)}{\mu - \lambda x} = \frac{\pi(0,0)}{1-\rho x}. \tag{193}$$

Evaluating at $x = 1$ with $P(1,1) = 1 - \pi_0 = \rho$ (using $\pi_0 = 1-\rho$ just established) yields $\pi(0,0) = \rho(1-\rho)$. ☐

Proof of Theorem 3. Stage 1: Standard form and the integrating factor.

For any fixed $y \neq 0$, divide the reduced fundamental equation (184) through by the coefficient $\gamma_1 xy(x-y)$ of $\partial P/\partial x$. The factor y cancels against the overall factor y in the bracket multiplying P, putting the equation in standard first-order form $\frac{dP}{dx} + p(x;y)\,P = q(x;y)$ with

$$p(x;y) = \frac{(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy}{\gamma_1 x(x-y)}, \tag{194}$$

$$q(x;y) = \frac{\mu\,P_y(y)}{\gamma_1 xy} - \frac{\mu(1-y)\,\pi(0,0)}{\gamma_1 y(x-y)}. \tag{195}$$

Write the numerator of p as a polynomial in x,

$$N(x) = -\lambda_1 x^2 + \big[(\lambda_1+\lambda_2+\mu) - \lambda_2 y\big]x - \mu = -\lambda_1\big(x - x^*(y)\big)\big(x - x^+(y)\big),$$

which factors through the roots $x^*(y), x^+(y)$ of the Model A kernel (Equation (112)), with $\lambda_1 x^* x^+ = \mu$. The partial-fraction coefficients of $p = N(x)/\big[\gamma_1 x(x-y)\big]$ follow by residues:

$$\begin{aligned}
A &= -\lambda_1 && \text{(ratio of leading } x^2 \text{ terms)},\\
B &= \frac{N(0)}{0-y} = \frac{-\mu}{-y} = \frac{\mu}{y},\\
C &= \frac{N(y)}{y} = \frac{-(\lambda y-\mu)(y-1)}{y} = \frac{(1-y)(\lambda y - \mu)}{y},
\end{aligned} \tag{196}$$

where $N(y) = -\lambda y^2 + (\lambda+\mu)y - \mu = -(\lambda y - \mu)(y-1)$. Hence

$$p(x;y) = \frac{1}{\gamma_1}\left[-\lambda_1 + \frac{\mu/y}{x} + \frac{(1-y)(\lambda y-\mu)/y}{x-y}\right]. \tag{197}$$

Integrating term by term yields the integrating factor

$$\mu_I(x;y) = e^{-\lambda_1 x/\gamma_1}\cdot x^{\,\alpha(y)}\cdot (x-y)^{\,\beta(y)}, \tag{198}$$

with exponents $\alpha(y) = \mu/(\gamma_1 y) > 0$ and $\beta(y) = (1-y)(\lambda y - \mu)/(\gamma_1 y)$ as in (187). Since $\alpha > 0$, $\mu_I \to 0$ as $x \to 0^+$, so the boundary term $P(0,y)\mu_I(0;y)$ vanishes when we integrate $\frac{d}{dx}[P\cdot\mu_I] = q\cdot\mu_I$ over $(0,x]$:

$$P(x,y)\,\mu_I(x;y) = \int_0^x q(t;y)\,\mu_I(t;y)\,dt. \tag{199}$$

Stage 2: Determination of $P_y(y)$ by analyticity at $x = y$.

Unlike Models A and C_2, no probabilistic (Pollaczek–Khinchine) route is available here. One-way jockeying makes the effective service experienced by a Class-2 customer depend on the Class-1 queue contents at the (random) jockeying instants, so the Class-2 input is no longer a Poisson stream feeding an $M/G/1$ subsystem and no busy-period reduction applies (cf. the discussion of Model B in §7). We therefore determine $P_y(y)$ *purely analytically*, by imposing holomorphy of $P(\cdot, y)$ at the interior point $x = y$.

Under stability, $\mu - \lambda y > 0$ for all $y \in (0, 1)$, so

$$\beta(y) = \frac{(1-y)(\lambda y - \mu)}{\gamma_1 y} < 0, \qquad 0 < y < 1. \tag{200}$$

Hence $(x - y)^\beta \to \infty$ as $x \to y$: the integrating factor μ_I diverges at the interior point $x = y$, and the solution $P = \mu_I^{-1} \int q\, \mu_I\, dt$ is a $0 \cdot \infty$ form there that must be resolved. The key observation is that $P(\cdot, y)$, being a probability generating function, must be holomorphic on the open unit disc; the non-integer-power branch term $(x - y)^{|\beta|}$ in the general solution must therefore vanish.

Branch resolution. For $0 < t < x < y$ the base $(t - y) < 0$, so $(t - y)^\beta = e^{i\pi\beta}(y - t)^\beta$ and $(t - y)^{\beta-1} = e^{i\pi(\beta-1)}(y - t)^{\beta-1}$. Substituting these into $P = \mu_I^{-1} \int_0^x q\, \mu_I\, dt$, the common phase $e^{i\pi\beta}$ shared by $\mu_I^{-1}(x) \propto (x - y)^{-\beta}$ and the $\mathcal{I}_1$-integrand cancels, while the $\mathcal{I}_2$-integrand carries the *relative* phase $e^{i\pi(\beta-1)}/e^{i\pi\beta} = e^{-i\pi} = -1$, which converts the minus sign in (195) into a plus. The manifestly real form of the solution (199) on $0 < x < y$ is therefore

$$P(x, y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}\, x^{-\alpha}(y - x)^{-\beta}}{\gamma_1 y} \Big[P_y(y)\, \mathcal{I}_1(x, y) + (1 - y)\pi(0, 0)\, \mathcal{I}_2(x, y) \Big], \tag{201}$$

with $\mathcal{I}_1, \mathcal{I}_2$ as defined in (188)–(189).

Local analysis at $x = y$. The ODE (194)–(195) has a simple pole at $x = y$ with residue β in p and a leading singularity $-D/(x - y)$ in q, where $D = \mu(1 - y)\pi(0, 0)/(\gamma_1 y)$. The local model equation is

$$P'(x, y) + \frac{\beta}{x - y}\, P(x, y) = \frac{-D}{x - y} + (\text{analytic near } x = y). \tag{202}$$

A constant particular solution $P \equiv P_*$ requires $\beta P_* = -D$, giving

$$P(y, y) = P_* = -\frac{D}{\beta} = \frac{\mu\, \pi(0, 0)}{\mu - \lambda y} = \frac{\rho(1 - \rho)}{1 - \rho y}, \tag{203}$$

consistent with (193) (using $\pi(0, 0) = \rho(1 - \rho)$). The general local solution adds a multiple of the homogeneous mode $(x - y)^{-\beta} = (x - y)^{|\beta|}$. Since $|\beta| \notin \mathbb{Z}$ generically, this is a branch term: it vanishes at $x = y$ but is not analytic there. Holomorphy therefore forces its coefficient to zero.

No-branch condition and $P_y(y)$. As $x \to y^-$ the two integrals in (201) each split into a divergent part and a finite part. The divergent part of $\mathcal{I}_2$ (scaling like $(y - x)^\beta$) combines with the prefactor $(y - x)^{-\beta}$ to reproduce exactly the regular value (203), and the divergent part of $\mathcal{I}_1$ contributes an integer power $(y - x)^1$; both are analytic at $x = y$. The only non-analytic contribution is the prefactor $(y - x)^{-\beta} = (y - x)^{|\beta|}$ multiplying the *finite parts* of $\mathcal{I}_1, \mathcal{I}_2$. Eliminating this branch term requires

$$P_y(y)\, \mathrm{f.p.}\, \mathcal{I}_1(y, y) + (1 - y)\, \pi(0, 0)\, \mathrm{f.p.}\, \mathcal{I}_2(y, y) = 0, \tag{204}$$

where f.p. denotes the Hadamard finite part (the analytic continuation in the exponent parameter). Substituting $t = ys$ in (188)–(189) gives $\int_0^1 s^{\alpha-1}(1-s)^\beta e^{-cs}\, ds$ (resp. $s^\alpha(1-s)^{\beta-1}$) with $c = \lambda_1 y/\gamma_1$, to which we apply the Euler integral representation (11), read as its analytic continuation in the second parameter b beyond the convergence domain $b > 0$ (here $b = \beta + 1$, resp. $b = \beta$, with $\beta < 0$ by (200) — this is precisely the finite part):

$$\mathrm{f.p.}\, \mathcal{I}_1(y, y) = B(\alpha,\, \beta + 1)\, y^{\alpha+\beta}\, {}_1F_1\big(\alpha;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big), \tag{205}$$

$$\mathrm{f.p.}\, \mathcal{I}_2(y, y) = B(\alpha + 1,\, \beta)\, y^{\alpha+\beta}\, {}_1F_1\big(\alpha + 1;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big). \tag{206}$$

The common factor $y^{\alpha+\beta}$ cancels in (204). The Beta ratio reduces, via $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ and $\Gamma(\beta+1) = \beta\Gamma(\beta)$, to

$$\begin{aligned}\frac{B(\alpha+1,\,\beta)}{B(\alpha,\,\beta+1)} &= \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)}\cdot\frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta+1)}\\ &= \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)}\cdot\frac{\Gamma(\beta)}{\Gamma(\beta+1)} = \frac{\alpha}{\beta},\end{aligned}\tag{207}$$

and the prefactor combination simplifies, using $\alpha/\beta = \mu/\big[(1-y)(\lambda y-\mu)\big]$, as

$$\begin{aligned}(1-y)\left(-\frac{\alpha}{\beta}\right) &= -(1-y)\cdot\frac{\mu}{(1-y)(\lambda y-\mu)}\\ &= -\frac{\mu}{\lambda y-\mu} = \frac{\mu}{\mu-\lambda y}.\end{aligned}\tag{208}$$

Solving (204) for $P_y(y)$:

$$P_y(y) = \frac{\mu\,\pi(0,0)}{\mu-\lambda y}\cdot\frac{{}_1F_1\big(\alpha(y)+1;\,b^\star(y);\,-\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha(y);\,b^\star(y);\,-\lambda_1 y/\gamma_1\big)},\tag{209}$$

confirming (185). The prefactor equals the regular value $P(y,y)$ of (203), and the Kummer ratio is the boundary correction. As $y\to 0^+$ the argument $-\lambda_1 y/\gamma_1 \to 0$, so the ratio tends to 1 and $P_y(0) = \pi(0,0)$, consistent with $P_y(0) = \sum_{n_2}\pi(0,n_2)\cdot 0^{n_2} = \pi(0,0)$.

Stage 3: Recovery of $P(x,y)$.

Substituting (209) into the real-form expression (201) gives, for $0\le x\le y$,

$$\begin{aligned}P(x,y) = \frac{\mu\,e^{\lambda_1 x/\gamma_1}\,x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Bigg[&\frac{\mu\,\pi(0,0)}{\mu-\lambda y}\cdot\frac{{}_1F_1\big(\alpha+1;\,b^\star;\,-\frac{\lambda_1 y}{\gamma_1}\big)}{{}_1F_1\big(\alpha;\,b^\star;\,-\frac{\lambda_1 y}{\gamma_1}\big)}\cdot\mathcal{I}_1(x,y)\\ &+(1-y)\pi(0,0)\cdot\mathcal{I}_2(x,y)\Bigg],\end{aligned}\tag{210}$$

which is the statement of the theorem (Equation (186)). The no-branch condition (204) guarantees that this extends analytically across $x=y$, with diagonal value $P(y,y) = \rho(1-\rho)/(1-\rho y)$ as computed in (203). On $y < x \le 1$ the same formula holds with $\mathcal{I}_1,\mathcal{I}_2$ read as their finite-part continuations; $\pi(0,0) = \rho(1-\rho)$ from Corollary 4 makes the solution fully explicit. In the limit $\gamma_1\to 0^+$, the Kummer ratio collapses to the rational expression $x^*(y)(1-y)/(x^*(y)-y)$ of Model A, recovering formula (105). □

8.2 Mean queue lengths and mean waiting times

Total queue length from the diagonal

Setting $x=y=z$ in the reduced fundamental equation (184), the jockeying convection term $\gamma_1 xy(x-y)\partial_x P$ vanishes at $x=y=z$ (factor $(x-y)=0$), as does the boundary term $\mu(x-y)P_y(y)$. The remaining balance is identical to Model A's, so the same diagonal cancellation gives:

$$P(z,z) \;=\; \frac{\rho(1-\rho)}{1-\rho z},\tag{211}$$

the same formula as (145). Since $P(z,z) = \mathbb{E}[z^{N_1+N_2}] = \mathbb{E}[z^N]$ is the PGF of the total in-queue count N, the generating-function identity $\frac{d}{dz}P(z,z)\big|_{z=1} = \mathbb{E}[N_1]+\mathbb{E}[N_2]$ holds; differentiating by the quotient rule and evaluating at $z=1$:

$$\mathbb{E}[N_1]+\mathbb{E}[N_2] \;=\; \frac{d}{dz}\,\frac{\rho(1-\rho)}{1-\rho z}\bigg|_{z=1} \;=\; \frac{\rho(1-\rho)\cdot\rho}{(1-\rho z)^2}\bigg|_{z=1} \;=\; \frac{\rho^2(1-\rho)}{(1-\rho)^2} \;=\; \frac{\rho^2}{1-\rho}.\tag{212}$$

One-way jockeying does not alter the total mean queue length relative to Model A, since the diagonal PGF is unchanged.

Class-1 mean queue length

We extract $\mathbb{E}[N_1] = \frac{\partial P}{\partial x}\big|_{(1,1)}$ by first reducing the fundamental equation (184) at $y = 1$, and then differentiating the resulting marginal PGF at $x = 1$.

Step 1: marginal PGF $P(x,1)$. Evaluating (184) at $y = 1$, the bracket multiplying P becomes $(\lambda_1 + \mu)x - \mu - \lambda_1 x^2 = -\lambda_1(x-1)(x-1/\rho_1)$, the convection coefficient is $\gamma_1 x(x-1)$, and the right-hand side reduces to $\mu(x-1)P_y(1)$ (the $\pi(0,0)$ term carries the factor $(1-y) = 0$). Every term therefore shares the common factor $(x-1)$; cancelling it leaves the first-order linear ODE in x:

$$-\lambda_1(x - 1/\rho_1)\, P(x,1) \;+\; \gamma_1 x \,\frac{d}{dx}P(x,1) \;=\; \mu\, P_y(1). \tag{213}$$

The integrating factor for this ODE is $e^{-\lambda_1 x/\gamma_1}\, x^{\mu/\gamma_1}$ (noting $\lambda_1/\rho_1 = \mu$), and integrating from 0 to x (the boundary term vanishes since $0^{\mu/\gamma_1} = 0$):

$$P(x,1)\, e^{-\lambda_1 x/\gamma_1}\, x^{\mu/\gamma_1} \;=\; \frac{\mu\, P_y(1)}{\gamma_1} \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\mu/\gamma_1 - 1}\, dt. \tag{214}$$

Rearranging and writing $\alpha = \mu/\gamma_1$ and $\mathcal{J}(x) = \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\alpha-1}\, dt$:

$$P(x,1) \;=\; \frac{\mu\, P_y(1)}{\gamma_1} \cdot e^{\lambda_1 x/\gamma_1}\, x^{-\alpha}\, \mathcal{J}(x). \tag{215}$$

The constant $P_y(1)$ follows from $P(1,1) = 1 - \pi_0 = \rho$, which gives $P_y(1) = \rho\gamma_1 / \big(\mu\, e^{\lambda_1/\gamma_1}\, \mathcal{J}(1)\big)$.

Step 2: differentiate $P(x,1)$ at $x = 1$. Writing $P(x,1) = C \cdot f(x) \cdot \mathcal{J}(x)$ with $C = \mu P_y(1)/\gamma_1$ and $f(x) = e^{\lambda_1 x/\gamma_1}\, x^{-\alpha}$, the product rule together with Leibniz's rule ($d\mathcal{J}/dx = e^{-\lambda_1 x/\gamma_1}\, x^{\alpha-1}$, the integrand at the upper limit) gives:

$$\frac{d}{dx}P(x,1) \;=\; C\Big[f'(x)\, \mathcal{J}(x) + f(x) \cdot e^{-\lambda_1 x/\gamma_1}\, x^{\alpha-1}\Big]. \tag{216}$$

The logarithmic derivative $f'(x)/f(x) = \lambda_1/\gamma_1 - \alpha/x = (\lambda_1 x - \mu)/(\gamma_1 x)$, and the second product term simplifies as $f(x)\, e^{-\lambda_1 x/\gamma_1} x^{\alpha-1} = x^{-1}$, so:

$$\frac{d}{dx}P(x,1) \;=\; \frac{\mu\, P_y(1)}{\gamma_1} \left[e^{\lambda_1 x/\gamma_1}\, x^{-\alpha} \cdot \frac{\lambda_1 x - \mu}{\gamma_1 x} \cdot \mathcal{J}(x) \;+\; x^{-1}\right]. \tag{217}$$

Step 3: evaluate at $x = 1$.

$$\begin{aligned} \mathbb{E}[N_1] \;&=\; \frac{d}{dx}P(x,1)\bigg|_{x=1} \\ &=\; \frac{\mu\, P_y(1)}{\gamma_1} \left[e^{\lambda_1/\gamma_1} \cdot \frac{\lambda_1 - \mu}{\gamma_1} \cdot \mathcal{J}(1) \;+\; 1\right], \end{aligned} \tag{218}$$

where $\mathcal{J}(1) = \int_0^1 e^{-\lambda_1 t/\gamma_1}\, t^{\mu/\gamma_1 - 1}\, dt = (\gamma_1/\lambda_1)^{\mu/\gamma_1}\, \gamma(\mu/\gamma_1,\, \lambda_1/\gamma_1)$ and $\gamma(\cdot,\cdot)$ is the lower incomplete gamma function. The factor $(\lambda_1 - \mu) < 0$ (since $\rho_1 = \lambda_1/\mu < 1$), confirming that $e^{\lambda_1/\gamma_1}(\lambda_1 - \mu)\mathcal{J}(1)/\gamma_1 + 1 > 0$. No elementary simplification of (218) is known; the expression is evaluated numerically. As $\gamma_1 \to 0^+$ the Model A value (143) is recovered.

Class-2 mean queue length by subtraction

$$\mathbb{E}[N_2] \;=\; \big(\mathbb{E}[N_1] + \mathbb{E}[N_2]\big) - \mathbb{E}[N_1] \;=\; \frac{\rho^2}{1-\rho} - \mathbb{E}[N_1]. \tag{219}$$

Since jockeying defections reduce the class-1 queue, $\mathbb{E}[N_1]$ is smaller than the Model A value $\rho\rho_1/(1-\rho_1)$, so $\mathbb{E}[N_2]$ correspondingly exceeds the Model A value.

Mean waiting times via Little's Law

For Class 1, the entry rate into Queue 1 is exactly λ_1: every Class-1 arrival joins Queue 1, whether it later jockeys to Queue 2 or enters service. **Little's Law** applied to the Queue-1 waiting line therefore gives the genuine mean time in Queue 1,

$$\mathbb{E}[W_1] \;=\; \frac{\mathbb{E}[N_1]}{\lambda_1}. \tag{220}$$

For Class 2, Queue 2 receives not only the Poisson arrivals at rate λ_2 but also the jockeying transfers from Queue 1, whose long-run rate is $\gamma_1\,\mathbb{E}[N_1]$. **Little's Law** applied to the Queue-2 buffer therefore reads $\mathbb{E}[N_2] = \big(\lambda_2 + \gamma_1\,\mathbb{E}[N_1]\big)\,\mathbb{E}[W_2]$, so the mean sojourn in Queue 2 is

$$\mathbb{E}[W_2] \;=\; \frac{\mathbb{E}[N_2]}{\lambda_2 + \gamma_1\,\mathbb{E}[N_1]}, \tag{221}$$

which reduces to $\mathbb{E}[N_2]/\lambda_2$ only in the no-jockeying limit $\gamma_1 \to 0^+$. As $\gamma_1 \to 0^+$ both (220) and (221) recover the Model A values (148)–(149).

9 Analysis of Model C_2 $\big(\theta_1 > 0,\, \gamma_1 = \gamma_2 = \theta_2 = 0\big)$

In this section, we will derive $P(x, y)$ for this model where only true abandonments coming from Class-1 take place, while no abandonments in Class-2 or jockeying of any sort are taken into account.

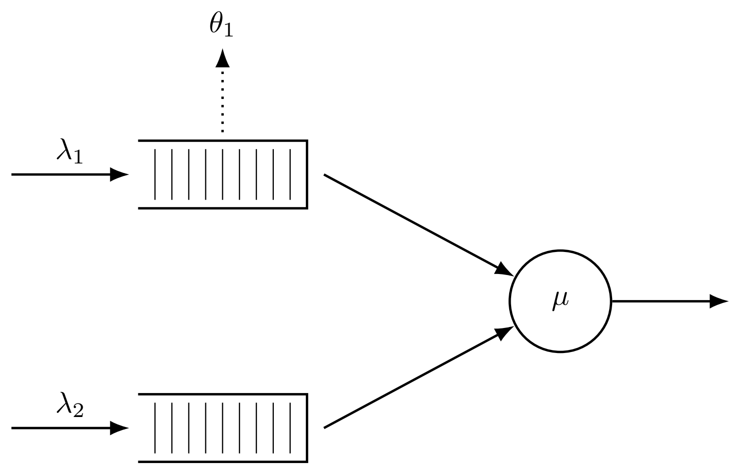

Figure 7: Queue layout for Model C_2. Class-1 customers abandon the system at rate θ_1 per waiting customer (dotted arrow, upward from Queue 1). Class-2 never abandons ($\theta_2 = 0$) and there is no jockeying.

The rationale behind using this combination of parameters is that, thanks to the fact that Class-2 still sees Poisson arrivals, the same sort of probabilistic arguments we used for Model A will hold. Moreover, we will see that the resulting equation when plugging this combination of parameters in our fundamental equation (Equation (226)) yields a *first-order linear ODE* in x and, $P_y(y)$ can be taken out of the integral, saving us from having to solve an equation of the Volterra family.

Theorem 4. *Consider a queueing system with* 2 *priority classes with Poisson arrival rates* λ_1 *and* λ_2*, where class-*1 *customers have non-preemptive priority over those of class-*2*, exponentially distributed service times of rate* μ *independently of the class, and per-customer abandonment of class-*1 *customers from the queue at rate* θ_1 *(with no class-*2 *abandonment and no jockeying). Under the stability condition*

$$\rho_2 \cdot {}_1F_1\Big(1;\, \tfrac{\mu}{\theta_1} + 1;\, \tfrac{\lambda_1}{\theta_1}\Big) \;<\; 1 \quad\iff\quad \lambda_2\,\mathbb{E}[B_C] \;<\; 1, \tag{222}$$

the probability generating function $P(x,y)$ of such a system is given by

$$P(x,y) \;=\; \frac{e^{\lambda_1 x/\theta_1}\,\mu\,\pi(0,0)\,(1-y)}{\theta_1\,x^{\mu/\theta_1}(1-x)^{\lambda_2(1-y)/\theta_1}\big(\widetilde{B}_C(\lambda_2(1-y))-y\big)} \times\Big[\widetilde{B}_C(\lambda_2(1-y))\,\mathcal{I}_1(x,y)-\big(1-\widetilde{B}_C(\lambda_2(1-y))\big)\,\mathcal{I}_2(x,y)\Big], \tag{223}$$

where

$$\mathcal{I}_1(x,y) \;=\; \int_0^x e^{-\lambda_1 t/\theta_1}\,t^{\,\mu/\theta_1-1}(1-t)^{\,\lambda_2(1-y)/\theta_1}\,dt,$$
$$\mathcal{I}_2(x,y) \;=\; \int_0^x e^{-\lambda_1 t/\theta_1}\,t^{\,\mu/\theta_1}(1-t)^{\,\lambda_2(1-y)/\theta_1-1}\,dt,$$

$\widetilde{B}_C(s)$ is the LST of the busy period of the $M/M/1+M$ subsystem of class-1 given by Equation (6), and $\pi(0,0)$ is given by Corollary 5 below.

Proof. The proof is structured in three stages.

Stage 1: Reduction to a first-order ODE in x and the integrating factor. Substituting $\theta_1>0$, $\gamma_1=\gamma_2=\theta_2=0$ into the general fundamental equation eliminates both convection terms in y, leaving a first-order linear ODE in x for each fixed $y\in[0,1]$ (Equation (226)). Partial-fraction decomposition of the ODE coefficient gives the integrating factor

$$\mu_I(x;y) \;=\; e^{-\lambda_1 x/\theta_1}\,x^{\alpha}\,(1-x)^{\beta(y)}, \qquad \alpha=\tfrac{\mu}{\theta_1}, \quad \beta(y)=\tfrac{\lambda_2(1-y)}{\theta_1},$$

which vanishes at $x=0$ and $x=1$ because $\alpha,\beta(y)>0$.

Stage 2: Determination of $P_y(y)$. Two independent routes are given in §§9.2–9.3:

- *Probabilistic (Pollaczek–Khinchine).* Because $\theta_2=\gamma_i=0$, the class-2 arrival stream remains Poisson; PASTA applies. Class-2 customers see an $M/G/1$ queue with effective service time B_C (the busy period of the class-1 $M/M/1{+}M$ subsystem), and the P-K formula yields $P_y(y)$ directly (Equation (232)).
- *Analytical (boundedness at $x=1$).* Since $\mu_I(1;y)=0$, the only way for $P(x,y)$ to remain bounded as $x\to1$ is $\int_0^1 q(t;y)\mu_I(t;y)\,dt=0$ (Equation (242)). Evaluating this integral via the Kummer–Euler identity (248) recovers the same $P_y(y)$.

Stage 3: Recovery of $P(x,y)$. With $P_y(y)$ known, the ODE is integrated against the integrating factor (§9.4), yielding the closed form (255) stated in the theorem. □

Corollary 5. *The limiting probability $\pi(0,0)$ of the joint state $(N_1,N_2)=(0,0)$ with the server busy, and the empty-system probability π_0, are given by*

$$\pi(0,0) \;=\; \frac{(\lambda_1+\lambda_2)\big(1-\lambda_2\,\mathbb{E}[B_C]\big)}{\mu\,\big(1+\lambda_1\,\mathbb{E}[B_C]\big)} \;=\; (\rho_1+\rho_2)\,\frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]}, \tag{224}$$

$$\pi_0 \;=\; \frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]}, \tag{225}$$

where $\mathbb{E}[B_C]$ is the mean busy period of the $M/M/1+M$ subsystem of class-1, given by Equation (7). In the limit $\theta_1\to0^+$, $\mathbb{E}[B_C]\to(\mu-\lambda_1)^{-1}$ and the expressions above recover the classical Model A results $\pi(0,0)=\rho(1-\rho)$ and $\pi_0=1-\rho$.

9.1 Setup: reduction to a first-order linear ODE in x

Plugging $\theta_1>0$, $\gamma_1=\gamma_2=\theta_2=0$ into the fundamental equation (41) (both partial-derivative terms collapse onto the θ_1 contribution) leaves

$$\big[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\big]\,P(x,y)+\theta_1\,xy(x-1)\,\frac{\partial}{\partial x}P(x,y) \;=\; \mu(x-y)\,P_y(y)-\mu x(1-y)\,\pi(0,0), \tag{226}$$

where the boundary function $P_y(y) = P(0,y) = \sum_{n_2=0}^{\infty} \pi(0,n_2)\, y^{n_2}$ is, as in Model A, the partial PGF restricted to states where the server is busy and the priority queue is empty ($N_1 = 0$).

For any fixed y, Equation (226) is a *first-order linear ODE in x* whose unknowns are the function $P(x,y)$ on $[0,1]^2$ and the boundary trace $P_y(y) = P(0,y)$. The strategy in what follows is in two stages:

1. Identify $P_y(y)$ — first by a probabilistic argument exploiting the $M/G/1$ structure seen by Class-2 (§9.2), then re-derived analytically from boundedness of $P(x,y)$ at $x = 1$ (§9.3).

2. Recover $P(x,y)$ by solving the ODE with $P_y(y)$ now known (§9.4).

We close the setup with the conditional-expectation decomposition that will drive the probabilistic derivation:

$$P_y(y) \;=\; \mathbb{P}(\text{busy},\, N_1 = 0) \cdot \mathbb{E}\big[y^{N_2} \,\big|\, \text{busy},\, N_1 = 0\big]\,. \tag{227}$$

9.2 Probabilistic derivation of $P_y(y)$

The idea is, again, to adopt the perspective of a Class-2 customer waiting in line while the server is busy. Unlike in the case of jockeying, the customers from class-1 truly abandon the system, leaving the Poisson arrival process of Class-2 undisturbed. Therefore, customers in the non-priority class are driven by an $M/G/1$ process, with Poisson arrivals at rate λ_2 and general effective service times B_C equivalent to the busy period of Class-1's $M/M/1+M$ queue with arrival rate λ_1, service rate μ and per-customer abandonment rate θ_1.

The LST $\widetilde{B}_C(s)$ and mean $\mathbb{E}[B_C]$ of this busy period are derived via the first-step analysis in §3.2; see Equations (6)–(7).

The stability condition required for the Class-2 subsystem is that of the $M/G/1$ defined above, i.e. $\lambda_2\, \mathbb{E}[B_C] < 1$:

$$\rho_2 \cdot {}_1F_1\!\left(1;\, \frac{\mu}{\theta_1} + 1;\, \frac{\lambda_1}{\theta_1}\right) \;<\; 1. \tag{228}$$

Compared to Model A, this stability condition is strictly weaker than the classical $\rho_1 + \rho_2 < 1$, which aligns with the fact that abandonments lower the load of the system: for all $\theta_1 > 0$, the series defining ${}_1F_1(1;\, \mu/\theta_1 + 1;\, \lambda_1/\theta_1)$ is termwise dominated by the geometric series for $1/(1-\rho_1)$, so ${}_1F_1(1;\, \mu/\theta_1 + 1;\, \lambda_1/\theta_1) < 1/(1-\rho_1)$. Abandonments in the priority queue therefore stabilise the non-priority queue.

With the LST and expectation of the effective service time seen by Class-2 customers, and thanks to the constant arrival rate λ_2 from the Poisson arrival process, we apply the Pollaczek–Khinchine formula:

$$L_2(y) \;\equiv\; \mathbb{E}\big[y^{N_2} \,\big|\, \text{busy},\, N_1 = 0\big] \;=\; \frac{(1-\lambda_2\mathbb{E}[B_C])\,(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y)) - y}. \tag{229}$$

Substituting (229) into (227) yields

$$P_y(y) \;=\; \mathbb{P}(\text{busy},\, N_1 = 0)\left[\frac{(1-\lambda_2\mathbb{E}[B_C])\,(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y)) - y}\right]. \tag{230}$$

By the definition of P_y, $P_y(0) = \pi(0,0)$. Evaluating (230) at $y = 0$,

$$\begin{aligned} P_y(0) \;=\; \pi(0,0) \;&=\; \mathbb{P}(\text{busy},\, N_1 = 0)\left[\frac{(1-\lambda_2\mathbb{E}[B_C])\cdot 1 \cdot \widetilde{B}_C(\lambda_2)}{\widetilde{B}_C(\lambda_2)}\right] \\ &=\; \mathbb{P}(\text{busy},\, N_1 = 0)\,(1-\lambda_2\mathbb{E}[B_C]), \end{aligned} \tag{231}$$

so that $\mathbb{P}(\text{busy},\, N_1 = 0) = \pi(0,0)/(1-\lambda_2\mathbb{E}[B_C])$. Substituting this back into (230), the $(1-\lambda_2\mathbb{E}[B_C])$ factor cancels and we are left with

$$P_y(y) \;=\; \pi(0,0)\,\frac{(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y)) - y}. \tag{232}$$

This is structurally the same expression as in the probabilistic argument for Model A (Equation (114)), with the only difference being the effective service time: B_C is now the busy period of an $M/M/1+M$ subsystem with abandonment rate θ_1, rather than the busy period of a pure $M/M/1$.

9.3 Analytical re-derivation of $P_y(y)$ via boundedness at $x = 1$

For an analytic counterpart of the previous derivation, we put (226) in standard form:

$$\frac{\partial P}{\partial x} + \underbrace{\frac{(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy}{\theta_1\, x(x-1)}}_{p(x;y)} P = \underbrace{\frac{\mu(x-y)\, P_y(y) - \mu x(1-y)\,\pi(0,0)}{\theta_1\, xy(x-1)}}_{q(x;y)}. \tag{233}$$

The numerator of $p(x;y)$ is the same polynomial we have seen in every model so far. With $x^*(y)$ and $x^+(y)$ as its roots (Equation (112)), we can write

$$(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy = -\lambda_1\big(x - x^*(y)\big)\big(x - x^+(y)\big), \tag{234}$$

with $x^*(y)\, x^+(y) = \mu/\lambda_1$. Partial fraction decomposition gives

$$\frac{-\lambda_1\big(x - x^*(y)\big)\big(x - x^+(y)\big)}{x(x-1)} = A + \frac{B}{x} + \frac{C}{x-1}, \tag{235}$$

where $A = -\lambda_1$ is read off from the ratio of leading coefficients of the quadratic terms in numerator and denominator, $B = \mu$ from setting $x = 0$ and using $\lambda_1\, x^* x^+ = \mu$, and $C = \lambda_2(1-y)$ from setting $x = 1$ and using $(1-x^*)(1-x^+) = -\lambda_2(1-y)/\lambda_1$. Thus

$$p(x;y) = \frac{1}{\theta_1}\left[-\lambda_1 + \frac{\mu}{x} + \frac{\lambda_2(1-y)}{x-1}\right]. \tag{236}$$

Integrating term by term,

$$\int p(x;y)\, dx = \frac{1}{\theta_1}\left[-\lambda_1 x + \mu \ln x + \lambda_2(1-y)\ln(1-x)\right], \tag{237}$$

which yields the integrating factor

$$\mu_I(x;y) = e^{-\lambda_1 x/\theta_1} \cdot x^{\alpha} \cdot (1-x)^{\beta(y)}, \tag{238}$$

where, in parallel with Model B_2, we set

$$\alpha = \frac{\mu}{\theta_1}, \qquad \beta(y) = \frac{\lambda_2(1-y)}{\theta_1}. \tag{239}$$

Under our stability assumptions both exponents are positive ($\alpha > 0$ always; $\beta(y) \geq 0$ for $y \in [0,1]$), so μ_I vanishes at both $x = 0$ and $x = 1$. From $\frac{d}{dx}[P \cdot \mu_I] = q \cdot \mu_I$, integration over $(0, x]$ reads

$$P(x,y)\,\mu_I(x;y) - P(0,y)\,\mu_I(0;y) = \int_0^x q(t;y)\,\mu_I(t;y)\, dt, \tag{240}$$

and the boundary term vanishes, yielding

$$P(x,y) = \frac{1}{\mu_I(x;y)} \int_0^x q(t;y)\,\mu_I(t;y)\, dt. \tag{241}$$

Now, in the same fashion as the root-cancellation argument used for Model A, we exploit boundedness of PGFs to determine $P_y(y)$ by observing the behaviour of $P(x,y)$ as $x \to 1$. Since $\mu_I(x;y) \to 0$, the only way to keep the right-hand side of (241) bounded is

$$\int_0^1 q(t;y)\,\mu_I(t;y)\, dt = 0. \tag{242}$$

Substituting the explicit forms of q and μ_I into (242) and using $(1-t)^b/(t-1) = -(1-t)^{b-1}$ to simplify,

$$\int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\alpha-1}(1-t)^{\beta(y)-1}\big[(t-y)\, P_y(y) - t(1-y)\,\pi(0,0)\big]\, dt = 0, \tag{243}$$

with α and $\beta(y)$ as in (239). Introducing the auxiliary integrals

$$\mathcal{J}_0(y) \;=\; \int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha-1}(1-t)^{\,\beta(y)-1}\, dt, \tag{244}$$

$$\mathcal{J}_1(y) \;=\; \int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha}(1-t)^{\,\beta(y)-1}\, dt, \tag{245}$$

and splitting the integrand, the boundedness condition (242) becomes

$$P_y(y)\big[\mathcal{J}_1(y) - y\,\mathcal{J}_0(y)\big] \;=\; (1-y)\,\pi(0,0)\,\mathcal{J}_1(y), \tag{246}$$

which gives

$$P_y(y) \;=\; \pi(0,0)\,(1-y)\,\frac{\mathcal{J}_1(y)}{\mathcal{J}_1(y) - y\,\mathcal{J}_0(y)}. \tag{247}$$

The continued-fraction representation (6) admits an equivalent integral form. Setting $a = \alpha = \mu/\theta_1$, $b = s/\theta_1$, $c = \lambda_1/\theta_1$, the Euler integral representation (11) identifies each of the integrals below as a Beta-function multiple of a ${}_1F_1$ value:

$$\widetilde{B}_C(s) \;=\; \frac{\displaystyle\int_0^1 t^{a}(1-t)^{b-1}e^{-ct}\,dt}{\displaystyle\int_0^1 t^{a-1}(1-t)^{b-1}e^{-ct}\,dt} \;=\; \frac{B(a+1,b)\,{}_1F_1(a+1;\,a+1+b;\,-c)}{B(a,b)\,{}_1F_1(a;\,a+b;\,-c)}. \tag{248}$$

That this ratio equals the continued fraction (6) follows from the three-term recurrence for ${}_1F_1$ in its first parameter [DLM], which is precisely the ratio recursion (**??**) derived in §3.2.

Setting $s = \lambda_2(1-y)$ — so that $b = \beta(y)$ matches the exponent in (239) — yields exactly the ratio of our $\mathcal{J}_i$:

$$\widetilde{B}_C\big(\lambda_2(1-y)\big) \;=\; \frac{\mathcal{J}_1(y)}{\mathcal{J}_0(y)}. \tag{249}$$

Dividing numerator and denominator of (247) by $\mathcal{J}_0(y)$ then recovers exactly Equation (232). The boundedness route, which uses no probabilistic structure, reaches the same Pollaczek–Khinchine form only through the Beta-integral identity (248), which is the analytic shadow of the $M/G/1$/busy-period decomposition used in §9.2.

9.4 Recovery of $P(x,y)$ from $P_y(y)$

With $P_y(y)$ now in hand, we return to the ODE (226) and substitute (232) into its right-hand side. Writing $\zeta(y) \equiv \widetilde{B}_C(\lambda_2(1-y))$ for brevity,

$$\begin{aligned}
\text{RHS} \;&=\; \mu(x-y)\,\pi(0,0)\,\frac{(1-y)\,\zeta(y)}{\zeta(y)-y} \;-\; \mu x(1-y)\,\pi(0,0)\\
&=\; \mu\,\pi(0,0)\,(1-y)\left[\frac{(x-y)\,\zeta(y)}{\zeta(y)-y} - x\right]\\
&=\; \mu\,\pi(0,0)\,(1-y)\,\frac{(x-y)\,\zeta(y) - x\big[\zeta(y)-y\big]}{\zeta(y)-y}\\
&=\; \mu\,\pi(0,0)\,y(1-y)\,\frac{x-\zeta(y)}{\zeta(y)-y},
\end{aligned}$$

which is structurally identical to the RHS found for Model A, though now with the more involved effective service time B_C. With $p(x;y)$ and $\mu_I(x;y)$ as in (236) and (238), the source $q(x;y)$ of the standard-form ODE now reads

$$q(x;y) \;=\; \frac{\mu\,\pi(0,0)\,(1-y)\,\big[x-\zeta(y)\big]}{\theta_1\,x(x-1)\,\big[\zeta(y)-y\big]}, \tag{250}$$

and the implicit solution (241) becomes

$$P(x,y) \;=\; \frac{1}{\mu_I(x;y)}\int_0^x q(t;y)\,\mu_I(t;y)\,dt. \tag{251}$$

To evaluate the integral, we use the partial-fraction identity

$$\frac{t-\zeta(y)}{t(t-1)} \;=\; \frac{\zeta(y)(t-1)+t\big(1-\zeta(y)\big)}{t(t-1)} \;=\; \frac{\zeta(y)}{t}+\frac{1-\zeta(y)}{t-1},$$

and define the auxiliary integrals

$$\mathcal{I}_1(x,y) \;=\; \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha-1}(1-t)^{\,\beta(y)}\,dt, \tag{252}$$

$$\mathcal{I}_2(x,y) \;=\; \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha}(1-t)^{\,\beta(y)-1}\,dt, \tag{253}$$

with α and $\beta(y)$ as in (239).

Note on the $\mathcal{J}$–$\mathcal{I}$ correspondence. The integrals $\mathcal{I}_1,\mathcal{I}_2$ are the partial (upper limit x) counterparts of the full integrals $\mathcal{J}_0,\mathcal{J}_1$ from (242). Evaluating at $x=1$ gives $\mathcal{I}_2(1,y)=\mathcal{J}_1(y)$ exactly. However, $\mathcal{I}_1(1,y)$ is *not* $\mathcal{J}_0(y)$: the former has the exponent $\beta(y)$ on $(1-t)$ while the latter has $\beta(y)-1$, which under the Euler representation (11) corresponds to incrementing the second Kummer parameter by one. Both families are evaluated by the same representation; the exponent shift is harmless because $\mathcal{I}_1$ and $\mathcal{I}_2$ always appear together in the combination $\zeta(y)\,\mathcal{I}_1-(1-\zeta(y))\,\mathcal{I}_2$, not individually.

Since $\pi(0,0)$ and $\zeta(y)$ come out of the integral,

$$\int_0^x q(t;y)\,\mu_I(t;y)\,dt \;=\; \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y)-y\big)}\Big[\zeta(y)\,\mathcal{I}_1(x,y) \;-\; \big(1-\zeta(y)\big)\,\mathcal{I}_2(x,y)\Big]. \tag{254}$$

Dividing by $\mu_I(x;y)$ and restoring $\zeta(y)=\widetilde{B}_C(\lambda_2(1-y))$, we arrive at the closed form

$$\begin{aligned} P(x,y) \;&=\; \frac{e^{\lambda_1 x/\theta_1}\,\mu\,\pi(0,0)\,(1-y)}{\theta_1\,x^{\alpha}(1-x)^{\beta(y)}\big(\widetilde{B}_C(\lambda_2(1-y))-y\big)} \\ &\quad\times\Big[\widetilde{B}_C(\lambda_2(1-y))\,\mathcal{I}_1(x,y)-\big(1-\widetilde{B}_C(\lambda_2(1-y))\big)\,\mathcal{I}_2(x,y)\Big], \end{aligned} \tag{255}$$

which is the statement of the theorem. This closed form is the product of a probabilistic derivation of $P_y(y)$ via Pollaczek–Khinchine, combined with an analytic integration of the ODE in x; the same result is reached from the purely analytic route of §9.3.

Proof of Corollary 5. The boundary balance at the empty state $(N_1,N_2)=(0,0)$ with server idle, equating the rate-out (arrivals at total rate $\lambda_1+\lambda_2$) and the rate-in (service completion from $(0,0)$ with the server busy at rate μ), reads

$$(\lambda_1+\lambda_2)\,\pi_0 \;=\; \mu\,\pi(0,0), \tag{256}$$

so $\pi(0,0)=(\rho_1+\rho_2)\,\pi_0$.

To fix π_0, we exploit the $M/G/1$ structure used in §9.2. Class-2 customers form an $M/G/1$ with arrival rate λ_2 and service time B_C, whose utilisation is $\lambda_2\,\mathbb{E}[B_C]$. The event "$M/G/1$ idle" coincides with the absence of any Class-2 customer from the system, so

$$\mathbb{P}(\text{no Class-2 present}) \;=\; 1-\lambda_2\,\mathbb{E}[B_C]. \tag{257}$$

Conditional on no Class-2 being present, the priority subsystem evolves freely as the same $M/M/1+M$ queue used to define B_C, with arrival rate λ_1, service rate μ and abandonment rate θ_1. The stationary idle fraction of that subsystem is given by the standard renewal-theoretic ratio of mean idle period $(1/\lambda_1)$ to mean cycle $(1/\lambda_1+\mathbb{E}[B_C])$:

$$\mathbb{P}(\text{Class-1 absent} \mid \text{no Class-2 present}) \;=\; \frac{1}{1+\lambda_1\,\mathbb{E}[B_C]}. \tag{258}$$

The event $\{\text{server idle}, N_1=N_2=0\}$ is precisely the intersection of these two, so

$$\pi_0 \;=\; \big(1-\lambda_2\,\mathbb{E}[B_C]\big)\cdot\frac{1}{1+\lambda_1\,\mathbb{E}[B_C]} \;=\; \frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]}, \tag{259}$$

which is (225). Substituting into (256) yields (224).

For the Model A limit, $\theta_1\to 0^+$ gives $\mathbb{E}[B_C]\to(\mu-\lambda_1)^{-1}$, hence $\lambda_1\,\mathbb{E}[B_C]\to\rho_1/(1-\rho_1)$ and $1+\lambda_1\,\mathbb{E}[B_C]\to 1/(1-\rho_1)$; substituting into (225) and (224) recovers $\pi_0=1-\rho$ and $\pi(0,0)=\rho(1-\rho)$. $\square$

9.5 Mean queue lengths and mean waiting times

Why the diagonal no longer simplifies

Setting $x = y = z$ in the fundamental equation (226), the abandonment convection $\theta_1 xy(x-1)\partial_x P$ does *not* vanish (since $(x-1)|_{x=z} \neq 0$ for $z \in (0,1)$). The diagonal equation reads:

$$\begin{aligned}&\big[(\lambda_1{+}\lambda_2{+}\mu)z^2 - \mu z - \lambda z^3\big]P(z,z) + \theta_1 z^2(z-1)\,\frac{\partial P}{\partial x}(z,z)\\ &\quad=\ -\mu z(1-z)\,\pi(0,0),\end{aligned} \tag{260}$$

and the unknown $\partial_x P(z,z)$ prevents a closed form for $P(z,z)$ and hence for the total $\mathbb{E}[N_1] + \mathbb{E}[N_2]$. Abandonment removes customers and reduces both means relative to Model A.

Class-2 mean queue length via $P(1,y)$

Setting $x = 1$ in the fundamental equation (226), the abandonment term $\theta_1 xy(x-1)\partial_x P$ vanishes identically, and the coefficient of $P(1,y)$ simplifies to $\lambda_2 y(1-y)$:

$$\lambda_2 y(1-y)\,P(1,y)\ =\ \mu(1-y)\big[P_y(y) - \pi(0,0)\big]. \tag{261}$$

Cancelling $(1-y)$ for $y \neq 1$ and substituting the known $P_y(y)$ (232):

$$P(1,y)\ =\ \frac{\mu\big[P_y(y) - \pi(0,0)\big]}{\lambda_2 y}\ =\ \frac{\mu\,\pi(0,0)}{\lambda_2}\cdot\frac{1-\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y)) - y}. \tag{262}$$

This is the marginal PGF of N_2. To differentiate at $y = 1$ (where the expression is 0/0), set $\varepsilon = 1 - y \to 0$ and use the LST expansion $\widetilde{B}_C(\lambda_2\varepsilon) = 1 - \lambda_2\mathbb{E}[B_C]\,\varepsilon + \frac{1}{2}\lambda_2^2\mathbb{E}[B_C^2]\,\varepsilon^2 + O(\varepsilon^3)$:

$$\begin{aligned}1 - \widetilde{B}_C(\lambda_2\varepsilon)\ &=\ \lambda_2\mathbb{E}[B_C]\,\varepsilon - \tfrac{1}{2}\lambda_2^2\mathbb{E}[B_C^2]\,\varepsilon^2 + O(\varepsilon^3),\\ \widetilde{B}_C(\lambda_2\varepsilon) - y\ &=\ (1 - \lambda_2\mathbb{E}[B_C])\,\varepsilon + \tfrac{1}{2}\lambda_2^2\mathbb{E}[B_C^2]\,\varepsilon^2 + O(\varepsilon^3),\end{aligned} \tag{263}$$

so that (writing $a = \lambda_2\mathbb{E}[B_C]$ and $c = \frac{1}{2}\lambda_2^2\mathbb{E}[B_C^2]$):

$$P(1,1-\varepsilon)\ =\ \frac{\mu\,\pi(0,0)}{\lambda_2}\cdot\frac{a\varepsilon - c\varepsilon^2 + O(\varepsilon^3)}{(1-a)\varepsilon + c\varepsilon^2 + O(\varepsilon^3)}\ =\ \frac{\mu\,\pi(0,0)}{\lambda_2}\cdot\frac{a - c\varepsilon}{(1-a) + c\varepsilon} + O(\varepsilon^2). \tag{264}$$

Differentiating $f(\varepsilon) = (a - c\varepsilon)/((1-a) + c\varepsilon)$ by the quotient rule:

$$f'(\varepsilon)\ =\ \frac{-c\big((1-a) + c\varepsilon\big) - (a - c\varepsilon)\cdot c}{\big((1-a) + c\varepsilon\big)^2}\ =\ \frac{-c}{(1 - a + c\varepsilon)^2}, \tag{265}$$

and since $d/dy\,P(1,y)\big|_{y=1} = -d/d\varepsilon\,P(1,1-\varepsilon)\big|_{\varepsilon=0}$:

$$\mathbb{E}[N_2]\ =\ \frac{d}{dy}P(1,y)\bigg|_{y=1}\ =\ \frac{\mu\,\pi(0,0)}{\lambda_2}\cdot\frac{c}{(1-a)^2}\ =\ \frac{\mu\,\pi(0,0)\,\lambda_2\,\mathbb{E}[B_C^2]}{2\,(1 - \lambda_2\mathbb{E}[B_C])^2}. \tag{266}$$

As $\theta_1 \to 0^+$, $\mathbb{E}[B_C] \to (\mu - \lambda_1)^{-1}$ and $\mathbb{E}[B_C^2] \to 2\mu/(\mu - \lambda_1)^3$; substituting recovers the Model A value $\mathbb{E}[N_2] = \rho\rho_2/((1-\rho_1)(1-\rho))$.

Class-1 mean queue length

We extract $\mathbb{E}[N_1] = \frac{\partial P}{\partial x}\big|_{(1,1)}$ by evaluating the theorem formula at $y = 1$ (limit) and then differentiating in x.

Step 1: marginal PGF $P(x,1)$. As $y \to 1$, the exponent $\beta(y) = \lambda_2(1-y)/\theta_1 \to 0$ so $(1-x)^{\beta(y)} \to 1$ for fixed $x \in (0,1)$. Using the LST expansion $(1-y)/(\widetilde{B}_C(\lambda_2(1-y)) - y) \to 1/(1 - \lambda_2\mathbb{E}[B_C])$ as $y \to 1$, and $\widetilde{B}_C(0) = 1$, the theorem formula gives:

$$P(x,1)\ =\ \frac{\mu\,\pi(0,0)}{\theta_1\,(1 - \lambda_2\mathbb{E}[B_C])}\cdot e^{\lambda_1 x/\theta_1}\,x^{-\mu/\theta_1}\int_0^x e^{-\lambda_1 t/\theta_1}\,t^{\mu/\theta_1 - 1}\,dt. \tag{267}$$

Step 2: differentiate $P(x,1)$ at $x=1$. Write $P(x,1) = C \cdot f(x) \cdot \mathcal{K}(x)$ with $C = \mu\pi(0,0)/(\theta_1(1-\lambda_2\mathbb{E}[B_C]))$, $f(x) = e^{\lambda_1 x/\theta_1}\, x^{-\mu/\theta_1}$, and $\mathcal{K}(x) = \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\mu/\theta_1 - 1}\, dt$. By the product rule and Leibniz's rule ($d\mathcal{K}/dx = e^{-\lambda_1 x/\theta_1}\, x^{\mu/\theta_1-1}$):

$$\frac{d}{dx}P(x,1) \;=\; C\Big[f'(x)\,\mathcal{K}(x) + f(x)\cdot e^{-\lambda_1 x/\theta_1}\, x^{\mu/\theta_1-1}\Big]. \tag{268}$$

The logarithmic derivative $f'(x)/f(x) = \lambda_1/\theta_1 - (\mu/\theta_1)/x = (\lambda_1 x - \mu)/(\theta_1 x)$, so:

$$\frac{d}{dx}P(x,1) \;=\; C\left[e^{\lambda_1 x/\theta_1}\, x^{-\mu/\theta_1} \cdot \frac{\lambda_1 x - \mu}{\theta_1 x} \cdot \mathcal{K}(x) \;+\; x^{-1}\right]. \tag{269}$$

Step 3: evaluate at $x=1$.

$$\begin{aligned} \mathbb{E}[N_1] \;&=\; \frac{d}{dx}P(x,1)\bigg|_{x=1} \\ &=\; \frac{\mu\,\pi(0,0)}{\theta_1\,(1-\lambda_2\mathbb{E}[B_C])}\left[e^{\lambda_1/\theta_1} \cdot \frac{\lambda_1 - \mu}{\theta_1} \cdot \mathcal{K}(1) \;+\; 1\right], \end{aligned} \tag{270}$$

where $\mathcal{K}(1) = \int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\mu/\theta_1-1}\, dt = (\theta_1/\lambda_1)^{\mu/\theta_1}\, \gamma(\mu/\theta_1,\, \lambda_1/\theta_1)$, with $\gamma(\cdot,\cdot)$ the lower incomplete gamma function. The factor $(\lambda_1 - \mu) < 0$ (since $\rho_1 < 1$), so the bracket is positive. No elementary simplification is known; $\mathbb{E}[N_1]$ is evaluated numerically. As $\theta_1 \to 0^+$, $\mathbb{E}[N_1]$ recovers the Model A value (143).

Little's Law with abandonment

The universal relation $\mathbb{E}[N_i] = \lambda_i\, \mathbb{E}[W_i]$ holds for any stable system. Here λ_i is the *arrival* rate of all class-i customers, including those that will eventually abandon, so $\mathbb{E}[W_i]$ is the mean time in queue averaged over both served and abandoned customers:

$$\mathbb{E}[W_1] \;=\; \frac{\mathbb{E}[N_1]}{\lambda_1}, \qquad \mathbb{E}[W_2] \;=\; \frac{\mathbb{E}[N_2]}{\lambda_2}. \tag{271}$$

Using the throughput $\mu(1-\pi_0)$ instead would give the conditional waiting time of customers who complete service only — a different object. As $\theta_1 \to 0^+$ both means recover the Model A values (143)–(149).

10 Comparison of Models A, B_2, and C_2

The three solved models represent the three tractable specializations of Model X: a pure baseline (A), a jockeying extension (B_2), and an abandonment extension (C_2). Each is obtained from Model X by zeroing all but one of the extra mechanisms, and each produces a closed-form or integral-form expression for $P(x,y)$. Table 3 collects the key structural facts; the remainder of this section traces the consequences.

Table 3: Structural comparison of Models A, B_2, and C_2.

	Model A	**Model** B_2	**Model** C_2
Extra mechanism	none	jockeying $1 \to 2$ ($\gamma_1 > 0$)	class-1 abandonment ($\theta_1 > 0$)
Stability condition	$\rho < 1$	$\rho < 1$	$\rho_2 \cdot {}_1F_1 < 1$
Fundamental equation	algebraic PGF equation	1st-order PDE in x, y	1st-order ODE in x
Pinning of $P_y(y)$	zero of kernel: $x = x^*(y)$	analyticity at $x = y$	boundedness at $x = 1$
Probabilistic route	yes (P-K with B)	no (jockeying breaks Poisson)	yes (P-K with B_C)
$P_y(y)$	rational in $x^*(y)$	Kummer ratio (209)	P-K / Kummer (232)
$P(x, y)$	closed form (105)	integral form (186)	integral form (255)
$\pi(0, 0)$	$\rho(1-\rho)$	$\rho(1-\rho)$	$(\rho_1+\rho_2)\pi_0$ via $\mathbb{E}[B_C]$
π_0	$1-\rho$	$1-\rho$	$> 1-\rho$ (abandonment increases idle time)
$P(z, z)$	$\rho(1-\rho)/(1-\rho z)$	$\rho(1-\rho)/(1-\rho z)$	no closed form
$\mathbb{E}[N_1] + \mathbb{E}[N_2]$	$\rho^2/(1-\rho)$	$\rho^2/(1-\rho)$	$< \rho^2/(1-\rho)$
$\mathbb{E}[N_1]$ closed form	$\rho\rho_1/(1-\rho_1)$	no	no

Model A: the rational baseline

Model A imposes no extra departure mechanism. Class-1 operates as an independent $M/M/1(\lambda_1, \mu)$ queue; class-2 sees an $M/G/1$ with service time equal to a class-1 busy period (by PASTA and the priority decomposition of §**??**). The Pollaczek–Khinchine formula (8) delivers $P_y(y)$ as a rational function of $x^*(y)$, and $P(x, y)$ is a ratio of polynomials in x, y, and $x^*(y)$. All performance measures — π_0, $\pi(0, 0)$, $\mathbb{E}[N_i]$, $\mathbb{E}[W_i]$ — are in closed form. Model A also furnishes the structural identities that persist across the hierarchy: $P(z, z) = \rho(1-\rho)/(1-\rho z)$ (total queue is $M/M/1$), $\pi_0 = 1-\rho$, and $\pi(0, 0) = \rho(1-\rho)$.

Effect of jockeying: Model A → Model B_2

Adding one-way jockeying ($\gamma_1 > 0$, $\gamma_2 = 0$) couples the two queues: a class-1 customer waiting in Queue 1 may defect to Queue 2 at rate γ_1. The consequences are asymmetric.

What is preserved. Jockeying transfers customers but does not remove them, so the total-customer process remains a standard $M/M/1(\lambda, \mu)$ birth–death chain. Consequently $P(z, z)$, π_0, $\pi(0, 0)$, and $\mathbb{E}[N]$ are identical to Model A for all $\gamma_1 > 0$.

What is destroyed. Jockeying breaks the per-class Poisson/renewal structure: a class-2 customer can no longer identify its effective service time with a class-1 busy period, because class-1 customers now arrive at Queue 2 from inside the system as well as from outside. The P-K route is therefore unavailable.

What is new. The fundamental PGF equation gains the convection term $\gamma_1 xy(x-y)\partial_x P$, which reduces it to a first-order ODE in x for each fixed y (since $\gamma_2 = 0$ kills the ∂_y term). The solution closes via an integrating factor with a branch singularity at the *interior* point $x = y$; the branch coefficient must vanish for P to be holomorphic, and this analyticity condition pins $P_y(y)$ as a Kummer ratio. The individual means $\mathbb{E}[N_1]$ and $\mathbb{E}[N_2]$ change with γ_1 (class-1 becomes shorter, class-2 longer), with no closed form for the split.

Effect of abandonment: Model $A \to$ Model C_2

Adding class-1 abandonment ($\theta_1 > 0$, $\theta_2 = 0$, $\gamma_i = 0$) allows each waiting class-1 customer to leave the system at rate θ_1.

What is preserved. Class-2 customers still see Poisson arrivals (abandonment only affects class-1; the class-2 arrival process is undisturbed). PASTA therefore still applies, and the $M/G/1$ subsystem for class-2 is intact — now with effective service time B_C (the busy period of the $M/M/1+M$ queue of §3.2) replacing B. The P-K formula (8) applies with B_C and delivers $P_y(y)$.

What is destroyed. Abandonment is a true departure: it reduces the total number of customers and breaks the conservation property. Consequently $P(z, z)$ no longer has the simple $M/M/1$ form (145), $\pi_0 > 1-\rho$, $\pi(0,0)$ requires $\mathbb{E}[B_C]$, and $\mathbb{E}[N] < \rho^2/(1-\rho)$. The stability condition is also generalized: $\rho < 1$ is no longer necessary; the relevant condition is $\lambda_2\,\mathbb{E}[B_C] < 1$, which is strictly weaker for all $\theta_1 > 0$.

What is new. The fundamental equation gains the convection term $\theta_1 xy(x-1)\partial_x P$, which already reduces it to a first-order ODE in x for each fixed y (with $\theta_2 = 0$ killing the ∂_y term). The integrating factor now has its singularity at the *boundary* $x = 1$ with a positive exponent, so μ_I vanishes there and mere boundedness of P forces $\int_0^1 q\,\mu_I\,dt = 0$. This determines $P_y(y)$ without any reference to holomorphy. Both $\mathbb{E}[N_1]$ and $\mathbb{E}[N_2]$ decrease relative to Model A as θ_1 increases: class-2 benefits indirectly because class-1 abandonments free the server sooner.

Structural comparison of Models B_2 and C_2

Both models reduce the fundamental equation to a first-order linear ODE in x and both close in confluent hypergeometric functions; yet the mechanism that pins $P_y(y)$ is structurally different. The contrast is as follows.

	Model C_2 ($\theta_1 > 0$)	**Model** B_2 ($\gamma_1 > 0$)
Convection coefficient	$\theta_1\, x(x-1)$	$\gamma_1\, x(x-y)$
Operative singular point	boundary $x = 1$	interior $x = y$
Exponent at singularity	$\beta(y) = \lambda_2(1-y)/\theta_1 > 0$ ($\mu_I \to 0$)	$\beta(y) = (1-y)(\lambda y - \mu)/(\gamma_1 y) < 0$ ($\mu_I \to \infty$)
Pinning principle	*boundedness*: $\int_0^1 q\,\mu_I\,dt = 0$	*analyticity*: branch coefficient $= 0$
$\pi(0,0)$	$M/G/1$+renewal (conservation broken)	$\rho(1-\rho)$ exactly (conservation holds)
Probabilistic backbone	class-2 is $M/G/1$; P-K with B_C	class-1 is $M/M/1$+M (reneging $= \gamma_1$); no P-K

In words: in Model C_2 the abandonment convection $\theta_1 x(x-1)$ places the singularity at the boundary $x = 1$ with a positive integrating-factor exponent, so μ_I vanishes there and mere boundedness forces the vanishing integral that determines $P_y(y)$. The Poisson character of the class-2 stream survives abandonment, so P-K delivers $P_y(y)$ probabilistically and an $M/G/1$ renewal argument delivers $\pi(0,0)$. In Model B_2, the jockeying convection $\gamma_1 x(x-y)$ places the singularity at the interior point $x = y$ with a negative exponent, so μ_I blows up, the homogeneous mode vanishes, and only the stronger requirement of holomorphy — the absence of the non-integer branch $(x-y)^{|\beta|}$ — pins $P_y(y)$. Jockeying destroys the per-class Poisson/renewal structure, so no P-K identity is available; what remains probabilistic is the total-customer $M/M/1$ behaviour and the $M/M/1$+M reading of the class-1 marginal, which is the origin of the ${}_1F_1$ factors.

Common Erlang-A structure

A common object appears in both B_2 and C_2:

$$ {}_1F_1\left(1;\, \frac{\mu}{r} + 1;\, \frac{\lambda_1}{r}\right), \tag{272} $$

with $r = \gamma_1$ (Model B_2) or $r = \theta_1$ (Model C_2). This is the Erlang-A utilisation factor: in C_2 it equals $\mu\,\mathbb{E}[B_C]$ (the mean class-1 busy period scaled by μ), while in B_2 it controls the Kummer ratio in $P_y(y)$ through the exponent $\alpha(y) = \mu/(\gamma_1 y)$. In both cases the function evaluates at a negative argument $(-\lambda_1/r)$ in C_2 and $(-\lambda_1 y/\gamma_1)$ in B_2, and the ratio ${}_1F_1(\alpha+1;\cdot;\cdot)/{}_1F_1(\alpha;\cdot;\cdot)$ acts as a boundary correction to the Model-A rational formula.

Model B_2 degenerates to Model A only in the singular limit $\gamma_1 \to 0^+$, in which $\alpha, \beta \to \infty$ and the Kummer ratio collapses to the rational expression $x^*(y)(1-y)/(x^*(y)-y)$ of Model A; this limit is not uniform, consistent with γ_1 multiplying the highest-order term of the ODE. Model C_2 degenerates to Model A uniformly as $\theta_1 \to 0^+$: $\mathbb{E}[B_C] \to (\mu - \lambda_1)^{-1}$, and all C_2 formulas recover their Model-A counterparts.

11 Results

11.1 Validation

Each theoretical result — two lemmas, three theorems, two corollaries, one approximation theorem, and two experiment-model theorems — is verified numerically against the exact stationary distribution obtained from a truncated CTMC solver ($N_{\max} = 40$). All models share base parameters $\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1.0$; the mechanism parameter (γ_1 or θ_1) is set to 0.5 where applicable.

Table 4 collects the verdict for every result. Figures 8–12 provide the visual evidence.

Table 4: Validation summary. All mathematical results are accepted; the PPGF approximation is conditionally accepted (accurate only for $\rho_1/\rho \geq 0.6$ and $\rho \leq 0.6$). Max errors reflect CTMC truncation, not formula error.

ID	**Result**	**Model(s)**	**Check**	**Max error**	**Verdict**
L1	Lemma 1	A, B, B_2, C_2	PGF eq. residual	$< 5 \times 10^{-9}$	Accept
L2	Lemma 2	A, B_2, C_2	PPGF dynamics residual	$< 2 \times 10^{-15}$	Accept
T-A	Theorem (Model A)	A	$P(x, y)$ on 5×5 grid	$< 4 \times 10^{-8}$	Accept
C-A	Corollary 2	A, B, B_2	π_0, $\pi(0, 0)$, 60 cfgs	$< 10^{-4}$	Accept
T-B2	Theorem (Model B_2)	B_2	$P(x, y)$ on 5×5 grid, $x < y$	$< 3 \times 10^{-7}$	Accept
T-C2	Theorem (Model C_2)	C_2	$P(x, y)$ on 5×5 grid	$< 5 \times 10^{-12}$	Accept
C-C2	Corollary 5	C_2	π_0, $\pi(0, 0)$, 60 cfgs	$< 10^{-4}$	Accept
T-S	Theorem 2	A	ε_∞ over (ρ_1, ρ_2)	0–100%	Conditional
E-B	Theorem 5 (i) — Model B_2^1	B_2^1	$P(x, y)$ on 5×5 grid	$< 3 \times 10^{-7}$	Accept
E-C	Theorem 5 (ii) — Model C_2^1	C_2^1	$P(x, y)$ on 5×5 grid	$< 3 \times 10^{-11}$	Accept

Structural lemmas

Figure 8 shows the maximum relative residual $|\text{LHS} - \text{RHS}|/|\text{RHS}|$ of each lemma's functional equation, evaluated on the exact CTMC distribution across all model variants. Both lemmas hold to better than 10^{-9}; Lemma 2 reaches floating-point precision ($\sim 10^{-15}$) because its recurrence involves only one discrete variable.

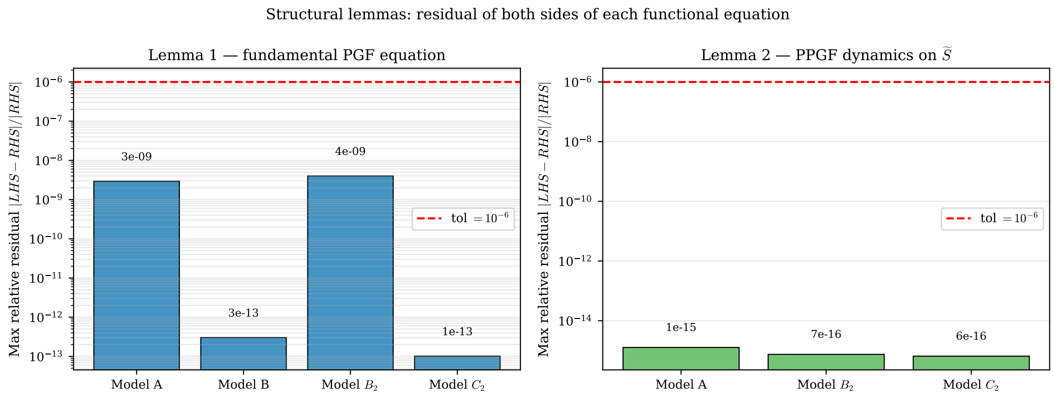

Figure 8: Lemma validation. *Left*: max relative residual of the fundamental PGF equation (Lemma 1) for each model variant on a 5×5 interior grid. *Right*: max relative residual of the PPGF dynamics recurrence (Lemma 2) over $n \in \{1, \ldots, 15\}$ and $y \in [0.05, 0.90]$. All values lie well below the 10^{-6} tolerance (red dashed line).

Closed-form and integral-form theorems

Figures 9 and 10 compare the three analytic formulas against the CTMC on a uniform grid, using identical parameters and the same error metric.

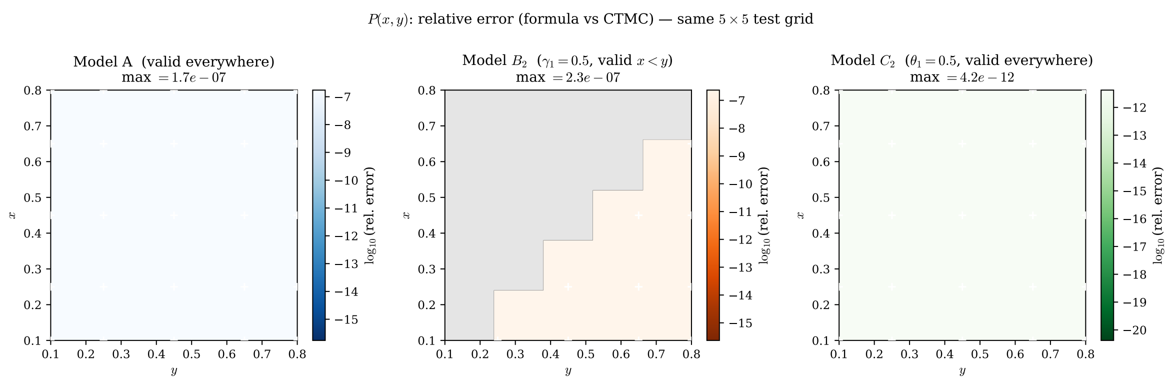

Figure 9: Relative error $|P_{\text{formula}}(x, y) - P_{\text{CTMC}}(x, y)|/|P_{\text{CTMC}}(x, y)|$ on the same 5×5 test grid for Models A, B_2, and C_2. Colour scales are anchored to the same range across all three panels. White crosses mark the 25 test points.

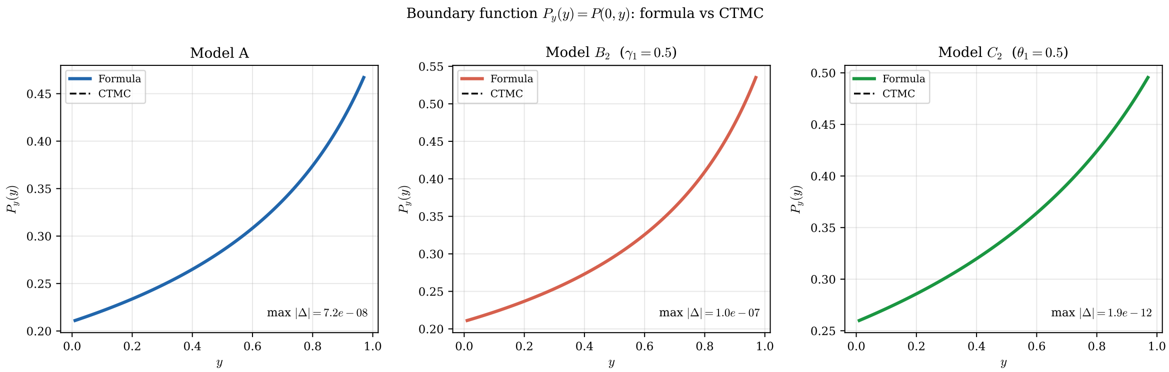

Figure 10: Boundary function $P_y(y) = P(0, y)$ for Models A, B_2, and C_2: analytic formula (solid) vs CTMC (dashed).

Corollaries

Figure 11 shows the scatter of formula vs CTMC values for π_0 and $\pi(0,0)$ across 60 random parameter configurations for each corollary. All points lie on the diagonal to within 10^{-4}.

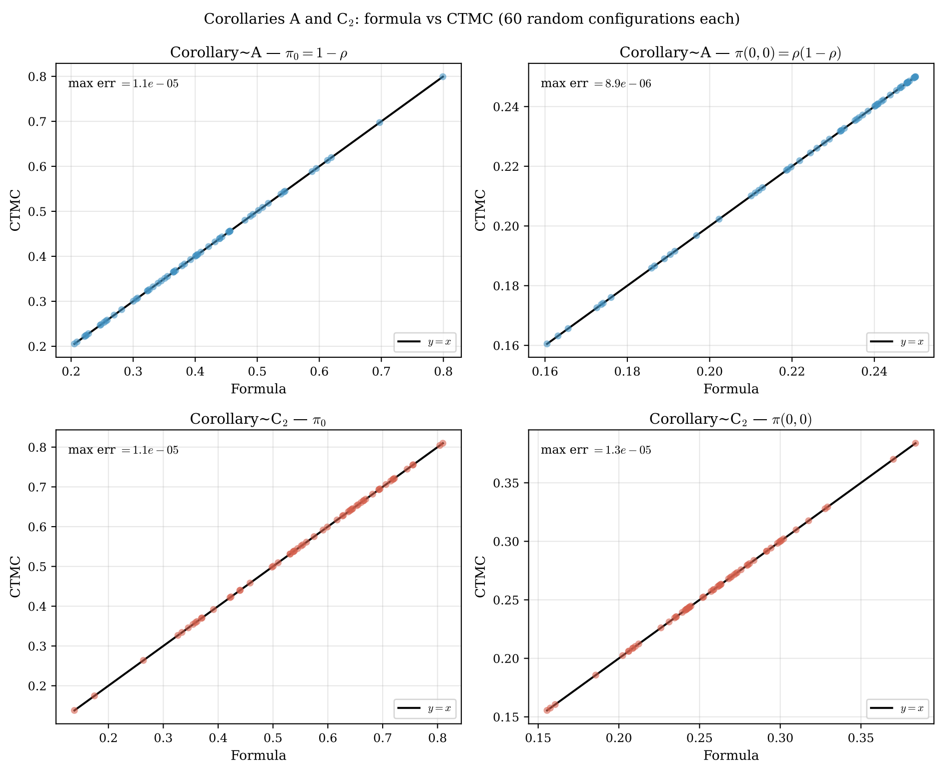

Figure 11: Corollaries A (blue) and C_2 (red): formula vs CTMC for π_0 (left column) and $\pi(0,0)$ (right column) over 60 random configurations each. Perfect agreement lies on the diagonal.

PPGF approximation (Theorem 2)

Figure 12 maps the maximum relative error $\varepsilon_\infty(\rho_1, \rho_2)$ over the parameter space.

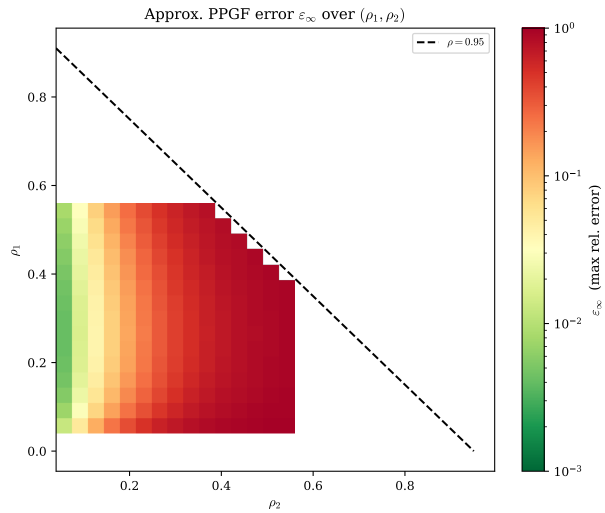

Figure 12: Max relative error ε_∞ of the approximate PPGF $\widetilde{P}_{\text{app}}(y, n) = (1-\rho)\rho[\widetilde{y}^*(y)]^n$ as a function of (ρ_1, ρ_2). Green: accurate ($\varepsilon_\infty < 3\%$); red: poor. The approximation is reliable only when the priority class carries most of the offered load ($\rho_1/\rho \gtrsim 0.6$) and the system is not heavily loaded ($\rho \lesssim 0.6$).

11.2 Mean queue lengths and mean waiting times

Figure 13 shows $\mathbb{E}[N_1]$, $\mathbb{E}[N_2]$, and the implied waiting times $\mathbb{E}[W_i] = \mathbb{E}[N_i]/\lambda_i$ as the mechanism parameter (γ_1 for Model B_2, θ_1 for Model C_2) varies from zero (Model A limit, dashed) to three.

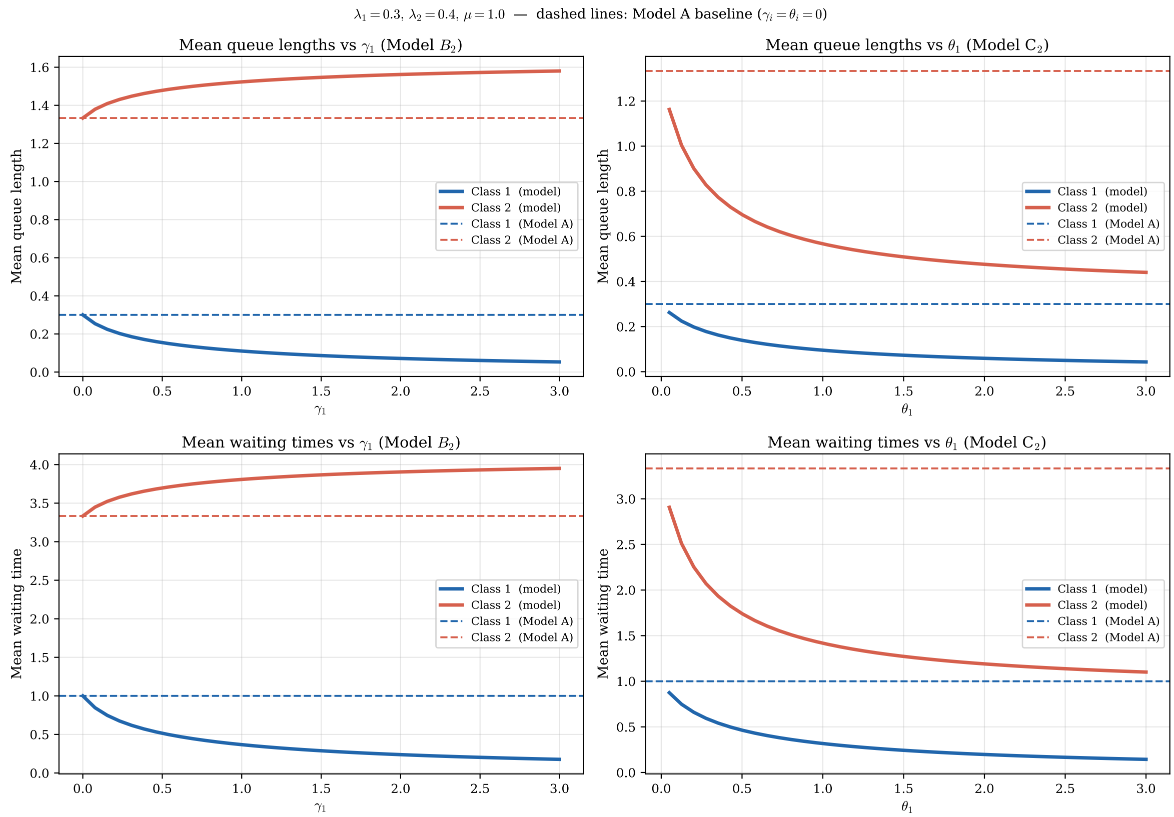

Figure 13: Mean queue lengths (top row) and mean waiting times (bottom row) for Models B_2 (left column) and C_2 (right column) as their respective mechanism parameters vary. Dashed lines indicate the Model A baseline.

For reference, Model A admits the closed-form expressions

$$\mathbb{E}[N_1] = \frac{\rho\rho_1}{1-\rho_1}, \qquad \mathbb{E}[N_2] = \frac{\rho\rho_2}{(1-\rho_1)(1-\rho)}, \tag{273}$$

with sum $\rho^2/(1-\rho)$ (the M/M/1 queue-length formula for the combined system). No analogous closed forms exist for the individual class means in Models B_2 and C_2.

11.3 Experiment models B_2^1 and C_2^1

The two experiment models possess algebraic fundamental equations. The lemma check therefore requires no differentiation of P: only the CTMC values of $P(x, y)$ and $P_y(y)$ are substituted into each side and the residual is measured.

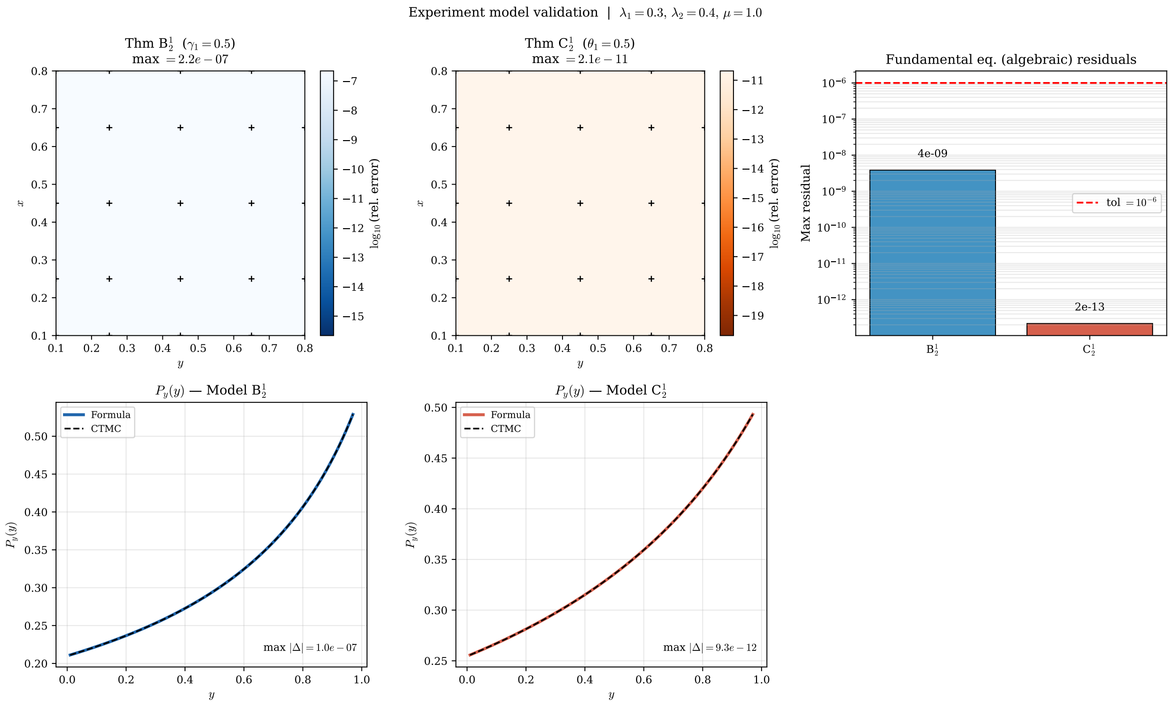

Figure 14: Validation of Models B_2^1 and C_2^1. *Top row, left and centre*: relative error $|P_{\text{formula}}(x, y) - P_{\text{CTMC}}(x, y)|/|P_{\text{CTMC}}(x, y)|$ on the 5×5 test grid. *Top row, right*: maximum relative residual of the algebraic fundamental equation for each model — both well below the 10^{-6} tolerance. *Bottom row*: boundary function $P_y(y) = P(0, y)$ comparing the rational closed-form formula against the CTMC.

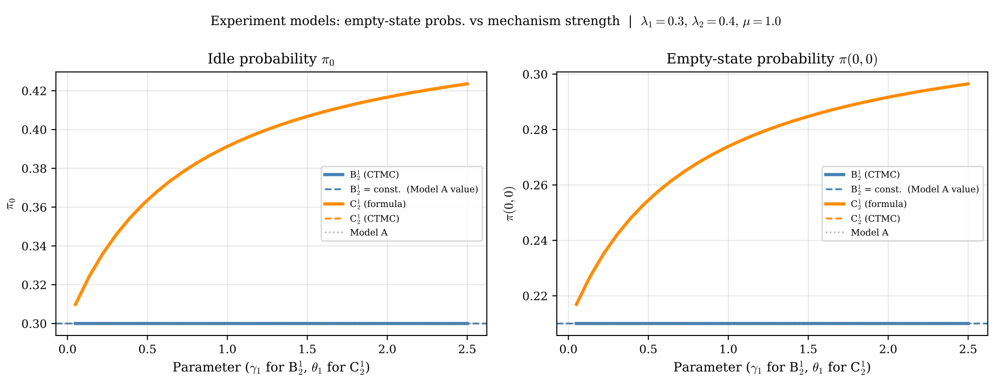

Figure 15: Empty-state probabilities π_0 and $\pi(0, 0)$ as the mechanism parameter increases from 0 to 2.5. *Left*: B_2^1 (blue) maintains $\pi_0 = 1 - \rho$ exactly for all γ_1 — jockeying conserves total customers. C_2^1 (orange) shows π_0 increasing with θ_1: abandonment clears the priority queue faster, raising the server's idle fraction. *Right*: same story for $\pi(0, 0)$. Solid orange = closed-form formula (303); dashed orange = CTMC. The two are indistinguishable (max error $< 10^{-4}$).

12 Experiment: models with a single active customer

The standard departure mechanisms studied so far — abandonment at rate $\theta_1 n_1$ (Model C_2) and jockeying at rate $\gamma_1 n_1$ (Model B_2) — produce *linear* departure rates proportional to the queue length. In the PGF equation this gives rise to derivative terms $xy(\cdot)\partial P/\partial x$, making the equation a first-order ODE and requiring the integrating-factor machinery of Sections 7–8.

This section explores a simpler variant: only the *first customer in line* (the head of Queue 1) can abandon or jockey, at a fixed rate independent of queue length. The aggregate departure rate is then $\theta_1 \cdot \mathbf{1}_{\{n_1 \geq 1\}}$ (not $\theta_1 n_1$), and the corresponding PGF contribution is *algebraic* rather than differential. The resulting fundamental equation can be solved directly by the kernel method, giving rational closed-form expressions for $P(x, y)$.

Models B_2^1 and C_2^1.

- **Model B_2^1:** $\gamma_2 = \theta_1 = \theta_2 = 0$, $\gamma_1 > 0$. Only the customer at the head of Queue 1 (the one that would be served next) may jockey to Queue 2, at rate γ_1. All other waiting Class-1 customers remain in Queue 1.
- **Model C_2^1:** $\gamma_1 = \gamma_2 = \theta_2 = 0$, $\theta_1 > 0$. Only the customer at the head of Queue 1 may abandon the system, at rate θ_1. All other waiting Class-1 customers are patient.

In both models the service discipline, arrival processes, and Class-2 dynamics are identical to Models B_2 and C_2 respectively.

12.1 Balance equations

Model B_2^1

The jockeying transition $(n_1, n_2) \to (n_1 - 1, n_2 + 1)$ occurs at the fixed rate γ_1 whenever $n_1 \geq 1$. The balance equations on S are:

$$\begin{aligned}&[\lambda_1 + \lambda_2 + \mu + \gamma_1]\,\pi(n_1, n_2) \\ &\quad= \mu\,\pi(n_1 + 1, n_2) + \lambda_1\pi(n_1 - 1, n_2) + \lambda_2\pi(n_1, n_2 - 1) + \gamma_1\pi(n_1 + 1, n_2 - 1)\end{aligned} \qquad n_1 \geq 1,\ n_2 \geq 1, \tag{274}$$

$$\begin{aligned}&[\lambda_1 + \lambda_2 + \mu + \gamma_1]\,\pi(n_1, 0) \\ &\quad= \mu\,\pi(n_1 + 1, 0) + \lambda_1\pi(n_1 - 1, 0)\end{aligned} \qquad n_1 \geq 1,\ n_2 = 0, \tag{275}$$

$$\begin{aligned}&[\lambda_1 + \lambda_2 + \mu]\,\pi(0, n_2) \\ &\quad= \mu\,\pi(1, n_2) + \gamma_1\pi(1, n_2 - 1) + \lambda_2\pi(0, n_2 - 1) + \mu\,\pi(0, n_2 + 1)\end{aligned} \qquad n_1 = 0,\ n_2 \geq 1, \tag{276}$$

$$[\lambda_1 + \lambda_2 + \mu]\,\pi(0, 0) = (\lambda_1 + \lambda_2)\pi_0 + \mu\,\pi(1, 0) + \mu\,\pi(0, 1), \tag{277}$$

$$(\lambda_1 + \lambda_2)\pi_0 = \mu\,\pi(0, 0). \tag{278}$$

The key structural difference from Model B_2 (Equation (24) with $\gamma_2 = \theta_i = 0$) is that the jockeying coefficient is γ_1 (constant) rather than $\gamma_1 n_1$ (linear in queue length).

Model C_2^1

The abandonment transition $(n_1, n_2) \to (n_1 - 1, n_2)$ occurs at the fixed rate θ_1 whenever $n_1 \geq 1$. The balance equations are:

$$\begin{aligned}&[\lambda_1 + \lambda_2 + \mu + \theta_1]\,\pi(n_1, n_2) \\ &\quad= (\mu + \theta_1)\pi(n_1 + 1, n_2) + \lambda_1\pi(n_1 - 1, n_2) + \lambda_2\pi(n_1, n_2 - 1)\end{aligned} \qquad n_1 \geq 1,\ n_2 \geq 1, \tag{279}$$

$$\begin{aligned}&[\lambda_1 + \lambda_2 + \mu + \theta_1]\,\pi(n_1, 0) \\ &\quad= (\mu + \theta_1)\pi(n_1 + 1, 0) + \lambda_1\pi(n_1 - 1, 0)\end{aligned} \qquad n_1 \geq 1,\ n_2 = 0, \tag{280}$$

$$\begin{aligned}&[\lambda_1 + \lambda_2 + \mu]\,\pi(0, n_2) \\ &\quad= (\mu + \theta_1)\pi(1, n_2) + \lambda_2\pi(0, n_2 - 1) + \mu\,\pi(0, n_2 + 1)\end{aligned} \qquad n_1 = 0,\ n_2 \geq 1, \tag{281}$$

$$[\lambda_1 + \lambda_2 + \mu]\,\pi(0, 0) = (\lambda_1 + \lambda_2)\pi_0 + (\mu + \theta_1)\pi(1, 0) + \mu\,\pi(0, 1), \tag{282}$$

$$(\lambda_1 + \lambda_2)\pi_0 = \mu\,\pi(0, 0). \tag{283}$$

In Model C_2, the coefficients $\mu + \theta_1(n_1 + 1)$ and $\theta_1 n_1$ grow with n_1. Here they are constants $\mu + \theta_1$ and θ_1 throughout; this is the departure point of the analysis.

12.2 Fundamental PGF equations

Lemma 3 (Fundamental equation for Model B_2^1)**.** *The joint PGF $P(x,y)$ satisfies*

$$\begin{aligned}&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2+\gamma_1y(x-y)\right]P(x,y)\\ &\quad=\quad[\mu(x-y)+\gamma_1y(x-y)]\,P_y(y)-\mu x(1-y)\,\pi(0,0).\end{aligned}\tag{284}$$

Proof. Multiply the interior equation (274) by $x^{n_1}y^{n_2}$ and sum over $n_1\geq 1$, $n_2\geq 1$. The jockeying term generates:

$$\begin{aligned}\text{Rate out: }\gamma_1\sum_{n_1\geq1,\,n_2\geq1}\pi(n_1,n_2)\,x^{n_1}y^{n_2}&=\gamma_1\big[P-P_y\big],\\ \text{Rate in: }\gamma_1\sum_{n_1\geq1,\,n_2\geq1}\pi(n_1+1,n_2-1)\,x^{n_1}y^{n_2}&=\gamma_1\,\tfrac{y}{x}\big[P-P_y\big]+(\text{boundary corrections}).\end{aligned}\tag{285}$$

The net contribution is $\gamma_1(1-y/x)(P-P_y)$. After repeating the procedure for the x-boundary (275), the y-boundary (276), and state (277), the boundary corrections cancel exactly (same algebra as in Lemma 1) and multiplying through by xy yields (284). □

Remark. Equation (284) contains *no derivative* of P. Contrast with Model B_2, whose fundamental equation has the term $\gamma_1xy(x-y)\partial P/\partial x$. The flat jockeying rate contributes the algebraic factor $\gamma_1y(x-y)$ to the coefficient of $P(x,y)$, reducing the functional equation to an algebraic one.

Lemma 4 (Fundamental equation for Model C_2^1)**.** *The joint PGF $P(x,y)$ satisfies*

$$\begin{aligned}&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2+\theta_1y(x-1)\right]P(x,y)\\ &\quad=\quad[\mu(x-y)+\theta_1y(x-1)]\,P_y(y)-\mu x(1-y)\,\pi(0,0).\end{aligned}\tag{286}$$

Proof. The abandonment term in (279) contributes:

$$\begin{aligned}\text{Rate out: }\theta_1\sum_{n_1\geq1,\,n_2\geq1}\pi(n_1,n_2)\,x^{n_1}y^{n_2}&=\theta_1\big[P-P_y\big],\\ \text{Rate in: }\theta_1\sum_{n_1\geq1,\,n_2\geq1}\pi(n_1+1,n_2)\,x^{n_1}y^{n_2}&=\theta_1x^{-1}\big[P-P_y\big]+(\text{boundary corrections}).\end{aligned}\tag{287}$$

The net contribution is $\theta_1(1-x^{-1})(P-P_y)$. Combining all boundary equations as before and multiplying by xy yields (286). □

Remark. Equation (286) likewise contains no derivative of P. The term replacing $\theta_1xy(x-1)\partial P/\partial x$ of Model C_2 is the algebraic factor $\theta_1y(x-1)$ in the coefficient of $P(x,y)$.

12.3 Closed-form solution via the kernel method

Both fundamental equations have the structure

$$D(x,y)\,P(x,y)\;=\;R(x,y)\,P_y(y)-\mu x(1-y)\,\pi(0,0),\tag{288}$$

where $D(x,y)$ is polynomial and $R(x,y)$ does not involve P. Since there is no derivative of P, this yields directly $P(x,y)=[R\,P_y-\mu x(1-y)\pi(0,0)]/D$ once $P_y(y)$ and $\pi(0,0)$ are known.

Step 1 — characteristic roots

For each model, the factor y in D cancels and the inner quadratic in x is:

$$\mathrm{B}_2^1:\quad \lambda_1x^2-\big(\lambda_1+\lambda_2(1-y)+\mu+\gamma_1\big)x+(\mu+\gamma_1y)=0,\tag{289}$$

$$\mathrm{C}_2^1:\quad \lambda_1x^2-\big(\lambda_1+\lambda_2(1-y)+\mu+\theta_1\big)x+(\mu+\theta_1)=0.\tag{290}$$

Both have the same form as Model A's quadratic (112), but with μ replaced by $\mu+\gamma_1 y$ (B_2^1) or $\mu+\theta_1$ (C_2^1). By Vieta's formulas, the product of the two roots is $(\mu+\gamma_1 y)/\lambda_1$ or $(\mu+\theta_1)/\lambda_1$ respectively. The negative-sign (inner) roots are:

$$x^*_{B_2^1}(y) = \frac{A_B(y) - \sqrt{A_B(y)^2 - 4\lambda_1(\mu+\gamma_1 y)}}{2\lambda_1}, \quad A_B(y) = \lambda_1 + \lambda_2(1-y) + \mu + \gamma_1, \tag{291}$$

$$x^*_{C_2^1}(y) = \frac{A_C - \sqrt{A_C^2 - 4\lambda_1(\mu+\theta_1)}}{2\lambda_1}, \quad A_C = \lambda_1 + \lambda_2(1-y) + \mu + \theta_1. \tag{292}$$

At $y=1$: $x^*_{B_2^1}(1) = 1$ and $x^*_{C_2^1}(1) = 1$ (both verified by direct substitution). At $\gamma_1 = 0$ (resp. $\theta_1 = 0$) both roots reduce to $x^*(y)$ of Model A (112).

Step 2 — boundary function $P_y(y)$

At $x = x^*_r(y)$ the denominator D vanishes. For P to be bounded, the numerator must also vanish. Solving for $P_y(y)$:

$$P_y^{B_2^1}(y) \;=\; \frac{\mu\, x^*_{B_2^1}(y)\,(1-y)\,\pi(0,0)}{\big(x^*_{B_2^1}(y) - y\big)\,(\mu+\gamma_1 y)}, \tag{293}$$

$$P_y^{C_2^1}(y) \;=\; \frac{\mu\, x^*_{C_2^1}(y)\,(1-y)\,\pi(0,0)}{(\mu+\theta_1 y)\, x^*_{C_2^1}(y) - y(\mu+\theta_1)}. \tag{294}$$

Setting $y = 0$: using $x^*_r(0)^2 \cdot \lambda_1 = \mu +$ (mechanism at $y=0$) and Vieta's product, one verifies $P_y(0) = \pi(0,0)$ in both cases, confirming the boundary condition.

Step 3 — joint PGF

Substituting (293) (resp. (294)) into (288) and simplifying, the factor $(x - x^*_r)$ cancels between numerator and denominator. Denoting $x^*_r = x^*_{B_2^1}$ or $x^*_{C_2^1}$, and x^+_r for the corresponding outer root:

$$P_{B_2^1}(x,y) \;=\; \frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\,\big(x^*_{B_2^1}(y) - y\big)\,\big(x^+_{B_2^1}(y) - x\big)}, \tag{295}$$

$$P_{C_2^1}(x,y) \;=\; \frac{\mu(\mu+\theta_1)\,(1-y)\,\pi(0,0)}{\lambda_1\,\big(x^+_{C_2^1}(y) - x\big)\,\big[(\mu+\theta_1 y)\,x^*_{C_2^1}(y) - y(\mu+\theta_1)\big]}. \tag{296}$$

Remark. Formula (295) is *structurally identical to Model A*:

$$P_A(x,y) \;=\; \frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\,(x^*_A(y) - y)\,(x^+_A(y) - x)}. \tag{297}$$

The only difference between Models A and B_2^1 is in the characteristic roots: replacing $\mu \to \mu + \gamma_1 y$ in the quadratic shifts the roots from x^*_A, x^+_A to $x^*_{B_2^1}, x^+_{B_2^1}$.

12.4 Empty-state probabilities

Model B_2^1

Jockeying conserves the total number of customers, so the same diagonal argument as in Model A (Equation (145)) gives $P(z,z) = \rho(1-\rho)/(1-\rho z)$, hence

$$\pi(0,0) \;=\; \rho(1-\rho), \qquad \pi_0 \;=\; 1-\rho. \tag{298}$$

Model C_2^1

Abandonment reduces the load, so conservation breaks and a different argument is needed. Setting $y = 1$ in (286): the factor $\mu x(1-y)$ on the right vanishes, and $R^{C_2^1}(x,1) = (\mu+\theta_1)(x-1)$. The left-hand polynomial in x factors as $-\lambda_1(x-1)(x-(\mu+\theta_1)/\lambda_1)$. Cancelling $(x-1)$:

$$P(x,1) \;=\; \frac{P_y(1)}{1-\rho_C\,x}, \qquad \rho_C \;\equiv\; \frac{\lambda_1}{\mu+\theta_1}. \tag{299}$$

This is a geometric marginal: Class-1's effective "service rate" is $\mu+\theta_1$ (completions plus first-customer abandonments). Evaluating at $x = 1$:

$$P(1,1) \;=\; \frac{P_y(1)}{1-\rho_C} \;=\; 1-\pi_0. \tag{300}$$

A second relation comes from setting $x = 1$ in (286) and using $D^{C_2^1}(1,y) = \lambda_2 y(1-y)$:

$$\lambda_2\,P(1,y) \;=\; \frac{\mu}{y}\big[P_y(y)-\pi(0,0)\big]. \tag{301}$$

Taking $y \to 1$ and using (300) and the idle balance $\lambda\pi_0 = \mu\pi(0,0)$:

$$\lambda_2(1-\pi_0) \;=\; \mu\big[P_y(1)-\pi(0,0)\big] \;=\; \mu\big[(1-\pi_0)(1-\rho_C)-\pi(0,0)\big]. \tag{302}$$

Substituting $\pi(0,0) = \lambda\pi_0/\mu$ and solving:

$$\pi_0 \;=\; \frac{(\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1}{\mu(\mu+\theta_1)+\lambda_1\theta_1}, \qquad \pi(0,0) \;=\; \frac{\lambda\big[(\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1\big]}{\mu\big[\mu(\mu+\theta_1)+\lambda_1\theta_1\big]}. \tag{303}$$

At $\theta_1 = 0$: $\pi_0 \to 1-\rho$ and $\pi(0,0) \to \rho(1-\rho)$, recovering Model A.
Stability requires $\pi_0 > 0$, i.e. $(\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1 > 0$, a *weaker* condition than $\rho < 1$.

12.5 Main results and comparison

Theorem 5 (Closed-form PGF for Models B_2^1 and C_2^1)**.** *Under stability, the joint PGF of the two-class non-preemptive priority queue is rational in both experiment models.*

(i) Model B_2^1. *With* $\pi(0,0) = \rho(1-\rho)$,

$$P_{B_2^1}(x,y) \;=\; \frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\,\big(x^*_{B_2^1}(y)-y\big)\,\big(x^+_{B_2^1}(y)-x\big)}, \tag{304}$$

where $x^*_{B_2^1}(y)$ and $x^+_{B_2^1}(y)$ are the two roots of (289)*. This has the* same rational form as Model A, *differing only in the characteristic roots ($\mu \to \mu+\gamma_1 y$ in the quadratic).*

(ii) Model C_2^1. *With $\pi(0,0)$ given by* (303)*,*

$$P_{C_2^1}(x,y) \;=\; \frac{\mu(\mu+\theta_1)\,(1-y)\,\pi(0,0)}{\lambda_1\,\big(x^+_{C_2^1}(y)-x\big)\,\big[(\mu+\theta_1 y)\,x^*_{C_2^1}(y)-y(\mu+\theta_1)\big]}, \tag{305}$$

where $x^*_{C_2^1}(y)$ and $x^+_{C_2^1}(y)$ are the roots of (290)*. At $\theta_1 \to 0^+$ the formula recovers Model A exactly.*

Table 5 summarises the structural differences.

Table 5: Key structural properties across all five models.

Property	**A**	$\mathbf{B}_2$	$\mathbf{B}_2^1$	$\mathbf{C}_2$	$\mathbf{C}_2^1$
Departure rate	—	$\gamma_1 n_1$	$\gamma_1 \cdot \mathbf{1}_{n_1 \geq 1}$	$\theta_1 n_1$	$\theta_1 \cdot \mathbf{1}_{n_1 \geq 1}$
PGF equation type	algebraic	ODE	**algebraic**	ODE	**algebraic**
$P(x, y)$ form	rational	integral+${}_1F_1$	**rational**	integral+${}_1F_1$	**rational**
Same form as Model A?	✓	$\times$	✓	$\times$	(similar)
$\pi(0, 0)$	$\rho(1-\rho)$	$\rho(1-\rho)$	$\rho(1-\rho)$	(224)	(303)
π_0	$1-\rho$	$1-\rho$	$1-\rho$	(225)	(303)
$P(z, z)$ closed form	yes	yes	yes	no	**via** (299)
Stability condition	$\rho < 1$	$\rho < 1$	$\rho < 1$	$\lambda_2 \mathbb{E}[B_C] < 1$	weaker than $\rho < 1$

The "first-customer-only" assumption converts both ODEs into algebraic equations. For B_2^1, the result is structurally identical to Model A — the only change is $\mu \to \mu + \gamma_1 y$ in the quadratic that defines the characteristic root. For C_2^1, the structure is similar but the extra factor $(\mu + \theta_1)$ in the numerator and the modified denominator reflect the broken conservation. In both cases all performance measures follow by direct differentiation of a rational function, with no quadrature required.

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